FILTERED LIE ALGEBROIDS AND PARABOLIC GEOMETRY

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Abstract. The filtered Lie algebroids introduced here generalize the filtered manifolds in Tohru Morimoto's study of those finite-type geometric structures known nowadays as parabolic geometries. This is a rich class generalizing conformal and projective structures. We propose a new abstraction of parabolic geometry based on Lie groupoids whose associated Lie algebroids are filtered. In particular, we present a novel prolongation procedure to recover the abstraction from an "underlying geometric structure", using semi-direct products in the Lie groupoid category. The construction is predicated on the observation that Lie groupoids with filtered Lie algebroids admit a semi-direct product structure whenever they also admit a Weyl gauge, an "oid" version of lifts of grading elements of a Lie algebra. A novel theme of our exposition is the parallel treatment of its infinitesimal and global aspects.

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This introduction is a work in progress. Some original sources may be missing. Help identifying these or any other corrections and criticism welcome.

1. Introduction

Parabolic geometry is an abstract framework that unifies a rich collection of finite-type geometric structures, such as conformal, projective, contact projective, CR, almost Grassmannian, almost quaternionic, conformal Lorenzian, almost Lagrangean, Lagrangean contact, Lie contact structures, finite-type n-plane fields, and many others [14, Chapter 4]. Among its important applications is the systematic derivation of many invariant differential operators for these geometries [28, 11, 29, 30]. This article is an outgrowth of our attempts to tame some of the theory's more technical aspects by approaching it from new points of view.

{firstfirst}

1.1. **The main idea.** According to Cartan's generalization of Klein's Elrangen program [26], geometric structures of finite type are viewed as "symmetries deformed by curvature". In the contemporary formulation, the object deformed is the principal bundle associated with some homogeneous space, G/H, as we recall in 1.2 below. The *parabolic* geometries arise when one chooses G to be semisimple, and $H \subset G$ to be a parabolic subgroup, objects in rich supply, in which case G/H is a *generalized flag manifold*. The alternative approach promoted here is to consider deformations of action groupoids, which in the transitive case have the form $G \times G/H$.

For the sake of concreteness, consider the conformal sphere S^n, a generalized flag manifold with action groupoid $\mathrm{SO}(n+1,n) \times S^n$. Now, differentiating conformal automorphisms, i.e., elements of $\mathrm{SO}(n+1,n)$, we obtain a natural Lie groupoid morphism $\mathrm{SO}(n+1,n) \times S^n \to \mathrm{CO}[TS^n]$ onto the frame groupoid $\mathrm{CO}[TS^n]$, consisting of conformal isomorphisms between tangent spaces. The kernel of this morphism is a Lie group bundle with commutative, simply-connected fibres. By arguments omitted here, it has a canonical identification with T^*S^n. A fundamental observation, following from general results presented in this article, is that this morphism splits, delivering an identification of $\mathrm{SO}(n+1,n) \times S^n$ with the semi-direct product $\mathrm{CO}[TS^n] \times T^*S^n$. This is significant, for in the "curved" case of an arbitrary conformal manifold M, where the action groupoid is gone, *a proxy for the action groupoid persists*, namely the semi-direct product $\mathrm{CO}[TM] \times T^*M$. This is what we call the *algebraic prolongation* of the conformal structure.

In the flat case, we can give the kernel T^*S^n the interpretation of all 2-jets of conformal automorphisms vanishing to first order. Relevant arguments can be found in [1]. This connects algebraic prolongation to the usual idea of introducing derivatives as new variables.

Prolongation allows us to recover, in the typical case, a parabolic geometry from it's "underlying geometric structure", such as the conformal structure above, also known as an *infinitesimal flag structure* [14]. For reasons explained at the end of 3.8, we prefer to call this the associated *Levi structure*.

Our version of prolongation is anticipated by the observation that natural candidates for parabolic geometries are certain Lie groupoids that *always* admit a semi-direct product structure. Indeed, splittings like the one above exist for a large class of Lie groupoids, even intransitive ones, provided they admit what we call a *Weyl gauge*. As a special case, we can show that the action groupoid of any generalized

flag manifold is a semi-direct product, a fact that does not seem to have been noticed before. Weyl gauges are discussed further in 1.4 below.

Algebraic prolongation methods for Cartan geometries, which are substitutes for the prolongation of PDEs, have a long and complicated history, some of which is recorded in [14, Appendix A]. The conceptually simple formulation of prolongation, along the lines just intimated, is the most important contribution of the present article. The idea of replacing principal bundles with Lie groupoids in the current context, dates back to Ehresmann's 1959 paper [19].[1]

{contemporary}

1.2. **The contemporary point of view.** In the principal bundle formulation, which originates in Ehresmann's earlier 1951 paper [18], a *Cartan geometry*, modeled on some homogeneous space G/H, is a smooth manifold M and a principal H-bundle P over M, equipped with: (i) A parallelism, understood as a V-valued one-form ω on P (the *connection*), required to be suitably compatible with the principal bundle structure; and (ii) A compatible identification of the vector space V with the infinitesimalization $\mathfrak{g}$ of G (the *model data*). Combining (i) and (ii), we can view ω as $\mathfrak{g}$-valued one-form, which is the object classically known as a *Cartan connection*. See, for example, [14, 6] for definitions (which will not be needed here).

In the case of a principal H-bundle $G \to G/H$, the Maurer-Cartan form provides a Cartan connection, and in the general case there is a notion of curvature, $d\omega - [\omega, \omega]_{\mathfrak{g}}$, which vanishes in the homogeneous case, by virtue of the Maurer-Cartan equations for G. Since it depends on the choice of model data (it involves the bracket on the Lie algebra $\mathfrak{g}$ of G) we refer to this as the *model curvature*; it's significance is that M is locally identifiable with the model G/H whenever the model curvature vanishes. (As explained later, an alternative notion of curvature, which does not depend on model data, will play a secondary role in this article.)

The connection and model data allow one to transfer certain structures from the model space, such as a G-invariant quadratic form on $\mathfrak{g}/\mathfrak{h}$, to structures on the underlying manifold M, for example a Riemannian metric, and this latter constitutes the "underlying geometric structure" of the Cartan geometry. In the parabolic case, the Lie algebra $\mathfrak{h}$ of H defines a filtration of $\mathfrak{g}$ which, among other things, implies that TM is non-trivially filtered in the typical case. A simple example is CR-geometry, where this filtration is defined by the contact structure $\mathcal{H} = T^{-1}M \subset T^{-2}M = TM$.

Different choices of ω may lead to the same underlying geometric structure, which leads to the first fundamental problem of the theory, how to choose ω so as to "normalize" the model curvature, for then ω (and its curvature, for example) encode basic invariants about the underlying geometry. The second fundamental problem is how to reconstruct an abstract Cartan geometry from the underlying geometric structure. Where this is possible one obtains a one-to-one correspondence between (isomorphism classes) of normalized Cartan geometries and the concrete geometries of interest, solving Cartan's Equivalence Problem for such those structures. The technical tool for making this reconstruction is known as *prolongation*, because of its roots in the procedure in the theory of partial differential equations (PDE's) of the same name.

[1] As far as we know, this paper predates the the introduction of Lie algebroids (essential in the present work) which in any case had not substantialy matured until the work of Jean Pradines in the late 1960's. Perhaps this explains why the groupoid approach did not gain traction at that time.

1.3. **The new approach.** In [24], Tohru Morimoto published a highly original attempt to systematize the construction and normalization of Cartan connections, applicable in the parabolic case. In it Morimoto formulates the notion of *filtered manifolds*, now a part of the contemporary formalism [14]. He also demonstrates the usefulness of separating the notions of *connection* and *model data*, a theme of our own work cited shortly, and the present exposition, but not emphasized more generally.

The present article revisits the construction and normalization of Cartan connections in parabolic geometry in the light of a new analysis of "firmly" filtered Lie algebroids, a natural generalization of Morimoto's filtered manifolds. In this recounting, Lie groupoids, whose corresponding infinitesimalization are filtered Lie algebroids, replace the principal bundles of the conventional approach, along the lines suggested in 1.1. Previous work of ours, reviewed in Section 5, shows how to view Lie algebroids or Lie groupoids as (possibly intransitive) "symmetries deformed by curvature", by equipping them with connections *without* model data [3, 4, 5, 7, 6] (but nevertheless called *Cartan connections*). In the groupoid case, the connection is a multiplicatively closed horizontal n-plane field, D, and in the infinitesimal case, it is a Koszul connection, ∇, suitably compatible with the Lie algebroid structure. The type of symmetry, if it exists, is not specified *a priori*, and the natural definition of curvature (the usual one for ∇) does not depend on a notion of model data. Here we introduce, for the first time in the "oid" context, a notion of compatible model data, and of model curvature, in which the model is specified ahead of time, as in the standard approach. With these basic ingredients we can imitate the standard theory, with some novelties discussed further below,

Our primary source for learning about parabolic geometry has been Čap and Slovák's seminal monograph [14], which has significantly informed the present work. In particular, prolongation, as described here, while new, is closer to the procedure described in [14] than to the earlier scheme of Morimoto. {intro_prolong}

1.4. **Weyl gauges and semi-direct products.** The general considerations leading to the Lie groupoid splittings alluded to above can be previewed here by considering the special Lie *algebra* case of our Theorem 4.3:

Theorem. *Let* $\cdots \supset \mathfrak{a}^{-1} \supset \mathfrak{a}^0 \supset \mathfrak{a}^1 \supset \mathfrak{a}^2 \supset \cdots$ *be a filtration*[2] *of some finite-dimensional Lie algebra* $\mathfrak{a}$*, and assume* $\mathrm{gr}(\mathfrak{a}^0)$ *admits a grading element* E*, necessarily lying in* $\mathrm{gr}(\mathfrak{a})_0$*. Moreover, assume that* E *has a lift* $\epsilon \in \mathfrak{a}^0$ *such that*

{invariance}

$$[\mathfrak{a}, \epsilon] \subset \mathfrak{a}^1. \tag{1}$$

Then $\mathfrak{a}^j \subset \mathfrak{a}$ *is an ideal for all* $j \geqslant 0$ *and* $\mathfrak{a}$ *is isomorphic to the semi-direct product* $\mathfrak{a}/\mathfrak{a}^1 \oplus_{\mathrm{ad}} \mathrm{gr}(\mathfrak{a}^1)$*, the isomorphism depending on the choice of* ϵ*.*

Recall here that an element E of a graded Lie algebra $\mathfrak{g} = \oplus_j \mathfrak{g}_j$ is a *grading element* if $[E, X] = jX$, whenever $X \in \mathfrak{g}_j$. Note that the theorem requires only a grading element of $\mathrm{gr}(\mathfrak{a}^0)$ and not $\mathrm{gr}(\mathfrak{a})$. When the additional invariance condition (1) is satisfied, we call ϵ a *Weyl gauge*.

If the condition about ideals is satisfied, necessary for the existence of a Weyl gauge, we say that $\mathfrak{a}$ is *firmly filtered*. This is a strong condition, but the right one

[2]See 2.1 for precisely what we mean here.

for our general setting, as we shall see already for homogeneous spaces G/H and other examples (Section 2). The theorem above has the following corollary:

> *A filtered Lie algebra $\mathfrak{a}$ admits a Weyl gauge if and only if it is firmly filtered and* $\mathrm{gr}(\mathfrak{a}^0)$ *admits an $\mathfrak{a}$-invariant grading element E.*

Weyl gauges have straightforward generalizations to filtered Lie algebroids and Lie groupoids, even intransitive ones. For us, a parabolic geometry is a firmly filtered transitive Lie groupoid, together with suitably compatible model data (Definition 3.5). The semisimplicity of the model data is enough to guarantee the existence of Weyl gauges, by a generalization of the italicized statement above. In the above super-simplification of the general scenario, $\mathfrak{a}/\mathfrak{a}^1$, together with its representation $\mathrm{gr}(\mathfrak{a}^1)$, play the role of the corresponding Levi structure.

The name "Weyl gauge" comes from the Weyl structures/connections in the standard theory of parabolic geometry [14], which generalize Weyl's own connections in conformal geometry. It may be there exists a correspondence between the new semi-direct product description and the constructions in [14] but this is not obvious to us. Here Weyl gauges ground the entire theory (and make sense before introducing any assumptions about semisimplicity) while in [14] prolongation is first developed independently of Weyl structures. We note that their semi-direct product structure leads to an explicit characterization of all parabolic geometries — indeed, of all transitive filtered Cartan geometries, a mild generalization (Theorem 7.5).

1.5. **Infinitesimal geometry.** An obvious advantage of the Lie groupoid approach is a direct relationship between global and infinitesimal representations of the geometry, namely a natural generalization of the relationship between Lie groups and Lie algebras. Indeed, many constructions in the present development are first conceptualized at the infinitesimal level, where they are easier to understand (and how we generally discovered them) and later globalized. There is even the possibility, not pursued here, of completely reducing the global theory to the infinitesimal one, using the Lie groupoid versions of Lie I and Lie II [15], provided one is happy to ignore certain topological distinctions between geometries blurred by restricting to the case of source-simply-connected Lie groupoids.

In the principal bundle approach, it is conventional to view something called the the *adjoint tractor bundle* A, as the corresponding infinitesimal object. For example, the (model) curvature of a parabolic geometry is there understood as an A-valued two-form. This seems natural because tractor bundles are associated bundles. In the Lie groupoid point-of-view the analogue of associated bundles are Lie groupoid representations, but there is no canonical representation of a Lie groupoid on its Lie algebroid. Stated differently, the Atiyah Lie algebroid of a principal bundle is not an associated bundle; it only becomes one (is identifiable with A) *after* choosing a Cartan connection. This essential difference between the two approaches has a related manifestation. As we recall in Section 5, a Cartan connection D on a Lie groupoid immediately infinitesimalizes to a Koszul connection ∇ on the corresponding Lie algebroid, without model data playing any role, but it seems this connection never appears explicitly in the Cartan geometry literature (outside of our own work); rather the focus is on the so-called tractor connection on A, which depends on its associated bundle structure. While the tractor connection has an analogue in the present approach, the infinitesimal Cartan connection ∇ and the adjoint tractor connection are different, except in the flat case. See 9.6(3).

{role}

1.6. **The role of model data.** Important ingredients of the present approach are the "oid" versions of classical Cartan connections, D and ∇ mentioned above. In [4] we outlined an ambitious program for constructing such Cartan connections for general geometric structures of finite type, without the need to prescribe a flat model *a priori*, using a direct PDE prolongation procedure. While some simple structures are amenable to this approach, the program has largely been unsuccessful. In the case of parabolic geometries, the specification of model data *a priori* appears to be essential: It provides a hook into the rich structure theory of semi-simple Lie algebras, simplifying the normalization process which is intractable in the earlier PDE approach.

An interesting feature of the current approach is that model data appears initially as a notion independent of the Lie groupoid Cartan connection D, but with which it must be compatible. Specifically, model data is a Lie algebra bundle structure on the underlying Lie algebroid that extends the canonical one on the kernel of the anchor; compatibility means invariance with respect to a self-representation of the Lie algebroid defined by D. As we were surprised to discover in the parabolic case, the model data *uniquely* determines the connection D, assuming a compatible such connection does exist (Theorem 10.1). A (global) parabolic geometry is accordingly encoded by just two structures *on its infinitesimalization*: the filtration and the compatible model data. This phenomenon is apparently peculiar to the parabolic case, and not true for all Cartan geometries, even filtered ones.

1.7. **Outline.** Section 2 introduces filtered Lie algebroids and the firmly filtered special case, and gives examples pertinent to the application to parabolic geometry. Section 3 formulates the Lie groupoid abstraction of parabolic geometries, and their reductions, Levi structures, and states the main theorems on existence, normalization and uniqueness, whose proofs are laid out in the remaining sections.

Section 4 generalizes the notion of Weyl gauges and the the semi-direct product structure theorem stated in 1.4 above. Ambiguities in the choice of gauge leads to the notion of *Weyl gauge transformations*, relevant to the uniqueness theorem for parabolic geometry, but these transformations are introduced in Section 3 under the guise of Lie groupoid conjugations, which are more concrete. So Section 4 concludes with a proof that these amount to same thing.

Section 5, which is logically independent of the preceding section, is mostly a summary our aforementioned work on Cartan connections on Lie groupoids and algebroids, in the absence of model data. Model data in the "oid" context is introduced in Section 6. A crucial technical result established in Section 6 is a method for constructing model data compatible with a given Cartan connection (Proposition 6.2).

Section 7 applies the semi-direct product theorems of Section 4 to give a characterization of transitive filtered Cartan geometries, a certain generalization of parabolic geometries: All such geometries are obtained by prolongation (semi-direct products) using a Levi structure and the following additional data: a splitting of the underlying filtered manifold, a splitting of the anchor of the Levi structure's Lie algebroid, and a *rho tensor*, which will be familiar from the standard theory.

Section 8 introduces *ψ-deformations*, which are a natural collection of transformations of a Lie groupoid or Lie algebroid which we can use to relate different choices

of model data, via push-forward. Unlike Weyl gauge transformations, these are necessarily *not* morphisms. They are defined by objects ψ which we view as sections of the kernel of $J^1\mathcal{G} \to \mathcal{G}$, where $J^1\mathcal{G}$ is the groupoid of one-jets of the underlying Lie groupoid $\mathcal{G}$ of the parabolic geometry, an interpretation that helped us to see how a compatible Cartan connection should transform to ensure compatibility with the ψ-deformed model data. This section concludes by establishing a relationship between ψ-deformations and Weyl gauge transformations that we need for uniqueness results.

Section 9 lays out the algebraic machinery of Kostant [21] regarding the adjoint representation of a parabolic subalgebra's reductive Levi subalgebra on the graded semisimple Lie algebra that contains it. This is the crucial algebraic ingredient for normalizing model curvature, which can be viewed as a 2-chain in a related complex of vector bundles. Section 9.3 onwards is mostly a reworking of parts of [14, §3.1]. The main point is Proposition 9.4, expressing the change in model curvature upon ψ-deforming the model data, in terms of a cochain complex differential. We conclude with a brief discussion of tractor bundles, so as to relate our Cartan connections on Lie algebroids with adjoint tractor connections.

In Section 10 we provide the remaining details needed to prove the results stated in Section 2.

Audience. We assume readers are familiar with Lie groupoid/algebroid basics — say, what is briskly summarized in the first two lectures of [15]. Other sources include [13, 23, 17] and the book [22]. In addition, parts of [6] describing the generalization of the adjoint action to Lie groupoids, may be helpful.

With the assistance of [10], we streamline most of what we need about semisimple Lie algebras and their parabolic subalgebras. What is ultimately needed in applications is non-trivial and deep, but also comprehensively summarized in [14].

Blanket assumptions. All manifolds are smooth, which means C^∞ or real-analytic, and have connected components sharing a common dimension. With the exception of Lie groupoids, manifolds are Hausdorff. All manifolds are second countable. In particular, all transitive Lie groupoids are automatically locally trivial (source maps are both surjective and submersive) [27, Proposition A.3] and so are gauge groupoids (groupoids defined by a principal bundle).

The kernel of the anchor of a Lie algebroid, which is variously known as the isotropy, vertex bundle, or adjoint bundle, is here called the *anchor kernel*.

2. Filtered Lie algebroids

{introduction}

In this section we introduce filtered Lie algebroids and the important special case of *firmly* filtered Lie algebroids, and give examples which include observations foreshadowing aspects of parabolic geometry to be presented in Section 3.

{filtered}

2.1. **Definition.** In this article, a filtration

$$\cdots A^{-1} \supset A^0 \supset A^1 \supset A^2 \supset \cdots$$

of some vector space or bundle A will always be understood to satisfy $\cap_j A^j = 0$ and $\cup_j A^j = A$. All filtrations are descending. If A is a vector bundle, then we call the filtration a *Lie algebroid filtration* if A is a Lie algebroid and:

F1. The induced filtration $\cdots \Gamma(A^{-1}) \supset \Gamma(A^0) \supset \Gamma(A^1) \supset \Gamma(A^2) \supset \cdots$ of the section space $\Gamma(A)$ is a filtration of Lie algebras, meaning $[X, Y] \in \Gamma(A^{j+k})$ for $X \in \Gamma(A^j)$ and $Y \in \Gamma(A^k)$.

F2. The kernel of the anchor contains A^0. {fa2}

A filtered Lie algebroid A is *firmly filtered* if A^0 and the filtration of A^0 are invariant under the canonical representation of A on its anchor kernel, i.e., if $\Gamma(A^j) \subset \Gamma(A)$ is an ideal for all $j \geqslant 0$.

Proposition. Let A be a filtered Lie algebroid. Then:

(1) The associated graded vector bundle $\mathrm{gr}(A)$ is a bundle of graded Lie algebras (a smooth field of graded Lie brackets). {b1}

(2) If A is firmly filtered, then $\mathrm{gr}(A)$ is the Lie algebra fibre-wise direct sum of its negative and non-negative parts, i.e., $\oplus_{i<0} \mathrm{gr}(A)_i$ and $\oplus_{i\geqslant 0} \mathrm{gr}(A)_i$ are fibre-wise ideals of $\mathrm{gr}(A)$. {rt2}

Remark. If the fibres of $\mathrm{gr}(A)$ are pairwise isomorphic Lie algebras, then $\mathrm{gr}(A)$ is a bona fide Lie algebra bundle, by, e.g., [22, Theorem 6.4.5]; this will be the case in all the examples discussed in this article. If A is firmly filtered, transitive, and *integrable*, then $\mathrm{gr}(A^0)$ must be a Lie algebra bundle, because the canonical representation of A on A^0 will drop to a representation of A on $\mathrm{gr}(A^0)$, and the latter integrates to a representation of the transitive (and so locally trivial) integrating Lie groupoid $\mathcal{G}$ having simply-connected source fibres. These comments notwithstanding, local triviality of Lie algebroid gradings will not play an essential role here.

Proof. Let $X \in \Gamma(A^j)$ and $Y \in \Gamma(A^k)$ be given. To prove (1), it suffices to show that

$$[fX, Y] \equiv f[X, Y] \mod \Gamma(A^{j+k+1})$$

for any smooth function f on the base manifold. If $\#$ denotes the anchor, then the necessary and sufficient condition is that

$$df(\#Y)X \in \Gamma(A^{j+k+1}).$$

If $k \geqslant 0$ then this is holds by F2. If instead $k \leqslant -1$, then $j + k + 1 \leqslant j$, implying $A^j \subset A^{j+k+1}$ and the condition is still satisfied.

Suppose A is firmly filtered. Then it is clear that $\oplus_{i<0} \mathrm{gr}(A)_i$ and $\oplus_{i\geqslant 0} \mathrm{gr}(A)_i$ are fibre-wise subalgebras of $\mathrm{gr}(A)$. To prove (2), it suffices to show that sections of one of these commute with sections of the other. To this end, suppose $X \in \Gamma(A^k)$ and $Y \in \Gamma(A^j)$, with $k < 0$ and $j \geqslant 0$. Then $j + k + 1 \leqslant j$, which implies $A^j \subset A^{j+k+1}$. Since A is firmly filtered, $\Gamma(A^j) \subset \Gamma(A)$ is an ideal, so that $[X, Y] \in \Gamma(A^j) \subset \Gamma(A^{j+k+1})$. It follows that $[\mathrm{gr}_j(X), \mathrm{gr}_k(Y)] = 0$. □

Notation (Bounds on degree)**.** Suppose A is a *graded* vector space or vector bundle. Then, we write $k \leqslant A$ if $A_i = 0$ for all $i < k$ and $A \leqslant k$ if $A_i = 0$ for all $i > k$. For example, if $A = A_{-2} \oplus A_{-1} \oplus A_0$, then we write $-2 \leqslant A \leqslant 0$. We write $|A| = k$ if $-k \leqslant A \leqslant k$ and $A_{-k} \neq 0 \neq A_k$.

If A is merely *filtered*, then we write $A \leqslant k$ if $\mathrm{gr}(A) \leqslant k$, etc. For example, for filtered A, we write $|A| = 1$ if $\mathrm{gr}(A) = \mathrm{gr}(A)_{-1} \oplus \mathrm{gr}(A)_0 \oplus \mathrm{gr}(A)_1$ with $\mathrm{gr}(A)_{\pm 1} \neq 0$.

The remainder of the section is devoted to examples of filtered Lie algebroids. All but the first class of examples are always firm.

2.2. **Filtered Lie algebras and Lie algebra bundles.** Evidently, every filtered Lie algebra, and more generally every filtered Lie algebra bundle, is a filtered Lie algebroid.

2.3. **Filtered manifolds.** For any Lie algebroid filtration of a tangent bundle TM, we have $TM \leqslant -1$, by Axiom F2. These filtrations are trivially firm. A smooth manifold M whose tangent bundle is so structured is called a *filtered manifold*, a notion introduced and analyzed by Morimoto [24], and appearing prominently in [14]. The anchor $\#\colon A \to M$ of any *transitive* filtered Lie algebroid A pushes the filtration down to a filtration of TM. That is, the base of any transitive filtered Lie algebroid is a filtered manifold.[3]

{yt}
{trivial}

2.4. **Trivial filtrations.** Suppose A is any Lie algebroid whose anchor kernel has constant rank (e.g., is transitive). Choosing A^0 to be this kernel, and defining $A^j = A$ for $j < 0$ and $A^j = 0$ for $j > 0$, we get a firm Lie algebroid filtration with $-1 \leqslant A \leqslant 0$ that we call the *trivial filtration* of A.

{eg2}

2.5. **Filtered Klein geometries.** For a more interesting class of examples, let G be a Lie group with Lie algebra $\mathfrak{g}$ and $P \subset G$ a closed subgroup. Furthermore, suppose that $\mathfrak{g}$ admits a P-invariant filtration such that $\mathfrak{g}^0$ is the Lie algebra of P. In particular, as recalled in 3.2, a canonical filtration of this kind exists whenever G/P is a generalized flag manifold. Write $M = G/P$ and let $A = \mathfrak{g} \times M$ be the action algebroid. Then we obtain a filtration of A, as a vector bundle, by defining $A^j(m) = \mathrm{Ad}_g(\mathfrak{g}^j) \times \{m\}$, where $g \in G$ is any element satisfying $m = g \cdot m_0$ and m_0 is the left coset of P containing the identity of G. It is not hard to see that A is a firmly filtered Lie algebroid (Proposition 5.5).

Significantly, $\mathrm{gr}(A)$, which will be a graded Lie algebra bundle, is *not* modeled on $\mathrm{gr}(\mathfrak{g})$ but modeled on the ordinary direct sum of the graded Lie algebras $\oplus_{i<0} \mathrm{gr}(\mathfrak{g})_i$, re-equipped with *negative* bracket, and $\oplus_{i\geqslant 0} \mathrm{gr}(\mathfrak{g})_i$. So $\mathrm{gr}(A)$ has *two* different, although related, Lie algebra bundle structures: the one inherited from the filtered Lie algebroid, and a "flat" one inherited from $\mathfrak{g}$. On A^0 they coincide.

Note that the filtration of A is firm, irrespective of whether $\mathfrak{g}_j \subset \mathfrak{g}$ is an ideal for any j!

{projective}

2.6. **Projective structures.** Let $A \subset J^2(TM)$ be the abstract homogeneous PDE you solve to find the infinitesimal isometries of some projective structure on M. Then A is a subalgebroid mapped isomorphically onto $J^1(TM)$ by the projection $J^2(TM) \to J^1(TM)$, whose kernel intersects A in the image A^1 of the natural injection $j\colon T^*M \to S^2(T^*M) \otimes TM$ defined by

$$j(\alpha)(V_1, V_2) = \alpha(V_1)V_2 + \alpha(V_2)V_1.$$

See [4, §5.9]. If we define A^0 to be the anchor kernel of A, and $A^{-1} = A$, we get a firm Lie algebroid filtration of A with $|A| = 1$. The corresponding graded Lie algebra bundle is

$$\mathrm{gr}(A) = TM \oplus \mathfrak{gl}(TM) \oplus T^*M,$$

where TM and T^*M are commutative, $\mathfrak{gl}(TM) \oplus T^*M$ is the Lie algebra bundle semi-direct product, and TM and $\mathfrak{gl}(TM) \oplus T^*M$ are ideals, by Proposition 2.1(2).

[3] One can drop the requirement of transitivity, but only at the expense of dropping our requirement $\cap_j A_j = 0$ in the definition of a filtration.

While not immediately obvious, $\mathrm{gr}(A)$ has a second, natural, *semisimple,* Lie algebra bundle structure, modeled on $\mathfrak{sl}(n+1,\mathbb{R})$, $n=\dim(M)$, and compatible with the grading. The bracket for this structure, which we will denote T, extends the existing bracket on $\mathrm{gr}(A^0)=\mathfrak{gl}(TM)\oplus T^*M$ with the new relations

{projective_bracket}

$$T(C,V)=CV,\quad T(V,\alpha)=j(\alpha)V, \tag{1}$$

for sections C, V and α of $\mathfrak{gl}(TM)$, TM and T^*M respectively. Significantly, this bracket coincides with the flat one referenced in 2.5, when M is the projective sphere, viewed as a homogeneous $\mathrm{SL}(n+1,\mathbb{R})$-space.

The fact that the Lie algebra $\mathfrak{sl}(n+1,\mathbb{R})$ for the full symmetry of the homogeneous model persists in some form in the "curved" case is a surprising but crucial one, and is shared by all parabolic geometries.

{conformal}

2.7. **Conformal structures.** Let $B\subset J^1(TM)$ be the abstract homogeneous PDE you solve to find the infinitesimal isometries of a conformal structure on M [4, Section 12]. Then B is a transitive subalgebroid whose anchor kernel $\mathfrak{co}(TM)\subset\mathfrak{gl}(TM)$ consists of the infinitesimal conformal automorphisms of tangent spaces. Now B acts on TM via restricted adjoint action, and so acts on T^*M. Regarding T^*M as a commutative Lie algebra bundle, we construct, in the category of Lie algebroids, the semi-direct product $A=B\oplus_{\mathrm{ad}}T^*M$ (what we shall come to call the *algebraic prolongation* of B). Then, if we put $A^1=T^*M$, $A^0=\mathfrak{co}(TM)\oplus T^*M$ (the anchor kernel of A), and $A^{-1}=A$, then A becomes a firmly filtered Lie algebroid with $|A|=1$. The corresponding graded Lie algebra bundle is

$$\mathrm{gr}(A)=TM\oplus\mathfrak{co}(TM)\oplus T^*M,$$

where $\mathfrak{co}(TM)\oplus T^*M$ is the semi-direct product, and TM and $\mathfrak{co}(TM)\oplus T^*M$ are ideals.

Again, there is also a semisimple Lie algebra bundle structure on $\mathrm{gr}(A)$ compatible with the grading, in this case modeled on $\mathfrak{so}(n+1,1)$, and coinciding with the flat one of 2.5 when M is the conformal sphere, viewed as a homogeneous $\mathrm{SO}(n+1,1)$-space. After redefining the injection j above,

$$j(\alpha)(V_1,V_2)=\alpha(V_1)V_2+\alpha(V_2)V_1-\sigma(V_1,V_2)\sigma^{-1}(\alpha),$$

now a map $j\colon T^*M\to(S^2(T^*M)\otimes TM)\cap(T^*M\otimes\mathfrak{co}(TM))$, the semisimple bracket T extends the existing bracket on $\mathrm{gr}(A^0)=\mathfrak{co}(TM)\oplus T^*M$, with the same additional relations (1) above. Here σ is any metric in the conformal class.

3. Parabolic geometries. Part I

{parabolic_summary}

In this section we give our definition of parabolic geometries à la Lie groupoids and outline our main results about them. Proofs appear after various restatements given in Section 10.

{filtered_groupoids}

3.1. **Filtered Lie groupoids.** We will call a Lie groupoid $\mathcal{G}$ over M *filtered* if its associated Lie algebroid A is filtered. Now $\mathcal{G}$ acts on elements of the anchor kernel of A, and so on elements of A^0, although this action need not preserve A^0 in general. This action will be denoted Ad. The Lie groupoid is *firmly filtered* if A^j is $\mathcal{G}$-invariant

under Ad, for all $j \geqslant 0$, a slightly stronger condition than firmness of the filtration of A.

{canonical_filtrations_}

3.2. **Parabolic subalgebras and canonical filtrations.** Let $\mathfrak{g}$ be a finite-dimensional semisimple Lie algebra over $\mathbb{C}$ or $\mathbb{R}$. Then $\mathfrak{p} \subset \mathfrak{g}$ is *parabolic* if its Killing polar $\mathfrak{p}^\perp$ is a nilpotent subalgebra.[4] Recall these subalgebras are intimately connected with filtrations of $\mathfrak{g}$: Define $\mathfrak{g}^0 = \mathfrak{p}$ and $\mathfrak{g}^1 = \mathfrak{p}^\perp$, and inductively define $\mathfrak{g}^j = [\mathfrak{g}^1, \mathfrak{g}^{j-1}]$, for $j \geqslant 2$, and $\mathfrak{g}^j = \{X \in \mathfrak{g} \mid [X, \mathfrak{g}^1] \subset \mathfrak{g}^{j+1}\}$, for $j \leqslant -1$. Then:

Proposition.

(1) The above definitions make $\mathfrak{g}$ into a filtered Lie algebra with $|\mathfrak{g}| \leqslant k$ for some $k \geqslant 0$, and this is called the *canonical filtration* associated with $\mathfrak{p}$.

{lqq}

(2) In any filtration of $\mathfrak{g}$ defined by a grading $\mathfrak{g} = \oplus_{j\in\mathbb{Z}}\mathfrak{g}_j$, $\mathfrak{g}^0 = \oplus_{j\geqslant 0}\mathfrak{g}_j$ is a parabolic subalgebra. Moreover, this filtration is the precisely the associated canonical filtration, provided $\mathfrak{g}_1$ generates $\mathfrak{g}^1 = \oplus_{j\geqslant 1}\mathfrak{g}_j$.

An elementary proof of this known result is given in Appendix A. Part (2) is useful for identifying parabolic subalgebras in examples.

Another crucial significance of a canonical filtration of semisimple $\mathfrak{g}$ is that passing to the associated grading incurs no essential loss of information, as $\mathrm{gr}(\mathfrak{g})$ turns out to be isomorphic to $\mathfrak{g}$ (Lemma A.2).

Remark. In fact, for a parabolic $\mathfrak{p}$, $\mathfrak{p}^\perp$ is the nilpotent radical of $\mathfrak{p}$, i.e., $\mathfrak{p}/\mathfrak{p}^\perp = \mathfrak{g}^0/\mathfrak{g}^1$ is the reductive Levi quotient of $\mathfrak{p}$. We won't need this fact here, but it explains the name "Levi structure" introduced below.

{parabolic_}

3.3. **Generalized flag manifolds.** Suppose that G is a semisimple Lie group and $P \subset G$ a parabolic subgroup. By definition, this means P has a parabolic Lie algebra $\mathfrak{p} \subset \mathfrak{g}$ and that the associated canonical filtration of $\mathfrak{g}$ is P-invariant under Ad (it is automatically $\mathfrak{p}$-invariant under ad). The quotient $M = G/P$ is known as a *generalized flag manifold*. The projective and conformal spheres are examples. Generalized flag manifolds constitute the flat models of parabolic geometries.

The P-invariance of the canonical filtration leads to a firm filtering of the action algebroid $A := \mathfrak{g} \times M$, as described in 2.5. It is easy to see that the action groupoid $\mathcal{G} := G \times M$ is also firmly filtered. Of particular relevance here are two structures on $\mathcal{G}$, to be generalized below: (i) The canonical horizontal n-plane field D on $\mathcal{G}$, tangent to the foliation $G \times \{m\}$, $m \in M$ (and its infinitesimalization, the flat Koszul connection ∇ on the trivial bundle A) — an example of a Lie groupoid (algebroid) *Cartan connection*; and (ii) The trivial Lie algebra bundle structure on the infinitesimalization $A = \mathfrak{g} \times M$ of $\mathcal{G}$ — an example of *model data*.

{preview}

3.4. **Cartan connections and model data.** Let $\mathcal{G}$ be an arbitrary Lie groupoid. As we detail in Section 5, a horizontal n-plane field D on $\mathcal{G}$ is a *Cartan connection* if it is suitably compatible with the Lie groupoid structure [6], [4, Appendix A]. It's infinitesimalization, called an *(infinitesimal) Cartan connection*, is a Koszul connection ∇ on the corresponding Lie algebroid A, suitably compatible with the Lie algebroid structure [3]. Curvature of ∇ is the obstruction to symmetry: When ∇ is flat, A is a (possibly twisted) action algebroid and, under certain technical conditions, $\mathcal{G}$ is a

[4] Readers may understand parabolic subalgebras as those containing a Borel (maximal solvable) subalgebra. That, over $\mathbb{C}$ or $\mathbb{R}$, this is equivalent to the stated definition is proven in [10].

(possibly twisted) action groupoid. For details and variations on these results, see Section 5.

Associated with any Cartan connection D is a representation $\bar{D}$ of $\mathcal{G}$ on A, extending the canonical action Ad of $\mathcal{G}$ on the anchor kernel of A, and we say D is *compatible* with a filtration of $\mathcal{G}$ (in the sense above) if the filtration of A is $\bar{D}$-invariant. The filtration of $\mathcal{G}$ is necessarily firm in that case (Proposition 5.4).

Similarly, there is associated with any infinitesimal Cartan connection ∇, a representation $\bar{\nabla}$ of A on itself, extending the canonical representation of A on the anchor kernel. If (A, ∇) is the infinitesimalization of some $(\mathcal{G}, D)$ as above, then $\bar{\nabla}$ is the infinitesimalization of the preceding global representation $\bar{D}$. In any case, it is related to ∇ by a simple formula, $\bar{\nabla}_X Y = \nabla_{\#Y} X + [X, Y]$, useful in computations. Here # denotes the anchor.

A smooth field τ of Lie brackets on a Lie algebroid A is *model data* if $\tau(X, Y) = [X, Y]$ whenever X and Y both lie in the kernel of the anchor of A. In particular, if A is transitive, any Lie algebra bundle structure on A extending the canonical Lie algebra bundle structure of the anchor kernel will be model data, and without further comment all model data in the present article is understood to be of this type.

Model data τ is *compatible* with a filtration of A if this is also a filtration with respect to τ, i.e., $\tau(X, Y) \in A^{j+k}$ whenever, $X \in A^j$ and $Y \in A^k$ have a common base point. We say τ is compatible with a Cartan connection D if τ is $\bar{D}$-invariant.

{parabolic_defined}

3.5. **Parabolic geometries defined.** A *parabolic geometry* is a transitive, firmly filtered, Lie groupoid $\mathcal{G}$ over some smooth manifold M, with associated Lie algebroid denoted A, and filtration-compatible model data τ, such that:

{parabolic2}

(1) The Lie algebra model for τ is semisimple.
(2) A^0 is the kernel of the anchor of A and a fibre-wise parabolic subalgebra of (A, τ). The filtration of A is, fibre-wise, the associated canonical filtration.
(3) There exists a Cartan connection D on $\mathcal{G}$ compatible with τ.

{parabolic1}
{parabolic4}

Proposition (Propositions 10.1, Theorem 10.2, Corollary 10.3)**.**

(4) Assuming M is connected, it suffices to check that (2) holds over a single point $m \in M$.
(5) A τ-compatible connection D is necessarily compatible with the filtration of A, and strongly compatible with τ.

An infinitesimal Cartan connection ∇ on a Lie algebroid A is *strongly compatible* with model data τ if it is compatible with τ, and if $\bar{\nabla}_Z X = \tau(Z, X)$ whenever Z lies in the anchor kernel of A. A global Cartan connection D is *strongly compatible* with τ if its infinitesimalization ∇ is. For more on this notion, see 6.1.

Parabolic geometries are generalized by *filtered Cartan geometries*, introduced in Section 7. However, the following result is apparently peculiar to the parabolic case:

Theorem (Theorem 10.2)**.** *In any parabolic geometry, a compatible Cartan connection D is uniquely determined by the model data τ.*

One may view this fact as mirroring the classical result that every Riemannian metric is compatible with a unique torsion free TM-connection.

{regularity _}

3.6. **Regularity.** Let A be any *transitive* filtered Lie algebroid. In that case the anchor $\#\colon A \to TM$ induces an isomorphism $\oplus_{i<0}\operatorname{gr}(A)_i \cong \operatorname{gr}(TM)$, and model data τ compatible with the filtration is said to be *regular* if the induced bracket on $\operatorname{gr}(TM)$ is minus one times the canonical bracket (the one induced by the Jacobi-Lie bracket on vector fields).[5] It is not hard to show that the canonical model data associated with any filtered Klein geometry, and in particular the canonical model data associated with a generalized flag manifold, is regular (Proposition 5.5).

A parabolic geometry is *regular* if the associated model data τ is regular.

{reduction _}

3.7. **Reduction.** The concrete manifestations of parabolic geometries (their "underlying geometric structures") are typically obtained by a applying a reduction procedure. Let $(\mathcal{G},\tau)$ be a parabolic geometry and D the compatible Cartan connection.

Proposition (Proposition 10.4)**.** In a parabolic geometry $(\mathcal{G},\tau)$ let $\mathcal{G}^1 \subset \mathcal{G}$ denote the source-connected subgroupoid integrating A^1. Then $\mathcal{G}^1$ is a bundle of simply-connected nilpotent Lie groups; it is also a normal subgroupoid of $\mathcal{G}$, so that $\mathcal{G}/\mathcal{G}^1$ is a Lie groupoid, also transitive.

Remark. It follows from Remark 3.2 that that A^0/A^1 is fibre-wise reductive, so that $\mathcal{G}/\mathcal{G}^1$ has reductive vertex groups.

Since we suppose $\mathcal{G}$ is firmly filtered, we have a grading-preserving representation, Ad, of $\mathcal{G}$ on $\operatorname{gr}(A^0)$. Moreover, it is easy to see that $\mathcal{G}^1$ acts trivially here (Proposition 4.3(2)). So Ad descends to a representation of $\mathcal{H} := \mathcal{G}/\mathcal{G}^1$ on $\operatorname{gr}(A^0)$. This canonical representation is extended by a representation of $\mathcal{H}$ on $\operatorname{gr}(A)$ induced by the representation $\bar{D}$ of $\mathcal{G}$ on A, which we denote by R^D.

The observations of the preceding paragraph carry over to the canonical, infinitesimal representation, ad, of A on A^0, delivering a canonical representation of $B := A/A^1$ on $\operatorname{gr}(A^0)$ and an extension R^∇ of this representation to one of B on $\operatorname{gr}(A)$, which are simply infinitesimalizations of the global ones: $R^\nabla = dR^D$.

If we write $T := \operatorname{gr}_0(\tau)$, then $(\operatorname{gr}(A), T)$ is a graded Lie algebra bundle with the same semisimple Lie algebra model as (A,τ) (by Lemma A.2). By transitivity, we have $\oplus_{i<0}\operatorname{gr}(A)_i \cong \operatorname{gr}(TM)$, so we obtain a representation of $\mathcal{H}$ on $\operatorname{gr}(TM)$. On the other hand, we can use the Killing form of T, or any fixed scalar multiple, to identify $\operatorname{gr}(A^1) = \oplus_{i>0}\operatorname{gr}(A)_i$ with $(\oplus_{i<0}\operatorname{gr}(A)_i)^* \cong \operatorname{gr}(TM)^* \cong \operatorname{gr}(T^*M)$ (see Lemma A.3). Under this identification, the representation R^D of $\mathcal{H}$ on $\operatorname{gr}(T^*M)$ must be the dual of the one on $\operatorname{gr}(TM)$, because the Killing form of T will be R^D-invariant, on account of the $\bar{D}$-invariance of τ. Finally, we notice that $A^0/A^1 = B^0$, where B^0 is the anchor of B.

In summary, we have

{soldering}

$$\operatorname{gr}(A) \cong \operatorname{gr}(TM) \oplus B^0 \oplus \operatorname{gr}(T^*M), \tag{1}$$

supporting the semisimple Lie algebra bundle structure T (c.f. Examples 2.6 and 2.7) and with $\mathcal{H}$ acting on each component.

[5] The minus sign requirement is connected to the convention adopted here, and in most Lie groupoid literature, that brackets on the Lie algebras of Lie groups are defined by *right*-invariant vector fields.

Remark. By our preceding Remark, B^0 is a fibre-wise reductive Levi subalgebra of the parabolic $B^0 \oplus \mathrm{gr}(T^*M)$.

{levi_}

3.8. **Levi structures.** A *Levi structure* consists of a filtered, transitive, Lie groupoid $\mathcal{H}$, whose Lie algebroid B has anchor kernel B^0 and satisfies $B \leqslant 0$ (see Notation 2.1), together with a semisimple Lie algebra bundle structure T on

{fug_}

$$\mathrm{gr}(TM) \oplus B^0 \oplus \mathrm{gr}(T^*M) \tag{1}$$

and a representation R of $\mathcal{H}$ on the same, such that:

(2) R preserves the obvious grading of (1).
(3) R extends the canonical representation of $\mathcal{H}$ on the anchor B^0 of its infinitesimalization B, while its restrictions to representations on $\mathrm{gr}(TM)$ and $\mathrm{gr}(T^*M)$ are dual.
(4) T is R-invariant.
(5) T extends the canonical Lie algebra bundle structure on B^0, i.e., $T(C, C') = [C, C']$, while T and R are compatible, in the sense that $T(C, X) = dR_C X$ and $T(C, Z) = -(dR_C)^* Z$. Here $C, C' \in \Gamma(B^0)$, $X \in \Gamma(\mathrm{gr}(TM))$, and $Z \in \Gamma(\mathrm{gr}(T^*M))$.
(6) The grading of (1) is compatible with T.
(7) The natural pairing of $\mathrm{gr}(TM)$ and $\mathrm{gr}(T^*M)$ coincides with that induced by some multiple of the Killing form of T.

{levi3_}

{levi6}

{killing_duality}

Notice that (6) and Proposition 3.2(2) ensure that $B^0 \oplus \mathrm{gr}(T^*M)$ is fibre-wise T-parabolic. It is not required that T restrict to the canonical Lie algebra bundle structure on $\mathrm{gr}(TM)$ (when TM gets the filtration pushed forward by the anchor of B). If this restriction is the *negative* of the canonical one, then the geometry is *regular*.

Proposition. Let $\mathcal{H}$, T and B assume the meanings of 3.7, and let $R := R^D$ be the grading-preserving representation of $\mathcal{H}$ on $\mathrm{gr}(TM) \oplus B^0 \oplus \mathrm{gr}(T^*M)$ described there. Then $(\mathcal{H}, T, R)$ is a Levi structure called the *Levi reduction* of $(\mathcal{G}, \tau)$.

Proof. Our observations under 3.7 establish everything that is not obvious, with the exception of property (5), which follows from the strong compatibility of τ and D. □

If B in the definition of a Levi structure is an arbitrary Lie algebroid (not necessarily integrated by some Lie groupoid $\mathcal{H}$) and R now denotes some *infinitesimal* action of B on (1), and we replace dR in (5) with R, then the preceding conditions define an *infinitesimal Levi structure* and, as we will see later, the immediately preceding proposition has an infinitesimal analogue. We have already encountered a simple example of an infinitesimal Levi structure, namely conformal geometry. It is easy to see this example globalizes:

Example (Conformal structures)**.** Define $B \subset J^1(TM)$ and T as in 2.7 and filter B trivially (see 2.4). If we let $\mathcal{H} \subset J^1(M \times M)$ denote the Lie groupoid of one-jets of local diffeomorphisms preserving the conformal structure to first order, and let $\mathcal{H}$ act on $\mathrm{gr}(TM) = TM$ by restricted adjoint action, and extend this to an action R on (1) in the obvious way, then $(\mathcal{H}, T, R)$ becomes a Levi structure.

In the next example the base manifold M is non-trivially filtered:

Example (n-plane fields with growth vector $(n, n(n+1)/2)$)**.** Consider, on a smooth manifold of dimension $n(n+1)/n$, an n-plane field $\mathcal{D} \subset TM$ that is maximally non-integrable. That is, at every $m \in M$, local sections of $\mathcal{D}$ and their Jacobi-Lie brackets, generate T_mM. Unlike contact structures, such an n-plane field is a geometric structure of finite type, admitting a canonical Cartan connection, as Bryant already showed in his Ph.D. thesis when $n = 3$ [8, 9].

The quotient $TM/\mathcal{D}$ and $\wedge^2\mathcal{D}$ have the same rank, and, by maximal non-integrability, the morphism $V_1 \wedge V_2 \mapsto [V_1, V_2] \bmod \mathcal{D}\colon \wedge^2 \mathcal{D} \to TM/\mathcal{D}$ is surjective, and whence an isomorphism. Taking $T^{-1}M = \mathcal{D}$ and $T^{-2}M = TM$, we make TM into a filtered Lie algebroid with $-2 \leqslant TM \leqslant -1$ and, on account of the isomorphism, have the identification $\mathrm{gr}(TM) \cong \wedge^2\mathcal{D} \oplus \mathcal{D}$ with grading-compatible bracket satisfying

$$[X_1, X_2] = X_1 \wedge X_2; \qquad X_1, X_2 \in \Gamma(\mathcal{D}). \tag{8}$$

{brillig}

To construct a Levi structure from this data, equip $\mathcal{D} \oplus (\mathbb{R} \times M) \oplus \mathcal{D}^*$ with a real quadratic form $(\,\cdot\,,\,\cdot\,)$, of signature $(n+1, n)$, by declaring $\mathcal{D}$ and $\mathcal{D}^*$ to be isotropic, $\mathcal{D} \perp (\mathbb{R} \times M) \perp \mathcal{D}^*$, $(X, Z) = X(Z)$, and $(f_1, f_2) = f_1 f_2$; here X lies in $\mathcal{D}$, Z in $\mathcal{D}^*$ and f_1, f_2 are smooth real functions. Then, it is not hard to show that the Lie algebra bundle of infinitesimal isometries of $(\,\cdot\,,\,\cdot\,)$, which has semisimple model $\mathfrak{so}(n+1, n)$, can be identified with

$$\wedge^2 \mathcal{D} \oplus \mathcal{D} \oplus \mathfrak{gl}(\mathcal{D}) \oplus \mathcal{D}^* \wedge^2 \mathcal{D}^* \cong \mathrm{gr}(TM) \oplus \mathfrak{gl}(\mathcal{D}) \oplus \mathrm{gr}(T^*M), \tag{9}$$

{lab}

with the filtration-compatible Lie bracket T generated by these relations:

$$\begin{gathered}
T(X_1, X_2) = -X_1 \wedge X_2, \qquad T(C, X) = R_C X := CX, \\
T(C, X_1 \wedge X_2) = R_C X_1, \wedge X_2 + X_1 \wedge R_C X_2 = R_C(X_1 \wedge X_2), \\
T(Z_1, Z_2) = Z_1 \wedge Z_2, \qquad T(C, Z) = R_C Z := -C^* Z, \\
T(C, Z_1 \wedge Z_2) = R_C Z_1 \wedge Z_2 + Z_1 \wedge R_C Z_2 = R_C(Z_1 \wedge Z_2), \\
T(X, Z) = Z \otimes X, \\
T(X_1 \wedge X_2, Z_1 \wedge Z_2) = Z_1(X_1) Z_2 \otimes X_2 + Z_2(X_2) Z_1 \otimes X_1 \\
-Z_2(X_1) Z_1 \otimes X_2 - Z_1(X_2) Z_2 \otimes X_1, \\
T(X_1 \wedge X_2, Z) = Z(X_1) X_2 - Z(X_2) X_1, \;\; T(X, Z_1 \wedge Z_2) = Z_1(X) Z_2 - Z_2(X) Z_1.
\end{gathered}$$

Here X, X_1, X_2 lie in $\mathcal{D}$, while Z, Z_1, Z_2 lie in $\mathcal{D}^*$, and C in $\mathfrak{gl}(\mathcal{D})$. We note that the fibre-wise parabolic subalgebra $\mathfrak{gl}(\mathcal{D}) \oplus \mathcal{D}^* \oplus \wedge^2\mathcal{D}^*$ is the stabilizer of the isotropic subbundle $\mathcal{D}^* \subset \mathcal{D} \oplus (\mathbb{R} \times M) \oplus \mathcal{D}^*$.

The frame groupoid $\mathcal{H}$ of $\mathcal{D}$ acts on each summand in (9) in the obvious way, delivering what is easily seen to be a Levi structure for this example (only (7) requires a little work). Note we have made choices that ensure the sign in the first relation is opposite to that in (8), to ensure regularity.

Unfortunately for us, the analogue of a *global* Levi structure is called an "infinitesimal flag structure" in [14]. Moreover, from the present point of view, "infinitesimal flag structure" seems a more apt name for the action algebroids described in 2.5, specialized to the parabolic case. To avoid all confusion, we eschew the term here.

{existence}

3.9. **Prolonging a Levi structure to a parabolic geometry.**

Theorem (Theorem 10.5)**.** *Every Levi structure $(\mathcal{H}, T, R)$ is the reduction of some parabolic geometry $(\mathcal{G}, \tau)$. If the Levi structure is regular, then the parabolic geometry can be chosen to be regular.*

The model data τ is not uniquely defined by the Levi structure, but determined up to choices of a suitable tensor $\mathsf{P} \in \Gamma(\mathrm{gr}(T^*M) \otimes \mathrm{gr}(T^*M))$ called the *rho tensor*, splitting of the anchor $\#\colon B \to TM$, and splitting of the filtration of TM. Here B is the Lie algebroid of $\mathcal{H}$. Suitable normalization of this data is addressed in 3.10 below.

The key idea in the theorem's proof is to observe that every parabolic geometry $(\mathcal{G}, \tau)$ has a semi-direct product structure, which follows from the guaranteed existence of *Weyl gauges*, a notion introduced in Section 4. Specifically, the representation R^D of $\mathcal{G}/\mathcal{G}^1$ on $\mathrm{gr}(T^*M)$, defined by the τ-compatible Cartan connection D, integrates to an action of $\mathcal{G}/\mathcal{G}^1$ on $\exp(\mathrm{gr}(T^*M))$, the Lie group bundle with simply-connected nilpotent model that integrates the nilpotent Lie algebra bundle $\mathrm{gr}(T^*M)$; and one obtains, after fixing a gauge, an isomorphism

$$\mathcal{G} \cong \mathcal{G}/\mathcal{G}^1 \times_{R^D} \exp(\mathrm{gr}(T^*M)),$$

where $\times_{R^D}$ indicates semi-direct product. This follows from Theorem 4.3, given later.

To recover the underlying Lie groupoid $\mathcal{G}$ associated with some parabolic geometry having $(\mathcal{H}, T, R)$ as reduction, we accordingly define $\mathcal{G} = \mathcal{H} \times_R \exp(\mathrm{gr}(T^*M))$, which we call the *algebraic prolongation* of the Levi structure. The more technical task is to reconstruct suitable model data τ such that $\mathrm{gr}_0(\tau) = T$. In fact, we state and prove a more general result, Theorem 7.5, characterizing arbitrary *filtered Cartan geometries*. This result includes an explicit description of τ and a compatible Cartan connection D, in terms of the aforementioned rho tensor P and splittings.

{normalization}

3.10. **Curvature and normalization.**

While the curvature of the infinitesimal Cartan connection ∇ is an obstruction to symmetry, and makes sense without the prescription of any model data, *model curvature* is more amenable to normalization. By definition, the model curvature of a parabolic geometry $(\mathcal{G}, \tau)$, with Lie algebroid A and compatible Cartan connection D, is the section κ of $\wedge^2 T^*M \otimes A$ defined by

$$\kappa(\#X, \#Y) = \operatorname{tor} \bar{\nabla}(X, Y) - \tau(X, Y); \qquad X, Y \in \Gamma(A),$$

where ∇ is the infinitesimalization of D. Here $\#$ is the anchor and

$$\operatorname{tor} \bar{\nabla}(X, Y) = \bar{\nabla}_X Y - \bar{\nabla}_X Y - [X, Y] = \nabla_{\#Y} X - \nabla_{\#X} Y + [X, Y].$$

That κ is well defined follows from the strong compatibility of τ and D. Significantly, $\kappa = 0$ implies $\operatorname{curv} \nabla = 0$ (Proposition 9.1). That is, κ is also an obstruction to symmetry, but of the kind prescribed *a priori* by the choice of the Lie algebra modeled by τ. The model curvature has the interpretation of a difference between a "deformed" Lie bracket, defined by the Cartan connection (see Theorem 5.3), and a bona fide Lie bracket, prescribed by the model data. The model curvature is the Lie algebroid analogue of "curvature", as that term is generally understood in the Cartan geometry literature.

As we have observed, there is considerable freedom in the choice of model data τ appearing in the existence Theorem 3.9, although we will always have $\mathrm{gr}_0(\tau) = T$. Since $A^0 \subset A$ is fibre-wise parabolic, we have $A^0 \perp A^1$, where $\perp$ is with respect to the Killing form of τ. Therefore, fixing some multiple of this form, as we did for T in 3.7 and 3.8, we can use it to identify $(A/A^0)^*$ with A^1. But, by transitivity, $A/A^0 \cong TM$, because A^0 is the anchor kernel (3.5(2)), and we conclude $T^*M \cong A^1$. In this way, we may view κ as a section of $\wedge^2 A^1 \otimes A$.

As the bracket τ makes the vector bundle A into a representation of the Lie algebra bundle A^1, $\wedge^2 A^1 \otimes A$ appears in a natural chain complex

{chain1_}

$$0 \xleftarrow{\partial^\tau} A \xleftarrow{\partial^\tau} A^1 \otimes A \xleftarrow{\partial^\tau} \wedge^2 A^1 \otimes A \xleftarrow{\partial^\tau} \cdots . \tag{1}$$

Over each point $m \in M$, this is the classical chain complex associated with the representation ad^τ of the Lie algebra $A^1|_m$ on $A|_m$. Following [14], we will refer to ∂^τ as the *Kostant codifferential*. Basic facts about such complexes immediately carry over to the case of Lie algebra bundle representations and we use such facts without further comment. By definition (c.f., [14, p. 262]),

{form1}

$$\begin{aligned}\partial^\tau(Y_0 \wedge \cdots \wedge Y_n \otimes X) = &\sum_{i=0}^{n} (-1)^{i+1} Y_0 \wedge \cdots \wedge \widehat{Y_i} \wedge \cdots \wedge Y_n \otimes \tau(Y_i, X) \\ &+ \sum_{i<j} (-1)^{i+j} \tau(Y_i, Y_j) \wedge Y_0 \wedge \cdots \wedge \widehat{Y_i} \wedge \cdots \widehat{Y_j} \wedge \cdots \wedge Y_n \otimes X.\end{aligned} \tag{2}$$

For example, $\partial^\tau(Y_0 \wedge Y_1 \otimes X) = Y_0 \otimes \tau(Y_1, X) - Y_1 \otimes \tau(Y_0, X) - \tau(Y_0, Y_1) \otimes X$. Curiously, using the strong compatibility of τ and ∇, we can express ∂^τ in terms of the Cartan connection ∇ instead:

$$\begin{aligned}\partial^\tau(Y_0 \wedge \cdots \wedge Y_n \otimes X) = &\sum_{i=0}^{n} (-1)^{i+1} Y_0 \wedge \cdots \wedge \widehat{Y_i} \wedge \cdots \wedge Y_n \otimes \bar{\nabla}_{Y_i} X \\ &+ \sum_{i<j} (-1)^{i+j} [Y_i, Y_j] \wedge Y_0 \wedge \cdots \wedge \widehat{Y_i} \wedge \cdots \widehat{Y_j} \wedge \cdots \wedge Y_n \otimes X.\end{aligned}$$

Model data τ in a parabolic geometry is *normal* if $\partial^\tau \kappa = 0$. Our proof of the following is an adaption of that of [14, Propositions 3.1.10, 3.1.13]:

Theorem (Theorem 10.6)**.** *Let $(\mathcal{G}, \tau)$ be a regular parabolic geometry. Then there exists normal model data τ' such that $(\mathcal{G}, \tau')$ is a regular parabolic geometry with the same Levi reduction as $(\mathcal{G}, \tau)$.*

{uniqueness}

3.11. **Uniqueness up to Weyl gauge transformations.** The inner automorphisms of an arbitrary Lie groupoid are those given by conjugation by bisections. When $\mathcal{G}$ is firmly filtered, a special role is played by those inner automorphisms generated by bisections of $\mathcal{G}^1 \subset \mathcal{G}$, the bundle of simply-connected nilpotent Lie groups integrating the totally intransitive Lie subalgebroid $A^1 \subset A$, where a bisection is the same thing as a section. We call such automorphisms *Weyl gauge transformations*, for reasons explained in 4.4. By definition, there is, for each such transformation, some $Z \in \Gamma(A^1)$ such that each arrow $g \in \mathcal{G}$, beginning at some $m \in M$, is mapped to $\exp(Z(gm))g\exp(-Z(m))$. We give the same name to the infinitesimal analogues,

namely those automorphisms of A of the form

$$X \mapsto e^{\mathrm{ad}_Z} X := X + [Z, X] + \frac{1}{2!}[Z, [Z, X]] + \cdots \quad ,$$

for some $Z \in \Gamma(A^1)$. Both global and infinitesimal Weyl gauge transformations cover the identity of the base M, and e^{ad_Z} is filtration-preserving, by F1.

Proposition (Proposition **??**)**.** If $(\mathcal{G}, \tau)$ is a (regular) parabolic geometry, and τ' the push-forward of τ under a Weyl gauge transformation, then $(\mathcal{G}, \tau')$ is another (regular) parabolic geometry.

Example. Let A be a filtered Lie algebroid with $|A| = 1$. Then the infinitesimal gauge transformations have the form $X \mapsto X + [Z, X]$, for some $Z \in \Gamma(A^1)$.

INCOMPLETE SECTION

4. Weyl gauges and the semi-direct product structure

{semi-direct}

We now return to the general study of filtered Lie algebroids introduced in Section 2, our main goal being the generalization of Theorem 1.4 to Lie algebroids and Lie groupoids. The idea of the proof of Theorem 1.4 is to show that the isotropy B of ϵ is a complement for $\mathfrak{g}^1$ in $\mathfrak{g}$, and that B preserves the grading of $\mathfrak{g}$ determined by ϵ. We suppress details, as we will provide them for the generalizations.

The groupoids and algebroids in this section are not required to be transitive and semisimplicity plays no role.

{splittings}

4.1. **Splittings of a filtered Lie algebra.** If $\mathfrak{g}$ is a finite-dimensional filtered Lie algebra, then the existence of a grading element $E \in \mathrm{gr}(\mathfrak{g})$ has very strong consequences for the Lie algebra. For, choosing any $\epsilon \in \mathfrak{g}^0$ such that $\mathrm{gr}_0(\epsilon) = E$, one can show, using induction and $\cap_j \mathfrak{g}^j = 0$, that the maps

$$X \mapsto [\epsilon, X] - jX \colon \mathfrak{g}^j \to \mathfrak{g}^{j+1}$$

are well-defined projections, whose kernels constitute a grading of $\mathfrak{g}$ for which ϵ is a grading element; this in turn gives rise to an isomorphism $\Phi_\epsilon \colon \mathrm{gr}(\mathfrak{g}) \to \mathfrak{g}$ which must in fact be a Lie algebra isomorphism. When we alter the choice of ϵ, the isomorphism Φ_ϵ changes in a way made precise in (2) below. {form1}

We call $\Phi_\epsilon \colon \mathrm{gr}(\mathfrak{g}) \to \mathfrak{g}$ an *E-splitting* of the filtration. With E and ϵ fixed as above, there is evidently a one-to-one correspondence between E-splittings and the affine space $\epsilon + \mathfrak{g}^1$, because $\Phi_{\epsilon'}(E) = \epsilon'$ for all $\epsilon' \in \epsilon + \mathfrak{g}^1$.

Now $\mathfrak{g}^1$ is a nilpotent Lie algebra, integrated by a simply connected nilpotent Lie group, which we denote by $\exp(\mathfrak{g}^1)$; nilpotency ensures the exponential map is a diffeomorphism. Part (1) of the lemma below shows that $\exp(\mathfrak{g}^1)$ acts freely and transitively on the set of E-splittings by constructing a concrete realization G^1. A proof of this known result appears in Appendix B. {form1}

Lemma. Let $\epsilon \in \mathfrak{g}^0$ be a lift of some grading element $E \in \mathrm{gr}(\mathfrak{g})$. Let G^0 be *any* Lie group integrating $\mathfrak{g}^0$, and let $G^1 \subset G^0$ be the connected Lie subgroup with Lie algebra $\mathfrak{g}^1$. Then:

(1) Under the adjoint action, G^1 acts freely and transitively on the affine space $\epsilon + \mathfrak{g}^1 \subset \mathfrak{g}^0$, and consequently on E-splittings of $\mathfrak{g}$. In particular, G^1 is simply connected (i.e., $G^1 \cong \exp(\mathfrak{g}^1)$) so that the infinitesimal adjoint action of $\mathfrak{g}^1$ on $\mathfrak{g}$ has a unique globalization $\mathrm{Ad}\colon G^1 \to \mathrm{Aut}(\mathfrak{g})$ extending the adjoint action just mentioned. {splittings1}

(2) For any $\epsilon_1, \epsilon_2 \in \epsilon + \mathfrak{g}^1$, we have $\Phi_{\epsilon_2} = \mathrm{Ad}_g \circ \Phi_{\epsilon_1}$, where $g \in G^1$ is the unique element for which $\epsilon_2 = \mathrm{Ad}_g\, \epsilon_1$. {splittings2}

4.2. **Weyl gauges for filtered Lie algebroids.** A *Weyl gauge* for a filtered Lie algebroid A will be any section ϵ of A^0 such that $E = \mathrm{gr}_0(\epsilon)$ is a fibre-wise grading section of the bundle of Lie algebras $\mathrm{gr}(A^0)$, and {weyl}

$$[X, \epsilon] \in A^1 \text{ for all } X \in A. \tag{1}$$ {w2}

If A is firmly filtered, then the canonical action of A on its anchor kernel determines an action ad of A on $\mathrm{gr}(A^0)$. Equation (1) then holds if and only if E is ad-invariant. That is, in the firm case, Weyl gauges exist if and only if an ad-invariant fibre-wise grading section of $\mathrm{gr}(A^0)$ exists, and any lift to A^0 of such a section is a Weyl gauge. In fact, according to the corollary stated below all Weyl gauges arise in this way.

Note that we do not require that E be a fibre-wise grading section of $\mathrm{gr}(A)$, only of $\mathrm{gr}(A^0)$ (but see also 6.3).

Theorem. *Let A be a filtered Lie algebroid admitting a Weyl gauge ϵ, and let $B \subset A$ denote the isotropy of ϵ under the canonical representation of A on its anchor kernel, i.e., the kernel of the map $X \mapsto [X, \epsilon]\colon A \to A^1$. Then:*

(2) *A is firmly filtered, and in particular, $A^1 \subset A$ is an ideal (i.e., $\Gamma(A^1) \subset \Gamma(A)$ is an ideal).* {pq1}

(3) *A is the internal semi-direct product of B and A^1.* {pq2}

(4) *The grading $A^0 = \oplus_{i \geqslant 0} \mathfrak{a}_i$ associated with ϵ is invariant under the canonical representation of B. In particular, the splitting $\Phi_\epsilon\colon \mathrm{gr}(A^1) \to A^1$ intertwines the representation of $A/A^1 \cong B$ on $\mathrm{gr}(A^1)$ with the representation of B on A^1, so that A is the external semi-direct product of A/A^1 and $\mathrm{gr}(A^1)$.* {pq3}

(5) *Under the identification $A \cong A/A^1 \oplus_{\mathrm{ad}} \mathrm{gr}(A^1)$ of (4), the Weyl gauge ϵ takes the form $\epsilon = (E, 0)$, for some section E of $A^0/A^1 \subset A/A^1$ that commutes under Lie algebroid bracket with all sections of A/A^1; and $\mathrm{ad}_E\, Y = jY$ for any $Y \in \mathrm{gr}(A^0)_j$.* {pq4}

Note that identification $A \cong A/A^1 \oplus_{\mathrm{ad}} \mathrm{gr}(A^1)$ transfers to $A/A^1 \oplus_{\mathrm{ad}} \mathrm{gr}(A^1)$ the following filtration: $(A/A^1 \oplus_{\mathrm{ad}} \mathrm{gr}(A^1))^i = (A^i/A^0) \oplus_{\mathrm{ad}} (\oplus_{j \geqslant \max\{1,i\}} \mathrm{gr}(A)_j)$.

Example (Projective geometry)**.** In example 2.6 the Lie algebroid $A \subset J^1(TM)$ acts on TM in the usual way, $\mathrm{ad}_{J^1 V} W = [V, W]$, which gives us associated representations of A on $\mathfrak{gl}(TM)$ and T^*M. On each component of $\mathrm{gr}(A^0) = \mathfrak{gl}(TM) \oplus T^*M$ the infinitesimal adjoint action of A is given by these representations, which shows that $E = -\mathrm{id} \in \mathfrak{gl}(TM) \subset \mathrm{gr}(A^0)$ is ad-invariant. Consequently, any lift ϵ of $-\mathrm{id}$ through $A^0 \to A^0/A^1 = \mathfrak{gl}(TM)$ is a Weyl gauge for A. According to the theorem, each such gauge defines an identification of A with the semi-direct product $J^1(TM) \oplus_{\mathrm{ad}} T^*M$.

Proof of theorem. The isotropy $B \subset A$ is a subalgebroid, by Jacobi's identity. We first prove

$$A = B \oplus A^1, \tag{6}$$ {y7}

as vector bundles. It suffices to show that the restriction r of $X \mapsto [X, \epsilon]$ to a map $A^1 \to A^1$ is an isomorphism. However, by F1, this restriction is filtration-preserving, and the corresponding map $\mathrm{gr}(A^1) \to \mathrm{gr}(A^1)$ sends $\xi \in \mathrm{gr}(A^1)_j$ to $j\xi$, because $\mathrm{gr}_0(\epsilon)$ is a fibre-wise grading section of $\mathrm{gr}(A^0)$. As this is obviously an isomorphism, r must be too.

Next we prove the grading $A^0 = \oplus_{i \geqslant 0} \mathfrak{a}_i$ associated with ϵ is invariant under the canonical representation of B. To that end, suppose $Y \in \Gamma(\mathfrak{a}_j)$ for some $j \geqslant 0$ and let $X \in \Gamma(B)$ be arbitrary. The Jacobi identity for the bracket on A gives us

$$[X, [\epsilon, Y]] = [[X, \epsilon], Y] + [\epsilon, [X, Y]].$$

On the other hand, applying ad_X to both sides of the identity $[\epsilon, Y] = jY$, we have

$$[X, [\epsilon, Y]] = j \,\mathrm{ad}_X Y.$$

Combining these two results, we obtain

$$[[X, \epsilon], Y] + [\epsilon, \mathrm{ad}_X Y] = j \,\mathrm{ad}_X Y.$$

But $[X, \epsilon] = 0$, because $X \in \Gamma(B)$. We conclude that $\mathrm{ad}_X Y \in \Gamma(\mathfrak{a}_j)$.

Having proven, in particular, that B preserves the filtration of A^0, to establish (2) it suffices, in view of (6), to show that the filtration of A^0 is invariant under the canonical action of itself, which follows immediately from F1.

Claim (3) follows from (2) and (6). The first part of (4) has already been addressed, and the rest now follows easily. Part (5) is now also clear. □ {ew1}

Corollary. *A filtered Lie algebroid A admits a Weyl gauge if and only if it is firmly filtered and $\mathrm{gr}(A^0)$ admits an* ad*-invariant fibre-wise grading section E.*

Proof. Given a firmly filtered Lie algebroid A with ad-invariant fibre-wise grading section E of $\mathrm{gr}(A^0)$, any lift of E to a section $\epsilon \in \Gamma(A^0)$ is immediately seen to be a Weyl gauge. The converse statement follows from the theorem. □

If A is a filtered Lie algebroid, then A^1, which lies in the anchor kernel of A, is a nilpotent Lie algebra bundle, and is accordingly integrated by a bundle of simply-connected nilpotent Lie groups, denoted $\exp(A^1)$.

Remark. By simple-connectivity, the adjoint action of A^1 on A^0 integrates to an action Ad of $\exp(A^1)$ on A^0. We have the identity $\mathrm{Ad}_{\exp(Z)} X = e^{-\mathrm{ad}_Z} X$, for any $Z \in \Gamma(A^1)$ and $X \in \Gamma(A^0)$, but the right-hand side makes sense for any $X \in \Gamma(A)$.

Proposition. Let A be a filtered Lie algebroid admitting a Weyl gauge, and $\mathcal{G}$ any Lie groupoid integrating A. Then the source-connected Lie subgroupoid $\mathcal{G}^1 \subset \mathcal{G}$ integrating A^1 is a bundle of simply-connected nilpotent Lie groups isomorphic to $\exp(A^1)$.

Proof. We prove the first part of the conclusion, as the second part follows from the first, by Lie I for Lie groupoids. By axiom F2, A^0 and A^1 lie in the kernel of the anchor of A, and so both are bundles of Lie algebras (totally intransitive). The source-connectedness of $\mathcal{G}^1$ now implies that $\mathcal{G}^1$ is totally intransitive (its source and target fibres coincide) so that these fibres are connected Lie groups. These Lie groups are subgroups of the fibres of the source-connected subgroupoid $\mathcal{G}^0 \subset \mathcal{G}$ integrating A^0 (also a bundle of Lie groups). With a Weyl gauge ϵ in hand, we may apply Lemma 4.1(1) fibre-wise to show that these groups are in fact simply-connected and nilpotent. □

{weyl2}

4.3. **Filtered Lie groupoids and Weyl gauges.** Let $\mathcal{G}$ be a filtered Lie groupoid over M. The filtration need not be firm; see 3.1 for definitions. We call $\epsilon \in \Gamma(A)$ a *Weyl gauge* for $\mathcal{G}$ if it is a Weyl gauge for A, and if for every arrow $g \in \mathcal{G}$ with source $m \in M$ we have

(1) {tui}
$$\mathrm{Ad}_g\, \epsilon(m) - \epsilon(gm) \in A^1(gm),$$

where gm denotes the target of g. If $\mathcal{G}$ is firmly filtered, then this condition holds if and only if $E = \mathrm{gr}_0(\epsilon)$ is $\mathcal{G}$-invariant under Ad, in which case the requirement 4.2(1) is automatic.

Proposition. Let $\mathcal{G}$ be a filtered Lie groupoid with Lie algebroid A and $\mathcal{G}^1 \subset \mathcal{G}$ the source-connected Lie subgroupoid integrating A^1.

(2) Under Ad, $\mathcal{G}^1$ acts trivially on $\mathrm{gr}(A^0)$. {xerox1}
(3) If $\mathcal{G}$ admits a Weyl gauge then $\mathcal{G}^1$ is a bundle of simply-connected nilpotent Lie groups isomorphic to $\exp(A^1)$. {xerox2}

Proof. Part (2) holds because A^1 already acts trivially on $\mathrm{gr}(A^0)$ by F1, and $\mathcal{G}^1$ has connected source-fibres. Part (3) is an immediate corollary of Proposition 4.2. □

To navigate a mild obstruction to obtaining semi-direct products in the groupoid case, we will say the filtration of $\mathcal{G}$ is *weakly firm* if A^1 is $\mathcal{G}$-invariant under Ad. Evidently a firm filtration of $\mathcal{G}$ is weakly firm.

Remark. Suppose $\mathcal{G}$ is filtered and admits a Weyl gauge ϵ. Then if $\mathcal{G}$ has connected source fibres, then we claim it is weakly firm. Indeed, ϵ will be a Weyl gauge for the Lie algebroid A of $\mathcal{G}$, so that $A^1 \subset A$ must be an ideal, by Theorem 4.2(2). Our claim follows now from the connectedness assumption.

Suppose $\mathcal{G}$ is firmly filtered, so that $\mathcal{G}/\mathcal{G}^1$ acts on $\mathrm{gr}(A^0)$, by (2). This action, also denoted Ad, preserves the Lie algebra bundle bracket. Now, let $\exp(\mathrm{gr}(A^1))$ denote the totally intransitive Lie groupoid with simply-connected fibres that integrates $\mathrm{gr}(A^1)$, a Lie algebra bundle with nilpotent Lie algebra model. Then the action just mentioned integrates to an action of $\mathcal{G}/\mathcal{G}^1$ on $\exp(\mathrm{gr}(A^1))$, also denoted Ad, preserving the totally intransitive Lie groupoid structure, i.e., it maps each (nilpotent) Lie group fibre isomorphically onto another Lie group fibre. We can accordingly define the semi-direct product $\mathcal{G}/\mathcal{G}^1 \times_{\mathrm{Ad}} \exp(\mathrm{gr}(A^1))$, and have the following globalization of Theorem 4.2.

Theorem. *Let $\mathcal{G}$ be a weakly firm, filtered Lie groupoid with Lie algebroid A. Define $\mathcal{G}^1 \subset \mathcal{G}$ as in the proposition above, and suppose $\mathcal{G}$ admits a Weyl gauge ϵ. Then, if $\mathcal{H} \subset \mathcal{G}$ denotes the isotropy subgroupoid of ϵ under the canonical representation of $\mathcal{G}$, then:*

(4) *$\mathcal{G}^1 \subset \mathcal{G}$ is normal,* {pr1}
(5) *$\mathcal{G} = \mathcal{G}^1\mathcal{H}$ is the internal semi-direct product of $\mathcal{H}$ and $\mathcal{G}^1$,* {pr2}
(6) *The grading $A^0 = \oplus_{i\geqslant 0}\mathfrak{a}_i$ associated with ϵ is invariant under the canonical representation of $\mathcal{H}$.* {pr3}
(7) *$\mathcal{G}$ is firmly filtered.* {pr4}
(8) *As a Lie groupoid, $\mathcal{G}$ is identifiable with the external semi-direct product $\mathcal{G}/\mathcal{G}^1 \times_{\mathrm{Ad}} \exp(\mathrm{gr}(A^1))$. Specifically, let $\Phi_\epsilon\colon \mathrm{gr}(A^1) \to A^1$ denote the splitting associated with ϵ, and let $\tilde{\Phi}_\epsilon\colon \exp(\mathrm{gr}(A^1)) \to \mathcal{G}^1$ be the morphism of totally intransitive Lie*

groupoids that integrates Φ_e. For $g \in \mathcal{G}/\mathcal{G}^1$, define $g^{\mathcal{H}} \in \mathcal{H}$ by the requirement $g^{\mathcal{H}}\mathcal{G}^1 = g$. Then the map

$$(g,k) \mapsto \tilde{\Phi}_\epsilon(k)g^{\mathcal{H}} \colon \mathcal{G}/\mathcal{G}^1 \times_{\mathrm{Ad}} \exp(\mathrm{gr}(A^1)) \to \mathcal{G}$$

is a Lie groupoid isomorphism. {pr5}

Proof. Since we assume $\mathcal{G}$ is weakly firm, $A^1 \subset A$ is invariant under the canonical representation Ad of $\mathcal{G}$ on $\mathcal{G}^1$. But, by the proposition, the exponential map for $\mathcal{G}^1$ is a diffeomorphism, and we will have $g\exp(Z)g^{-1} = \exp(\mathrm{Ad}_g Z)$ for any arrow $g \in \mathcal{G}$ beginning at the base point of each $Z \in A^1$. This proves (4) because the filtration is weakly firm.

To prove (5) we now show each arrow $g \in \mathcal{G}$, beginning at some $m \in M$, can be written uniquely as $g = kh$, with $k \in \mathcal{G}^1$ and $h \in \mathcal{G}$. Indeed, from (1) we have $\mathrm{Ad}_g\,\epsilon(m) \in \epsilon(gm) + A^1|_{gm}$. But, according to Lemma 4.1(1), the Lie group $\mathcal{G}^1_{gm}$ (the source=target fibre of $\mathcal{G}^1$ at $gm \in M$) acts freely and transitively on the affine space $\epsilon(gm) + A^1|_{gm}$. Therefore there exists a unique $k \in \mathcal{G}^1$ such that $\mathrm{Ad}_g\,\epsilon(m) = \mathrm{Ad}_k\,\epsilon(gm)$. But this means $\mathrm{Ad}_{k^{-1}g}\,\epsilon(m) = \epsilon(gm) = \epsilon(k^{-1}gm)$, i.e., $k^{-1}g \in \mathcal{H}$.

To prove (6), suppose $Y \in \Gamma(\mathfrak{a}_j)$ for some $j \geqslant 0$ and let $h \in \mathcal{H}$ be an arbitrary arrow beginning at some $m \in M$. Since the bracket on A^0 is $\mathcal{G}$-invariant under Ad, the grading identity $[\epsilon, Y](m) = jY(m)$ gives us $[\mathrm{Ad}_h\,\epsilon(m), \mathrm{Ad}_h\,Y(m)] = j\,\mathrm{Ad}_h\,Y(m)$. From the definition of $\mathcal{H}$, we conclude that $[\epsilon(hm), \mathrm{Ad}_h\,Y(m)] = j\,\mathrm{Ad}_h\,Y(m)$, establishing that $\mathrm{Ad}_h\,Y(m) \in \mathfrak{a}_j$, as required.

In view of the fact that $\mathcal{G} = \mathcal{G}^1\mathcal{H}$, and having established (6), it suffices, in proving (7), to show that the filtration of A^0 is invariant under the action of $\mathcal{G}^1$. But A^0 is already invariant under the action of A^1 by F1. Since $\mathcal{G}^1$ is source-connected, the filtration of A^0 is therefore invariant under the action of $\mathcal{G}^1$.

Part (8) is a rather straightforward consequence of the other parts, whose proof is omitted. □ {ew2}

Corollary. *A weakly firm, filtered Lie groupoid $\mathcal{G}$ with Lie algebroid A admits a Weyl gauge if and only if it is firmly filtered and* $\mathrm{gr}(A^0)$ *admits an* Ad*-invariant fibre-wise grading section E.*

Proof. Given an Ad-invariant fibre-wise grading section E of $\mathrm{gr}(A^0)$, any lift to a section of A^0 is a Weyl gauge for $\mathcal{G}$, as is immediately verified. The converse statement follows from the theorem. □

{gauge_transformations}

4.4. **Weyl gauge transformations.** Let A be a firmly filtered Lie algebroid over M and ϵ a Weyl gauge lifting some grading section E of $\mathrm{gr}(A^0)$. Then, applying Theorem 4.2, we obtain an isomorphism $\phi_\epsilon \colon (A/A^1) \oplus_{\mathrm{ad}} \mathrm{gr}(A^1) \xrightarrow{\sim} A$. A new Weyl gauge ϵ', lifting E, determines a second isomorphism $\phi_{\epsilon'} \colon (A/A^1) \oplus_{\mathrm{ad}} \mathrm{gr}(A^1) \xrightarrow{\sim} A$, so that $\phi_{\epsilon'} \circ \phi_\epsilon^{-1} \colon A \to A$ is a filtration-preserving Lie algebroid automorphism covering the identity on M, and mapping the gauge ϵ to ϵ'. We may construct automorphisms of a firmly filtered Lie groupoid in a similar way, using Theorem 4.3.

Theorem. *The automorphisms of firmly filtered Lie algebroids and Lie groupoids described above are precisely the Weyl gauge transformations defined in 3.11.*

Proof of the ***infinitesimal case****.* Let ϵ, ϵ', ϕ_ϵ and $\phi_{\epsilon'}$ have the meanings above. Then it is not hard to see, from the statement of Theorem 4.2, that $\phi_{\epsilon'} \circ \phi_\epsilon^{-1}$ is the unique Lie algebroid isomorphism φ covering the identity such that:

{gold1}

{gold2}

{gold3}

(1) $\varphi(B_\epsilon) = B_{\epsilon'}$, where B_ϵ denotes the isotropy of ϵ under the canonical representation of A on its anchor kernel.
(2) $\varphi(X) \equiv X \bmod A^1$, for all $X \in \Gamma(A)$.
(3) $\varphi(Y) = \Phi_{\epsilon'} \circ \Phi_\epsilon^{-1}(Y)$ for all $Y \in \Gamma(A^1)$, where $\Phi_\epsilon \colon \operatorname{gr}(A^1) \to A^1$ denotes the splitting determined by ϵ.

Now, a fibre-wise application of Lemma 4.1(1) show that there exists $Z \in \Gamma(A^1)$ such that $\epsilon' = \operatorname{Ad}_{\exp(Z)} \epsilon = e^{-\operatorname{ad}_Z}\epsilon$. See also Remark 4.2. To show $\phi_{\epsilon'} \circ \phi_\epsilon^{-1}$ is a Weyl gauge transformation, it suffices to show that the Weyl gauge transformation $\varphi := e^{-\operatorname{ad}_Z}$ satisfies the above conditions. But (1) follows immediately, by construction, because $e^{-\operatorname{ad}_Z}$ is a Lie algebroid morphism; (2) follows from the expansion $e^{-\operatorname{ad}_Z}X := X - [Z, X] + \frac{1}{2!}[Z, [Z, X]] - \cdots$ and the fact that $\Gamma(A^1) \subset \Gamma(A)$ is an ideal (A is firmly filtered); while (3) follows from the fibre-wise application of Lemma 4.1(2).

It remains to show *every* Weyl gauge transformation $e^{-\operatorname{ad}_Z}$ is of the form $\phi_{\epsilon'} \circ \phi_\epsilon^{-1}$ for some ϵ and ϵ'. But the above argument already shows this, if we choose ϵ to be any Weyl gauge and $\epsilon' := e^{-\operatorname{ad}_Z}(\epsilon)$. □

Proof of the ***global case****.* Let $\mathcal{G}$ be a firmly filtered Lie groupoid with Lie algebroid A, and let ϵ and ϵ' have the meanings above. Applying Theorem 4.3, we obtain Lie groupoid isomorphisms $\phi_\epsilon, \phi_{\epsilon'} \colon \mathcal{G}/\mathcal{G}^1 \times_{\operatorname{Ad}} \exp(\operatorname{gr}(A^1)) \to \mathcal{G}$. From the statement of the theorem, it is not too hard too see that $\phi_{\epsilon'} \circ \phi_\epsilon^{-1}$ is the unique Lie groupoid isomorphism φ covering the identity such that:

{silver1}

{silver2}

{silver3}

(4) $\varphi(\mathcal{H}_\epsilon) = \mathcal{H}_{\epsilon'}$, where $\mathcal{H}_\epsilon$ denotes the isotropy of ϵ under the canonical representation of $\mathcal{G}$ on the anchor kernel of A.
(5) $\varphi(g)\mathcal{G}^1 \equiv g\mathcal{G}^1$, for all $g \in \mathcal{G}$.
(6) $d\varphi(Y) = \Phi_{\epsilon'} \circ \Phi_\epsilon^{-1}(Y)$ for all $Y \in \Gamma(A^1)$, where $\Phi_\epsilon \colon \operatorname{gr}(A^1) \to A^1$ denotes the splitting determined by ϵ.

In convincing herself of this, the reader will keep in mind that $\mathcal{G}^1 \cong \exp(A^1)$ and $\exp(\operatorname{gr}(A^1))$ are source simply-connected, so that a groupoid morphism having one of these as domain is determined completely by its infinitesimalization.

Now choose $Z \in \Gamma(A^1)$ as in the infinitesimal argument, so that $\epsilon' = \operatorname{Ad}_{\exp(Z)} \epsilon = e^{-\operatorname{ad}_Z}\epsilon$. To show that $\phi_{\epsilon'} \circ \phi_\epsilon^{-1}$ is a Weyl gauge transformation, it suffices to show that the Weyl gauge transformation $\varphi := K_{\exp(Z)}$ satisfies the above conditions, where $K_{\exp(Z)} \colon \mathcal{G} \to \mathcal{G}$ denotes conjugation by the (bi)section $\exp(Z)$. To this end, suppose $h \in \mathcal{G}$ is an arrow beginning at some $m \in M$ and that $h \in \mathcal{H}_\epsilon$. Then $\operatorname{Ad}_h \epsilon(m) = \epsilon(hm)$. Applying $\operatorname{Ad}_{\exp(Z)}$ to both sides, we conclude that $\operatorname{Ad}_{K_{\exp(Z)}(h)} \epsilon'(m) = \epsilon'(m)$, showing $K_{\exp(Z)}(h) \in \mathcal{H}_{\epsilon'}$. We can reverse the roles of ϵ and ϵ' and repeat the argument, and conclude that $K_{\exp(Z)}(\mathcal{H}_\epsilon) = \mathcal{H}_{\epsilon'}$, i.e., (4) holds. The normality of $\mathcal{G}^1$ (Theorem 4.3(4)) means $g\mathcal{G}^1 = \mathcal{G}^1 g$, for any $g \in \mathcal{G}$. If $g \in \mathcal{G}$ is an arbitrary arrow, beginning at some $m \in M$, we compute, appealing to normality twice,

$$\begin{aligned} K_{\exp(Z)}(g)\mathcal{G}^1 &= \mathcal{G}^1 \exp(Z(gm)) g \exp(-Z(m)) \\ &= \mathcal{G}^1 g \exp(-Z(m)) = g \exp(-Z(m))\mathcal{G}^1 = g\mathcal{G}^1, \end{aligned}$$

which proves (5). Since $dK_{\exp(Z)} = e^{-\operatorname{ad}_Z}$, (6) already follows from the argument given in the infinitesimal case.[6]

To show every Weyl gauge transformation $K_{\exp(Z)}$ is of the form $\phi_{\epsilon'} \circ \phi_\epsilon^{-1}$ for some ϵ and ϵ', take ϵ to be any Weyl gauge, $\epsilon' := e^{-\operatorname{ad}_Z}(\epsilon)$, and repeat the preceding argument. □

5. Cartan connections

{cartan_section}

Cartan connections on Lie groupoids, introduced in [4, Appendix A], generalize to the intransitive case the classical Cartan connections introduced by Ehresmann [18] and in widespread use, when the latter notion is stripped of model data. The corresponding infinitesimal notion, Cartan connections on Lie algebroids, were introduced some time earlier, in [3]. The relationship with the classical notion is detailed in [6].

The meaning of curvature for these connections is different from the classical one. Curvature is still an obstruction to symmetry, but the kind of symmetry is not preordained by the specification of a flat model *a priori*.

With the exception of Proposition 5.1, which we presume is new, this section is a summary of definitions and results presented in the cited works and elsewhere. It is logically independent of Sections 2 and 4.

{cartan}

5.1. **Cartan connections on a Lie groupoid.** A *Cartan connection* on a Lie groupoid $\mathcal{G}$ over M is an n-plane field $D \subset T\mathcal{G}$ arising from some right inverse $\mathcal{G} \to J^1\mathcal{G}$ for the canonical projection $J^1\mathcal{G} \to \mathcal{G}$, which is necessarily uniquely determined by D. Here $J^1\mathcal{G}$ denotes the Lie groupoid of one-jets of bisections of $\mathcal{G}$. Of course, we are understanding the right inverse to be a Lie groupoid morphism. This is equivalent to requiring that $D \subset TM$ be multiplicatively closed, in the sense of [25, 16], but this interpretation will play no role here.

The usual adjoint action of a Lie group on its Lie algebra is generalized by an action of $J^1\mathcal{G}$ on the Lie algebroid A of $\mathcal{G}$, denoted Ad [6]. Composing with the right inverse $\mathcal{G} \to J^1\mathcal{G}$ of $J^1\mathcal{G} \to \mathcal{G}$ associated with D, we obtain a representation $\bar{D}$ of $\mathcal{G}$ on A. This representation extends the canonical one of $\mathcal{G}$ on elements of the anchor kernel of A (which we have also been denoting by Ad in Section 4).

The prototype of a Cartan connection is the horizontal n-plane field D on an action groupoid $G \times M$, for a smooth action of some Lie group G on M, which need not be transitive. In this case

{adjoint}

$$\bar{D}_{(g,m)}(\xi, m) = (\operatorname{Ad}_g \xi, g \cdot m). \tag{1}$$

Every Cartan connection D on a *transitive* Lie groupoid can be recovered from the representation $\bar{D}$:

Proposition. Suppose $\mathcal{G}$ is a Lie groupoid with a transitive Lie algebroid A. Then for every representation ρ of $\mathcal{G}$ on A, extending the canonical representation of $\mathcal{G}$ on the anchor kernel, there exists a unique Cartan connection D on $\mathcal{G}$ such that $\bar{D} = \rho$.

The infinitesimal version of this result, Proposition 5.2 below, already appears in [3], and has an easy proof. The proposition above is proven in Appendix C.

[6] The source of the minus sign is our convention of using right-invariant vector fields to define Lie brackets, so that $dK_{\exp(Z)}X = \operatorname{Ad}_{J^1 \exp(Z)} X = e^{-\operatorname{ad}_Z} X$, $X \in \Gamma(A)$.

{hsd}

5.2. **Infinitesimal Cartan connections.** Infinitesimalizing a Cartan connection, viewed as a Lie groupoid morphism $\mathcal{G} \to J^1\mathcal{G}$, we obtain a splitting $s\colon A \to J^1A$ of the exact sequence

$$0 \longrightarrow T^*M \otimes A \longrightarrow J^1A \longrightarrow A \longrightarrow 0$$

in the category of Lie algebroids. Here J^1A denotes the Lie algebroid of one-jets of sections of the Lie algebroid A of $\mathcal{G}$. In the category of vector bundles, splittings s of the above sequence are in one-to-one correspondence with Koszul connections ∇ on A; this correspondence is given by[7]

(1) {cspan}
$$sX = J^1X + \nabla X; \qquad X \in \Gamma(A).$$

When $s\colon A \to J^1A$ is a morphism of Lie algebroids, as is the case here, then ∇ is called a *Cartan connection* on A [3], or an *infinitesimal Cartan Connection* when we want to distinguish it from the corresponding global notion. Evidently, the notion makes sense for arbitrary Lie algebroids, not only those integrated by Lie groupoids. According to [3], ∇ is a Cartan connection on A if and only if

$$\nabla_V[X,Y] = [\nabla_V X, Y] + [X, \nabla_V Y] - \nabla_{\bar\nabla_X V} Y + \nabla_{\bar\nabla_Y V} X,$$

for all vector fields V on M and sections X, Y of A. Here $\bar\nabla$ denotes the associated A-connection on TM defined by

(2)
$$\bar\nabla_X V = \#\nabla_V X + [\#X, V],$$

an A-representation when $\bar\nabla$ is Cartan. Here $\#\colon A \to TM$ denotes anchor. Sometimes the following equivalent condition is more useful:

(3)
$$(\nabla_V \operatorname{tor} \bar\nabla)(X,Y) = \operatorname{curv} \nabla(\#X, V)Y - \operatorname{curv} \nabla(\#Y, V)X,$$

where

$$\operatorname{tor} \bar\nabla(X,Y) = \bar\nabla_X Y - \bar\nabla_Y X - [X,Y] = \nabla_{\#Y} X - \nabla_{\#X} Y + [X,Y].$$

Given a Cartan connection ∇ on some Lie algebroid A, we can compose the corresponding splitting $s\colon A \to J^1A$ with the adjoint action of J^1A on A, $\operatorname{ad}_{J^1X} Y = [X,Y]$, to obtain a self-representation $\bar\nabla$ of A. Explicitly, we have

(4) {zero}
$$\bar\nabla_X Y = \nabla_{\#Y} X + [X,Y].$$

The anchor $\#\colon A \to TM$ becomes $\bar\nabla$-equivariant.

The representation $\bar\nabla$ of A on itself coincides with the canonical representation of A on sections lying in the anchor kernel (denoted ad in Section 4). In the transitive case, ∇ can be recovered from $\bar\nabla$:

Proposition ([3, Proposition 6.1])**.** Suppose A is a transitive Lie algebroid. Then for every representation ρ of A on itself, extending the canonical representation of A on sections lying in the anchor kernel, there exists a unique Cartan connection ∇ on A such that $\bar\nabla = \rho$.

[7] According to our sign conventions, the inclusion $T^*M \otimes A \to J^1A$ induces the following map on sections: $df \otimes X \mapsto fJ^1X - J^1(fX)$.

{curvature}

5.3. **Curvature as obstruction to symmetry.** The *curvature* of ∇ is defined as usual as per Koszul connections. This notion of curvature is not to be confused with curvature as this is generally understood in the Cartan geometry literature, what we have called *model curvature*, whose Lie algebroid analogue was defined in 3.10. For further context, see [6]. The prototype of an infinitesimal Cartan connection is the trivial flat connection ∇ on an action algebroid $\mathfrak{g} \times M$, for some infinitesimal action of a Lie algebra $\mathfrak{g}$ on M. If we identify $\mathfrak{g}$ with the constant (∇-parallel) sections of $\mathfrak{g} \times M$, then its bracket is $\operatorname{tor} \bar{\nabla}(\,\cdot\,,\,\cdot\,)$. Indeed, let $\tau \in \Gamma(\wedge^2 A^* \otimes A)$ denote the trivial extension of the bracket on $\mathfrak{g}$ to a fibre-wise bracket on $A = \mathfrak{g} \times M$, and recall that the Lie algebroid bracket on $\mathfrak{g} \times M$ is given by

(1) {one}
$$[X, Y] = \nabla_{\#X} Y - \nabla_{\#Y} X + \tau(X, Y).$$

Then we can write

(2) {two}
$$[X, Y] = \bar{\nabla}_X Y - \bar{\nabla}_Y X - \tau(X, Y),$$

from which $\tau = \operatorname{tor} \bar{\nabla}$ immediately follows.

Theorem ([3])**.** *If M is simply connected, then every Lie algebroid A equipped with a flat Cartan connection ∇ is naturally isomorphic to an action algebroid $\mathfrak{g} \times M$ (possibly intransitive) where $\mathfrak{g} \subset \Gamma(TM)$ is the Lie algebra of ∇-parallel sections, equipped with bracket* $\operatorname{tor} \bar{\nabla}$.

More informally, we have:

> *A Lie algebroid equipped with a Cartan connection is an infinitesimal symmetry deformed by curvature; the torsion of the connection is the deformed symmetry Lie bracket.*

If $\operatorname{curv} \nabla = 0$ but M is not simply connected, then A is a Lie algebra action with monodromy twist [6]. Lie groupoids with Cartan connections, whose infinitesimalizations are flat, define pseudogroups of symmetries of M, and under an additional completeness assumption, twisted Lie group actions [7, 6]. In the transitive case, one can recover a locally homogeneous structure, assuming "geometric closure", if not global homogeneity up to cover [5].

Example (Riemannian geometry)**.** Associated with every Riemannian metric is a canonical Cartan connection ∇ on the isotropy subalgebroid[8] $A \subset J^1(TM)$ of the metric [3, 4], which is an infinitesimalization of a global connection D on the isotropy subgroupoid $\mathcal{G} \subset J^1(M \times M)$ of the metric [6]. It's significance is that the anchor induces a map $\Gamma(A) \to \Gamma(TM)$ establishing a one-to-one correspondence between ∇-parallel sections and infinitesimal isometries (Killing fields).

Under a natural identification of A with $TM \oplus \mathfrak{so}(TM)$, the Lie bracket is given by

$$\left[\begin{pmatrix} V_1 \\ C_1 \end{pmatrix}, \begin{pmatrix} V_2 \\ C_2 \end{pmatrix}\right] = \begin{pmatrix} [V_1, V_2] \\ [C_1, C_2] + \nabla^{TM}_{V_1} C_2 - \nabla^{TM}_{V_2} C_1 + \operatorname{curv} \nabla^{TM}(V_1, V_2) \end{pmatrix},$$

[8] This isotropy is the abstract homogeneous PDE you solve to find Killing fields (whether or not they exist).

where ∇^{TM} is the Levi-Cevita connection. The Cartan connection is given by

$$\nabla_U \begin{pmatrix} V \\ C \end{pmatrix} = \begin{pmatrix} \nabla^M_U V + CU \\ \nabla^M_U C + \operatorname{curv} \nabla^M(U,V) \end{pmatrix},$$

which has torsion

$$\text{(3)} \quad \{\text{torsion}\} \quad \operatorname{tor} \bar{\nabla} \left(\begin{pmatrix} V_1 \\ C_1 \end{pmatrix}, \begin{pmatrix} V_2 \\ C_2 \end{pmatrix} \right) = \begin{pmatrix} C_1 V_2 - C_2 V_1 \\ [C_1, C_2] - \operatorname{curv} \nabla^{TM}(V_1, V_2) \end{pmatrix}.$$

The curvature of ∇ is readily computed, vanishing precisely when M has constant sectional curvature s. From the formula for the torsion, we conclude that $\mathfrak{g}$ in the preceding theorem is the symmetry Lie algebra of the Riemannian sphere, plane or hyperbolic space, according to whether s is positive, zero, or negative.

{compat}

5.4. **Cartan connections compatible with a filtration.** An infinitesimal Cartan connection ∇ on a filtered Lie algebroid A is *compatible* with the filtration if the filtration is invariant under the associated representation $\bar{\nabla}$. Similarly, a Cartan connection D on a filtered Lie groupoid $\mathcal{G}$ with Lie algebroid A is *compatible* with the filtration provided the filtration of A is invariant under the associated representation $\bar{D}$, which in turn implies compatibility of the corresponding infinitesimal Cartan connection $\bar{\nabla}$.

In addition to the examples given in 2.1, the following result justifies our focus on *firmly* filtered Lie algebroids.

Proposition. If a transitive filtered Lie groupoid (Lie algebroid) admits a compatible (infinitesimal) Cartan connection, then it is firmly filtered.

Proof. Suppose ∇ is a Cartan connection on a transitive filtered Lie algebroid A. Then $\bar{\nabla}$ extends the canonical representation ad of A on its anchor kernel, which contains A^0. So $\operatorname{ad}_X Y := [X, Y] = \bar{\nabla}_X Y$ for $X \in A$ and $Y \in A^0$. If ∇ is compatible with the filtration, then A^0 and the filtration of A^0 must therefore be ad-invariant. The groupoid case follows immediately from the infinitesimal one. □

{homogeneous}

5.5. **Filtered Klein geometries.** We now prove some claims made in 2.5 regarding the homogeneous space $M = G/P$ described there. Let $\mathcal{G} = G \times M$ be the action groupoid, equipped with its canonical Cartan connection D, so that $\mathcal{G}$ acts on $A = \mathfrak{g} \times M$ according to 5.1(1). Since the filtration of $\mathfrak{g}$ is P-invariant, the filtration of A is $\bar{D}$-invariant.

Since A is transitive and A^0 is the anchor kernel, we have, as vector bundles,

$$\operatorname{gr}(A) \cong \operatorname{gr}(TM) \oplus \operatorname{gr}(A^0).$$

As in 5.3, we denote the fibre-wise Lie bracket on $A = \mathfrak{g} \times M$ coming from $\mathfrak{g}$ by τ, which coincides with the torsion of the infinitesimalization ∇ of D, namely the canonical flat Koszul connection on the trivial bundle $\mathfrak{g} \times M$. We note in passing that τ is evidently $\bar{D}$-invariant, and so $\bar{\nabla}$-invariant. When we want to emphasize we are using the bracket $T = \operatorname{gr}_0(\tau)$ when viewing $\operatorname{gr}(A)$, $\operatorname{gr}(A^0)$ or $\operatorname{gr}(TM)$ as Lie algebra bundles, we shall denote them by $\operatorname{gr}^T(A)$, $\operatorname{gr}^T(A^0)$ and $\operatorname{gr}^T(TM)$.

Proposition. The vector bundle filtration of $A = \mathfrak{g} \times M$ defined in 2.5 is a firm Lie algebroid filtration, and $\mathcal{G} = G \times M$ is a firmly filtered Lie groupoid. Furthermore,

equipping $\mathrm{gr}(A)$ with the corresponding Lie algebra bundle structure (Proposition 2.1), we have

$$\mathrm{gr}(A) = -\,\mathrm{gr}^T(TM) \oplus \mathrm{gr}^T(A^0) \quad \text{(Lie algebra bundle direct sum)}$$

where -$\mathrm{gr}^T(TM)$ is $\mathrm{gr}^T(TM)$ equipped with negative bracket, and $T = \mathrm{gr}_0(\tau)$.

Proof. Axiom F2 holds by construction. Let $X \in \Gamma(A^j)$ and $Y \in \Gamma(A^k)$ be given. Then we claim:

(1) {gamma} $\bar{\nabla}_X Y = [X, Y] \quad \text{if } k \geqslant 0$

(2) {delta} $\bar{\nabla}_X Y = \tau(X, Y) \quad \text{if } j \geqslant 0$

(3) {alpha} $[X, Y] = \tau(X, Y) \quad \text{if } j \geqslant 0 \text{ and } k \geqslant 0$

(4) {beta} $[X, Y] \equiv -\tau(X, Y) \mod A^{j+k+1} \quad \text{if } j < 0 \text{ and } k < 0$

(5) {blue} $[X, Y] \in \Gamma(A^{j+k+1}) \quad \text{if } j < 0 \text{ and } k \geqslant 0.$

Equation (1) follows from 5.2(4) and F2. If $j \geqslant 0$, then Equation 5.3(1) reads $[X, Y] = -\nabla_{\#Y} X + \tau(X, Y)$, because $\#A^j = 0$ in that case. This and 5.2(4) delivers (2). Claim (3) follows immediately from the preceding two.

If $j < 0$, then $j + k + 1 \leqslant k$, and so $A^k \subset A^{j+k+1}$. Similarly, if $k < 0$, then $A^j \subset A^{j+k+1}$. Since the filtration is $\bar{\nabla}$-invariant, (4) now follows from 5.3(2).

As already mentioned, if $j < 0$, then $A^k \subset A^{j+k+1}$. On the other hand, if additionally $k \geqslant 0$, then $[X, Y] = \bar{\nabla}_X Y$, by (1). So (5) is a consequence of the $\bar{\nabla}$-invariance of A^k.

To establish axiom F1 for filtered Lie algebroids, we need to show $[X, Y] \in \Gamma(A^{j+k})$. Equations (3) takes care of the case $j \geqslant 0$ and $k \geqslant 0$. Equation (4) takes care of the case $j < 0$ and $k < 0$ because $A^{j+k+1} \subset A^{j+k}$ is always true. The remaining case is $j < 0$ and $k \geqslant 0$ (and the case $j \geqslant 0$ and $k < 0$, which will follow by symmetry). But as $A^{j+k+1} \subset A^{j+k}$, this already follows from (5).

The last statement in of the theorem follows easily from (3), (4), and (5).

Since the filtration of A is $\bar{D}$-invariant, D is compatible with the filtration, and so firmness follows from Proposition 5.4. □

6. Model data

{modeldata}

Model data, defined in 3.4, generalizes the trivial Lie algebra bundle structure of an action algebroid $\mathfrak{g} \times M$.

{compat2}

6.1. **Strong compatibility.** Recall that model data τ is *compatible* with a filtration of A if this is also a filtration with respect to τ. And τ is compatible with an infinitesimal Cartan connection ∇ (Cartan connection D) if τ is $\bar{\nabla}$-invariant (respectively $\bar{D}$-invariant).

For the moment suppose that A is any Lie algebroid, which may or may not be filtered. In the presence of both model data τ, and an infinitesimal Cartan connection ∇, a Lie algebroid A acts on itself in *two* distinct ways: first, we have the representation $\bar{\nabla}$ of A on itself as a Lie algebroid; second we have the representation $X \mapsto \mathrm{ad}^\tau_X = \tau(X, \cdot\,)$ of A on itself as a Lie algebra bundle (a special kind Lie algebroid). If τ is compatible with ∇ (is $\bar{\nabla}$-invariant) and every element of the anchor

kernel of A acts the same way in both cases, then we say τ is *strongly compatible* with ∇. In symbols, the additional requirement is:

(1) {strong} $$\bar{\nabla}_Z X = \tau(Z, X) \text{ whenever } \#Z = 0,\ X \in \Gamma(A).$$

Here $\#\colon A \to TM$ denotes the anchor. We say τ is *strongly compatible* with a Cartan connection D if it is compatible with D (τ is $\bar{D}$-invariant) and strongly compatible with the infinitesimalization ∇ of D. Theorem 10.2 stated later shows that compatibility already implies strong compatibility, for model data relevant to parabolic geometry.

Example (filtered Klein geometries)**.** If A, $\mathfrak{g}$, and τ have the meanings given for filtered Klein geometries in 5.5, then τ is strongly compatible with the canonical flat connection ∇ on $A = \mathfrak{g} \times M$, by 5.5(2).

{constructing}

6.2. **Constructing Cartan connections compatible with model data.** In Section 7 we shall need the following important result, reducing, in the *transitive* case, the problem of constructing Cartan connections compatible with given model data, to the problem of constructing representations of a suitable subalgebroid/subgroupoid.

Proposition. ***Infinitesimal case.*** Let A be a transitive Lie algebroid that can be written as vector bundle sum $A = B \oplus A'$, where $B \subset A$ is a subalgebroid and $A' \subset A$ an ideal lying in the kernel of the anchor $\#\colon A \to TM$. Let τ be model data on A and suppose there exists a representation ρ^B of B on A, with respect to which τ is invariant, and such that ρ^B extends the canonical representation of $B \subset A$ on the anchor kernel of A:

(1) {ext1} $$\rho^B_X Z = [X, Z] \text{ whenever } \#Z = 0,\ Z \in \Gamma(A),\ X \in \Gamma(B).$$

Then there exists a unique Cartan connection ∇ on A, necessarily compatible with τ, such that

(2) {ext2} $$\nabla_{X+Y} W = \rho^B_X W + \tau(Y, W); \qquad X \in \Gamma(B),\ Y \in \Gamma(A'),\ W \in \Gamma(A).$$

If τ is compatible with a filtration of A, and ρ^B preserves the filtration, then ∇ is also compatible with the filtration. If, additionally,

(3) {ext3} $$\rho^B_Z W = \tau(Z, W) \text{ whenever } \#Z = 0,\ Z \in \Gamma(B),\ W \in \Gamma(A),$$

then τ is strongly compatible with ∇.

Global case. Suppose $\mathcal{G}$ is a Lie groupoid, with Lie algebroid A, that can be written $\mathcal{G} = \mathcal{G}'\mathcal{H}$ with $\mathcal{G}' \cap \mathcal{H} = M$ (groupoid of identity elements), where $\mathcal{G}' \subset \mathcal{G}$ is a totally intransitive, normal, Lie subgroupoid with simply-connected fibres. Let τ be model data on A and suppose there is a representation $\rho^{\mathcal{H}}$ of $\mathcal{H}$ on A, with respect to which τ is invariant, and such that $\rho^{\mathcal{H}}$ extends the canonical representation of $\mathcal{H} \subset \mathcal{G}$ on the anchor kernel of A:

(4) {pony4} $$\rho^{\mathcal{H}}_h Z = \mathrm{Ad}_h Z \text{ whenever } \#Z = 0,\ h \in \mathcal{H}.$$

Then, there exists a unique Cartan connection D on $\mathcal{G}$, necessarily compatible with τ, such that

(5) {ext17} $$\bar{D}_{kh} W = \mathrm{Ad}^\tau_k \rho^{\mathcal{H}}_h W,$$

for each arrow $h \in H$ beginning at some $m \in M$, $k \in \mathcal{G}^1|_{hm}$ and $W \in A|_m$. Here Ad^τ is unique representation of $\mathcal{G}'$ on A integrating the infinitesimal representation ad^τ of the Lie algebroid A' of $\mathcal{G}'$ on A defined by $\mathrm{ad}^\tau_Y W = \tau(Y, W)$. If τ is compatible with a filtration of A, and ρ^B preserves the filtration, then D is also compatible with the filtration.

Note that Ad^τ is well-defined on account of the hypothesis of simple-connectivity.

Proof. ***Infinitesimal case.*** Under the hypotheses of the theorem, define an A-connection ρ on A using the right-hand side of (2):

$$\rho_{X+Y} W = \rho^B_X W + \tau(Y, W); \qquad X \in \Gamma(B),\ Y \in \Gamma(A'),\ W \in \Gamma(A).$$

By Proposition 5.2, to show there exists a Cartan connection ∇ satisfying (2), it suffices to show that ρ is actually a representation, and, secondly, that ρ extends the canonical representation of A on sections of its anchor kernel:

(6) {gobbly}
$$\rho_W Z = [W, Z] \text{ whenever } \#Z = 0,\ W \in \Gamma(A).$$

If $W \in \Gamma(B)$, then (6) holds, by hypothesis (1). If $W \in \Gamma(A')$ and $\#Z = 0$, then $\rho_W Z = \tau(W, Z) = [W, Z]$, because both W and Z lie in the anchor kernel and τ is model data. Since $A = B \oplus A'$, this proves (6).

Since the restriction of $X \mapsto \rho_X$ to B is ρ^B, a representation, and the restriction of $Y \mapsto \rho_Y$ to A' is $Y \mapsto \tau(Y, \cdot)$, also a representation, to show ρ is itself a representation, it suffices to prove that

$$\rho_X \rho_Y W - \rho_Y \rho_X W - \rho_{[X,Y]} W = 0, \qquad W \in \Gamma(A),$$

whenever $X \in \Gamma(B)$ and $Y \in \Gamma(A')$. But in that case, $[X, Y] \in \Gamma(A')$, because $A' \subset A$ is an ideal, and we compute

$$\begin{aligned}
\rho_X \rho_Y W - \rho_Y \rho_X W - \rho_{[X,Y]} W &= \rho^B_X(\tau(Y, W)) - \tau(Y, \rho^B_X W) - \tau([X, Y], W) \\
&= \rho^B_X(\tau(Y, W)) - \tau(Y, \rho^B_X W) - \tau(\rho^B_X Y, W), \quad \text{by (1)} \\
&= 0, \quad \text{by the } \rho^B\text{-invariance of } \tau.
\end{aligned}$$

Proof of the remaining claims in infinitesimal case is straightforward and omitted.

Global case. Define a map $(g, W) \mapsto \rho_g W \in A$, from the submanifold of all $(g, W) \in \mathcal{G} \times A$ such that g is an arrow beginning at the base point of W, using the right-hand side of (5):

$$\rho_{kh} W = \mathrm{Ad}^\tau_k \rho^{\mathcal{H}}_h W; \qquad k \in \mathcal{G}',\ h \in \mathcal{H}.$$

To show that there exists a unique Cartan connection D on A satisfying (5), it suffices to show that ρ is a representation, and that ρ extends the canonical representation of $\mathcal{G}$ on the anchor kernel of A:

(7) {ext18}
$$\rho_g Z = \mathrm{Ad}_g Z \text{ whenever } \#Z = 0$$

and $g \in \mathcal{G}$ is an arrow beginning at the base point of Z.

By the hypothesis (4), (7) already holds for $g \in \mathcal{H}$. Since $\mathcal{G} = \mathcal{G}'\mathcal{H}$, to prove (7) in general, it suffices to prove

(8)
$$\mathrm{Ad}^\tau_k Z = \mathrm{Ad}_k Z \text{ whenever } \#Z = 0,\ k \in \mathcal{G}'.$$

In fact, since $\mathcal{G}'$ has connected fibres, is suffices to prove the infinitesimalization,

$$\operatorname{ad}^\tau_Y Z = [Y, Z] \text{ whenever } \#Z = 0,\ Y \in A'.$$

But this true because both Y and Z here lie in the anchor kernel, $\operatorname{ad}^\tau_Y Z := \tau(Y, Z)$, and τ is model data.

Since $\rho^{\mathcal{H}}$ and $\rho^{\mathcal{G}^1}$ are already representations, to show ρ itself is a representation, it suffices to show, for any arrow $h \in \mathcal{H}$ beginning at some $m \in M$, that

$$\rho^{\mathcal{H}}_h \rho^{\mathcal{G}^1}_k \rho^{\mathcal{H}}_{h^{-1}} W = \rho^{\mathcal{G}^1}_{hkh^{-1}} W,$$

where $k \in \mathcal{G}^1|_m$ and $W \in A|_{hm}$ are arbitrary. In fact, since $\mathcal{G}^1$ has connected fibres, it is enough to show the two Lie group homomorphisms $\mathcal{G}^1|_m \to \operatorname{Aut}(A|_m)$ below have the same derivative:

$$\begin{aligned} k &\mapsto \rho^{\mathcal{H}}_h \rho^{\mathcal{G}^1}_k \rho^{\mathcal{H}}_{h^{-1}} \\ k &\mapsto \rho^{\mathcal{G}^1}_{hkh^{-1}}. \end{aligned}$$

The readily computed derivatives are:

$$\begin{aligned} & X \mapsto \rho^{\mathcal{H}}_h \operatorname{ad}^\tau_X \rho^{\mathcal{H}}_{h^{-1}}(\,\cdot\,) = \rho^{\mathcal{H}}_h \tau(X, \rho^{\mathcal{H}}_{h^{-1}}(\,\cdot\,)) \\ \text{and} \quad & X \mapsto \operatorname{ad}^\tau_{\operatorname{Ad}_h X}(\,\cdot\,) = \tau(\operatorname{Ad}_h X, \,\cdot\,) = \tau(\rho^{\mathcal{H}}_h X, \,\cdot\,), \quad \text{by (4)}. \end{aligned}$$

Equality of these derivatives follows from the $\rho^{\mathcal{H}}$-invariance of τ.

Proof of the remaining claims of the proposition is elementary and omitted. □

{model_data_gradings}

6.3. **Gradings associated with model data.** Let us say that model data τ is *compatible* with a Weyl gauge $\epsilon \in \Gamma(A^0)$ if $E = \operatorname{gr}_0(\epsilon)$ is a fibre-wise grading element of $\operatorname{gr}^T(A)$, $T = \operatorname{gr}_0(\tau)$, and not just of $\operatorname{gr}(A^0) = \operatorname{gr}^T(A^0)$.

Let A be a filtered Lie algebroid admitting a Weyl gauge ϵ. Then, according to Theorem 4.2, we have an associated grading $A^0 = \oplus_{i \geqslant 0} \mathfrak{a}_i$ that is invariant under the action of the isotropy subalgebroid $B \subset A$ of ϵ. If ϵ is compatible with model data τ, that is itself compatible with the filtration of A, then this grading extends to a grading $A = \oplus_i \mathfrak{a}_i$, where $X \in \mathfrak{a}_j$ if and only if $\tau(\epsilon, X) = jX$. This defines a splitting isomorphism $\Phi_\epsilon \colon \operatorname{gr}^T(A) \to A$.

Proposition. ***Infinitesimal case.*** In the scenario above, let ∇ be any infinitesimal Cartan connection compatible with τ. Then:

(1) The grading $A = \oplus_i \mathfrak{a}_i$ is invariant under the representation $\bar{\nabla}$ of B on A. {u4}
(2) For any $X \in \Gamma(\mathfrak{a}_k)$, $k \geqslant 0$, and $Y \in \Gamma(\mathfrak{a}_j)$, $j \in \mathbb{Z}$, one has $\bar{\nabla}_X Y \in \Gamma(\mathfrak{a}_j \oplus \mathfrak{a}_{j+k})$. {u3}

Global case. If $\mathcal{G}$ is a weakly firm, filtered Lie groupoid with Lie algebroid A, ϵ is a Weyl gauge for $\mathcal{G}$ compatible with the model data τ, and D is a Cartan connection on $\mathcal{G}$ compatible with τ, then:

(3) The grading $A = \oplus_i \mathfrak{a}_i$ associated with ϵ is invariant under the representation $\bar{D}$ of the isotropy subgroupoid $\mathcal{H} \subset \mathcal{G}$ of ϵ. {uu4}

The reader is warned that B does not generally align itself with the complement $\oplus_{i \leqslant 0} \mathfrak{a}_i$ of $A^1 = \oplus_{i \geqslant 1} \mathfrak{a}_i$ in A defined by the grading. We may only conclude that B is the graph of some linear function $\oplus_{j \leqslant 0} \mathfrak{a}_i \to A^1$ that vanishes on $\mathfrak{a}_0$, which is necessarily contained in B.

Corollary. *If a (weakly firm) filtered Lie algebroid (resp., Lie groupoid) admits a Weyl gauge compatible with some model data τ, then any Cartan connection compatible with τ is compatible with the filtration.*

Proof of corollary. If ∇ is the Cartan connection in the Lie algebroid case, then $\bar{\nabla}$-invariance of the filtration of A follows from (1) and (2), and the semi-direct product structure of A (Theorem 4.2(3)).

For the Lie groupoid case and a global Cartan connection D, we know the filtration is $\bar{\nabla}$-invariant, where ∇ is the infinitesimalization of D, from the Lie algebroid case just proven. In particular, continuing in the notation of the proposition, the restriction of $\bar{\nabla}$ to a representation of A^1 on A is filtration-preserving. Therefore, the restriction of $\bar{D}$ to a representation of $\mathcal{G}^1$ is filtration-preserving, because $\mathcal{G}^1$ is source-connected. Full $\bar{D}$-invariance of the filtration follows from this fact, (3), and the semi-direct product structure of $\mathcal{G}$ (Theorem 4.3(5)). □

Proof of proposition. Our proofs of (1) and (2) are variations on an argument appearing in the proof of Theorem 4.2. Indeed, suppose that $Y \in \Gamma(\mathfrak{a}_j)$ for some j and let $X \in \Gamma(B)$ be arbitrary. Then, applying $\bar{\nabla}_X$ to both sides of the identity $\tau(\epsilon, Y) = jY$, and appealing to the $\bar{\nabla}$-invariance of τ, we have

$$\tau(\bar{\nabla}_X \epsilon, Y) + \tau(\epsilon, \bar{\nabla}_X Y) = j\bar{\nabla}_X Y.$$

But $\bar{\nabla}_X \epsilon = [X, \epsilon]$, because ϵ is in the anchor kernel, while $[X, \epsilon] = 0$, because $X \in \Gamma(B)$. We conclude that $\bar{\nabla}_X Y \in \Gamma(\mathfrak{a}_j)$. This proves (1).

Turning to (2), write $\bar{\nabla}_X Y = \sum_i W_i$ for some $W_i \in \Gamma(\mathfrak{a}_i)$, which implies

$$\tau(\epsilon, \bar{\nabla}_X Y) = \sum_i iW_i.$$

On the other hand, $\tau(\epsilon, Y) = jY$, which, after applying $\bar{\nabla}_X$ to both sides, and appealing to the $\bar{\nabla}$-invariance of τ, becomes

$$\tau(\bar{\nabla}_X \epsilon, Y) + \tau(\epsilon, \bar{\nabla}_X Y) = j \sum_i W_i.$$

As X and ϵ are sections of A^0, we have $\bar{\nabla}_X E = [X, \epsilon] = \tau(X, \epsilon) = -kX$. From this and the two preceding equations we deduce

$$\sum_i (i - j)W_i + k\tau(Y, X) = 0.$$

Since $A = \oplus_i \mathfrak{a}_i$ is a grading with respect to τ, the last term on the left-hand side is a section of $\Gamma(\mathfrak{a}_{j+k})$. Projecting onto each component $\mathfrak{a}_i$ of the grading $A = \oplus_i \mathfrak{a}_i$, we obtain $W_i = 0$ for all i, with the possible exceptions $i = j, j + k$, which shows $\bar{\nabla}_X Y \in \Gamma(\mathfrak{a}_j \oplus \mathfrak{a}_{j+k})$.

The proof of (3) is similar to that of (1) above, following from the $\bar{D}$-invariance of τ, the definition of the spaces $\mathfrak{a}_j$, and the fact that $\mathcal{H}$ fixes ϵ. □

7. Filtered Cartan geometries

{filtered_cartan}

Suppose $\mathcal{G}$ is a firmly filtered Lie groupoid, with Lie algebroid A, supporting a Cartan connection D compatible with the filtration, and model data τ compatible with the filtration. Then we call $(\mathcal{G}, D, \tau)$ a *filtered Cartan geometry* if:

(1) τ is strongly compatible with D.

(2) $\mathcal{G}$ admits a Weyl gauge compatible with τ. {fc2}

Because we assume $\mathcal{G}$ is firmly filtered, (2) just means $\mathcal{G}$ admits an Ad-invariant, fibre-wise grading section of $\mathrm{gr}^T(A)$, where $T := \mathrm{gr}_0(\tau)$ (Corollary 4.3). There is no requirement here that τ be semisimple, and semisimplicity plays no role in the current section.

If we only have a firmly filtered Lie algebroid A, and an infinitesimal Cartan connection ∇, compatible with the filtration and strongly compatible with τ, and if A admits a τ-compatible Weyl gauge, then (A, ∇, τ) is an *infinitesimal filtered Cartan geometry*.

In this section, we provide a general, explicit, description of *transitive*, filtered Cartan geometries, by exploiting theorems on semi-direct product structure established in Section 4. In this description, we specify the additional data needed to recover a transitive Cartan geometry from its Levi reduction. While we are mainly interested in the special parabolic case introduced in Section 3, the extra abstraction introduced here emphasizes Weyl gauges as the essential actor in the resulting characterization. {inflevi}

7.1. **Abstract Levi structures: Infinitesimal case.** The reduction procedure described for parabolic geometries in 3.7 generalizes to transitive filtered Cartan geometries. We begin with the infinitesimal case.

Let (A, ∇, τ) be an transitive infinitesimal filtered Cartan geometry. Since A is firmly filtered, the canonical representation ad of A on A^0, in which A^1 is acting trivially, descends to a representation ad of $B := A/A^1$ on $\mathrm{gr}(A^0)$. This canonical representation is extended by a representation of B on $\mathrm{gr}(A)$, induced by the representation $\bar{\nabla}$ of A on itself, and we denote this by R^∇. If we write $\mathfrak{a}_j := \mathrm{gr}(A)_j$ for $j > 0$, then using the transitivity we can write

C0. $\mathrm{gr}(A) \cong \mathrm{gr}(TM) \oplus B^0 \oplus (\oplus_{j>0}\mathfrak{a}_j)$.

In addition to the grading-preserving representation R^∇ on the above, we have the Lie algebra bundle bracket $T := \mathrm{gr}_0(\tau)$ which, on account of strong compatibility between τ and $\bar{\nabla}$, shows that

(1) {form_} $$T(C, V) = R^\nabla_C V, \quad T(C, Z) = R^\nabla_C Z, \quad T(C, C') = [C, C'],$$

whenever $C, C' \in \Gamma(B^0)$, $V \in \Gamma(\mathrm{gr}(TM))$, $Z \in \Gamma(\oplus_{j>0}\mathfrak{a}_j)$. The last equation holds because τ extends the canonical bracket on A^0. Note $R^\nabla Z$ does not actually depend on ∇, because it is determined by the canonical representation of A on A^1.

Motivated by the above, an *abstract infinitesimal Levi structure* will be a 4-tuple $(B, \oplus_{j>0}\mathfrak{a}_j, R, T)$, where:

(2) B is a transitive, filtered Lie algebroid over M with $B \leqslant 0$ and with B^0 as its anchor kernel.
(3) $\oplus_{j>0}\mathfrak{a}_j$ is a graded vector bundle over M.
(4) R is a a grading-preserving representation of B on C0 that extends the canonical action of B on B^0 {chh0}
(5) T is an R-invariant Lie algebra bundle structure on C0 compatible with the obvious grading and satisfying
$$T(C, V) = R_C V, \quad T(C, Z) = R_C Z, \quad T(C, C') = [C, C'],$$
whenever $C, C' \in \Gamma(B^0)$, $V \in \Gamma(\mathrm{gr}(TM))$, $Z \in \Gamma(\oplus_{j>0}\mathfrak{a}_j)$.

(6) As a graded Lie algebra bundle, C0 admits an R-invariant grading section, necessarily of the form $(0, E, 0)$, for some $E \in \Gamma(B^0)$.

Note that it is not required that T restrict to the canonical Lie algebra bundle structure on $\mathrm{gr}(TM)$. If this restriction is the *negative* of the canonical one, then the geometry is *regular*.

According to the immediately preceding observations, associated with every transitive infinitesimal filtered Cartan geometry is an infinitesimal Levi structure, called its *Levi reduction*, regular if the filtered Cartan geometry is regular.

{abstract_levi}

7.2. **Abstract Levi structures: Global case.**
An *abstract Levi structure* is a 4-tuple $(\mathcal{H}, \oplus_{j>0}\mathfrak{a}_j, R, T)$ where:

(1) $\mathcal{H}$ is a transitive, filtered Lie groupoid over M with Lie algebroid B, whose anchor kernel is B^0, and with $B \leqslant 0$.
(2) $\oplus_{j>0}\mathfrak{a}_j$ is a vector bundle over M.
(3) R is a grading-preserving representation of $\mathcal{H}$ on C0 that extends the canonical action of $\mathcal{H}$ on B^0.

{chh00}

(4) T an R-invariant Lie algebra bundle structure on C0 compatible with the grading, and such that
$$T(C, V) = dR_C V, \quad T(C, Z) = dR_C Z, \quad T(C, C') = [C, C'],$$
for all $C, C' \in \Gamma(B^0)$, $V \in \Gamma(\mathrm{gr}(TM))$, $Z \in \Gamma(\oplus_{j>0}\mathfrak{a}_j)$.
(5) As a graded Lie algebra bundle, C0 admits an R-invariant grading section, necessarily of the form $(0, E, 0)$, for some $E \in \Gamma(B^0)$.

Here dR is the infinitesimalization of the representation R. The geometry is *regular* if its infinitesimalization $(B, \oplus_{j>0}\mathfrak{a}_j, dR, T)$ is regular.

In analogy with our observations in 7.1, we can associate with each transitive filtered Cartan geometry $(\mathcal{G}, D, \tau)$ an abstract Levi structure $(\mathcal{G}/\mathcal{G}^1, \mathrm{gr}(A^1), R^D, \mathrm{gr}_0(\tau))$ that we call its *Levi reduction*. It is regular if $(\mathcal{G}, D, \tau)$ is. Here R^D is the representation of $\mathcal{G}/\mathcal{G}^1$ induced by the representation $\bar{D}$ of $\mathcal{G}$ on the Lie algebroid A of $\mathcal{G}$.

{consequences}

7.3. **Consequences of the semi-direct product structure.** Beginning our analysis with the infinitesimal case, suppose (A, ∇, τ) is a transitive infinitesimal filtered Cartan geometry such that A^0 is the anchor kernel, and let ϵ be a τ-compatible gauge. According to Theorem 4.2, we may suppose, without loss of generality, that A is the semi-direct product of a transitive, filtered Lie algebroid B, with $|B| \leqslant 0$ and anchor B_0, with a filtered Lie algebra bundle $\oplus_{j>0}\mathfrak{a}_j$, on which B is acting, an action that preserves the grading. Moreover, if we denote the action of B on $\oplus_{j>0}\mathfrak{a}_j$ by R, so that $A = B \oplus_R (\oplus_{j>0}\mathfrak{a}_j)$, then we know there exists a section E of B^0 such that:
$$[E, X] = 0 \quad \text{for all } X \in \Gamma(B)$$
$$R_E Y = jY \quad \text{for all } Y \in \Gamma(\mathfrak{a}_j),\ j > 0.$$
$$\epsilon = (E, 0) \in \Gamma(A).$$
The Weyl gauge determines a grading $A = \oplus_i \mathfrak{a}'_i$, where $\mathfrak{a}'_i$ is the kernel of $X \mapsto \tau(E, X) - iX \colon A^i \to A^{i+1}$.

In general, $\mathfrak{a}'_k \subset B \oplus 0$ does not hold for $k < 0$. Noting that the filtration of $A = B \oplus_R (\oplus_{j>0}\mathfrak{a}_j)$ is defined by $A^i = B^i \oplus_R (\oplus_{j \geqslant \max\{1,i\}})\mathfrak{a}_j)$, it is not hard to see,

from the identities above, that $\mathfrak{a}'_0 = B^0 \oplus 0$ and $\mathfrak{a}'_j = 0 \oplus \mathfrak{a}_j$ for $j > 0$. From this we may deduce that the splitting isomorphism $\Phi\colon \operatorname{gr}(A) \to A$ induced by the grading $A = \oplus_i \mathfrak{a}'_i$ has the form

$$\begin{aligned}\Phi\colon\ \operatorname{gr}(TM) \oplus B^0 \oplus (\oplus_{j>0}\mathfrak{a}_j) &\to B \oplus_R (\oplus_{j>0}\mathfrak{a}_j)\\ (V, C, Z) &\mapsto (C + \tilde{\sigma}(V), Z - \mathsf{P}(V)),\end{aligned}$$

for some $\mathsf{P} \subset \Gamma(\operatorname{gr}(T^*M) \otimes (\oplus_{j>0}\mathfrak{a}_j))$, and some vector bundle map $\tilde{\sigma}\colon\ \operatorname{gr}(TM) \to B$ satisfying

$$\tilde{\sigma}(\operatorname{gr}(TM)_k) \subset B^k \ \text{ and } \ \operatorname{gr}_k(\#\tilde{\sigma}(V)) = V \quad \text{for all } V \in \operatorname{gr}(TM)_k,\ k < 0.$$

Here $\#\colon B \to TM$ denotes the anchor. This is just linear algebra and an unpacking of the implicit identification $\mathfrak{a}_k \cong \operatorname{gr}(A)_k \cong \operatorname{gr}(TM)_k$ for $k < 0$.

The composite $\sigma = \# \circ \tilde{\sigma}$ satisfies

$$\sigma(\operatorname{gr}(TM)_k) \subset T^kM \ \text{ and } \ \operatorname{gr}_k(\sigma(V)) = V \quad \text{for all } V \in \operatorname{gr}(TM)_k,\ k < 0.$$

We call such $\sigma\colon\ \operatorname{gr}(TM) \to TM$, which is necessarily an isomorphism, a *splitting* of the filtered manifold M. The composite $s = \tilde{\sigma} \circ \sigma^{-1}\colon TM \to B$ is a right-inverse for the anchor $\#\colon B \to TM$, which we will call a *splitting of the anchor of* B. We may now rewrite our formula for Φ as

C1. $\quad \Phi(V, C, Z) = (C + s(\sigma(V)), Z - \mathsf{P}(V))$.

By construction, the model data τ is the push-forward of $T := \operatorname{gr}_0(\tau)$ under Φ:

C2. $\quad \tau(\Phi(\xi_1), \Phi(\xi_2)) = \Phi(T(\xi_1, \xi_2))$,

for ξ_1, $\xi_2 \in \operatorname{gr}(TM) \oplus B \oplus_R (\oplus_{j>0}\mathfrak{a}_j)$, and $(0, E, 0)$ is a fibre-wise grading section of $\operatorname{gr}(TM) \oplus B^0 \oplus (\oplus_j \mathfrak{a}_j)$, for the Lie bracket T. To complete our analysis, we derive a formula for the self-representation $\bar{\nabla}$ of A, which characterizes the connection ∇, because A is transitive (Proposition 5.2). Since Proposition 6.3 applies, the grading $A = \oplus_i \mathfrak{a}'_i$ (there denoted $\oplus_i \mathfrak{a}_i$) is $\bar{\nabla}_{(X,0)}$-invariant, for every section X of B. It follows that Φ intertwines some grading-preserving representation R of B on $\operatorname{gr}(A) = \operatorname{gr}(TM) \oplus B^0 \oplus (\oplus_{j>0}\mathfrak{a}_j)$ with the $\bar{\nabla}$-representation of $B \oplus 0$ on A:

$$\bar{\nabla}_{(X,0)}\Big(\Phi(V, C, Z)\Big) = \Phi\Big(R_X(V, C, Z)\Big); \quad X \in B.$$

It's not hard to see that R on the right is the same thing as R^∇, as defined in 7.1 above, under the identification $B \oplus 0 \cong A/A^1$. As a consequence we can write this as

(1) {groof}
$$\bar{\nabla}_{(X,0)}\Big(\Phi(V, C, Z)\Big) = \Phi(R_X V, [X, C], R_X Z),$$

where the "R" in $R_X Z$ has the meaning assigned to it in the beginning of the current discussion, and with

$$T(C, V) = R_C V, \quad T(C, Z) = R_C Z, \quad T(C, C') = [C, C'],$$

for all $C, C' \in \Gamma(B^0)$, $V \in \Gamma(\operatorname{gr}(TM))$, $Z \in \Gamma(\oplus_{j>0}\mathfrak{a}_j)$.

On the other hand, for any $Y \in \Gamma(\oplus_{j>0}\mathfrak{a}_j)$, we have $\Phi(0,0,Y) = (0,Y)$, so strong compatibility also leads to

(2) {cab0}

$$\bar{\nabla}_{(0,Y)}\Big(\Phi(V,C,Z)\Big) = \tau\Big(\Phi(0,0,Y),\, \Phi(V,C,Z)\Big) = \Phi\Big(T((0,0,Y),(V,C,Z))\Big)$$
$$= \Phi\Big((0,0,[Y,Z] - R_C Y) + T(Y,V)\Big); \quad Y \in \Gamma(\oplus_{j>0}\mathfrak{a}_j).$$

Combining this last with (1), we have, for any $(X,Y) \in \Gamma(B \oplus_R (\oplus_{j>0}\mathfrak{a}_j))$,

C3. $\bar{\nabla}_{(X,Y)}\Phi(V,C,Z) = \Phi\Big(R_X(V,C,Z) + T((0,0,Y),(V,C,Z))\Big)$,

whose right-hand side can also be written as

$$\Phi\Big((R_X V, [X,C], [Y,Z] + R_X Z - R_C Y) + T(Y,V)\Big)$$

.

The above analysis leads to the characterizations of transitive filtered Cartan geometries given next.

{inflevi_}

7.4. **Algebraic prolongation: Infinitesimal case.**
Let $(B, \oplus_{j>0}\mathfrak{a}_j, R, T)$ be an abstract infinitesimal Levi structure. The structure T makes $\oplus_{j>0}\mathfrak{a}_j$ into a graded Lie algebra bundle and the *algebraic prolongation* of the abstract infinitesimal Levi structure is the Lie algebroid semi-direct product $A = B \oplus_R (\oplus_{j>0}\mathfrak{a}_j)$. This is a firmly filtered Lie algebroid with $A^i = B^i \oplus_R (\oplus_{j \geqslant \max\{1,i\}}\mathfrak{a}_j)$.

Theorem (A characterization of transitive infinitesimal filtered Cartan geometries)**.** *Let $(B, \oplus_{j>0}\mathfrak{a}_j, R, T)$ be an abstract infinitesimal Levi structure, P be an arbitrary section of $\mathrm{gr}(T^*M) \otimes (\oplus_{j>0}\mathfrak{a}_j)$, $s\colon TM \to B$ any splitting of the anchor of B, and $\sigma\colon \mathrm{gr}(TM) \to TM$ some splitting of the filtered manifold M. Then:*

(1) *The map $\Phi\colon \mathrm{gr}(TM) \oplus B^0 \oplus (\oplus_j \mathfrak{a}_j) \to B \oplus_R (\oplus_{j>0}\mathfrak{a}_j)$ defined by C1 is invertible. Specifically, if $\pi\colon B \to B^0$ is defined by $\pi(X) := X - s(\#X)$, and if $(X,Y) = \Phi(V,C,Z)$, then $C = \pi X$, $V = \sigma^{-1}(\#X)$, and $Z = Y + \mathsf{P}(V)$.* {chh2}
(2) *The algebraic prolongation $A = B \oplus_R (\oplus_{j>0}\mathfrak{a}_j)$ is a transitive, infinitesimal, filtered Cartan geometry, with model data τ defined by C2 and Cartan connection ∇ uniquely determined by C3.* {cch3}
(3) *Every transitive, infinitesimal, filtered Cartan geometry (A, ∇, τ), with anchor kernel A^0, is, up to isomorphism, the algebraic prolongation of its Levi reduction, with Cartan connection ∇ and model data τ of the form just described.*

The section P is called the *rho tensor*.

Proof. It is a triviality that Φ is injective. A dimension count implies Φ must be an isomorphism. A substitution verifies the rest of (1).

That τ, defined by C2, is model data follows from the properties of T posited in (5) and the definition of the semi-direct product Lie bracket for $B^0 \oplus_R (\oplus_{j>0}\mathfrak{a}_j)$.

The isomorphism Φ intertwines the representation R of B on $\mathrm{gr}(TM) \oplus B^0 \oplus (\oplus_j \mathfrak{a}_j)$ with some representation $\rho^{B\oplus 0}$ of $B \oplus 0 \subset A$ on A:

(4) {ret1}

$$\rho^{B\oplus 0}_{(X,0)}\Phi(V,C,Z) = \Phi\Big(R_X((V,C,Z))\Big).$$

To apply Proposition 6.2 to A, with τ defined by C2, and $A' = A^1 = 0 \oplus (\oplus_{j>0}\mathfrak{a}_j)$, we need to establish 6.2(1). Here this means verifying that

$$\rho^{B\oplus 0}_{(X,0)}\Phi(0,C,Z) = [(X,0),\Phi(0,C,Z)],$$

for $X \in \Gamma(B)$, $C \in B^0$, $Z \in \Gamma(\oplus_{j>0}\mathfrak{a}_j)$, which is an easy computation following from (4), and the fact that the bracket on the right-hand side is the semi-direct product bracket. Similarly, since τ is defined by C2, the properties of T in 7.1(5) allow one to show that 6.2(3) also holds in the present case. We conclude that there exists a unique Cartan connection ∇ on A, strongly compatible with τ, such that

$$\bar{\nabla}_{(X,Y)}\Phi(V,C,Z) = \rho^{B\oplus 0}_{(X,0)}\Phi(V,C,Z) + \tau((0,Y),\Phi(V,C,Z)).$$

The formula C3 follows from (4), C2, and the identity $(0,Y) = \Phi(0,0,Y)$. This concludes the proof of (2). Part (3) of the theorem was already established in 7.3. □

{abstract_levi_}

7.5. **Algebraic prolongation: Global case.** Let $(\mathcal{H},\oplus_{j>0}\mathfrak{a}_j,R,T)$ be an abstract Levi structure. The vector bundle $\oplus_{j>0}\mathfrak{a}_j$ gets an R-invariant graded Lie algebra bundle structure by restricting T. This bundle has a necessarily nilpotent Lie algebra model, and we let $\exp(\oplus_{j>0}\mathfrak{a}_j)$ denote the nilpotent Lie group bundle with simply connected fibres that integrates $\oplus_{j>0}\mathfrak{a}_j$. The restriction of R to a representation of $\mathcal{H}$ on $\oplus_{j>0}\mathfrak{a}_j$, by fibre-wise Lie algebra derivations, integrates to a representation of $\mathcal{H}$ on $\exp(\oplus_{j>0}\mathfrak{a}_j)$ by fibre-wise Lie group isomorphisms; denoting this by R also, we have $R_h(\exp(Y)) = \exp(R_hY)$, for any arrow $h \in \mathcal{H}$ beginning at the base point of $Y \in \oplus_{j>0}\mathfrak{a}_j$, because exp is a diffeomorphism in this case.

The *algebraic prolongation* of the Levi structure is the Lie groupoid semi-direct product $\mathcal{G} = \mathcal{H} \times_R \exp(\oplus_{j>0}\mathfrak{a}_j)$. This is a filtered Lie groupoid with Lie algebroid $B \oplus_{dR} (\oplus_{j>0}\mathfrak{a}_j)$, the algebraic prolongation of Levi structure's infinitesimalization.

Additionally, the representation ad^T of the nilpotent Lie algebra bundle $\oplus_{j>0}\mathfrak{a}_j$ on $\mathrm{gr}(TM) \oplus B^0 \oplus (\oplus_{j>0}\mathfrak{a}_j)$ integrates to a representation of the Lie group bundle $\exp(\oplus_{j>0}\mathfrak{a}_j)$ on $\mathrm{gr}(TM) \oplus B^0 \oplus (\oplus_{j>0}\mathfrak{a}_j)$. We denote this representation by Ad^T.

Theorem (A characterization of transitive filtered Cartan geometries)**.**
Let $(\mathcal{H},\oplus_{j>0}\mathfrak{a}_j,R,T)$ be an abstract Levi structure, B the Lie algebroid of $\mathcal{H}$, P an arbitrary section of $\mathrm{gr}(T^*M) \otimes (\oplus_{j>0}\mathfrak{a}_j)$, $s\colon TM \to B$ any splitting of the anchor of B, and $\sigma\colon \mathrm{gr}(TM) \to TM$ some splitting of the filtered manifold M. Then:

{chh01}
{chh02}

(1) *The map $\Phi\colon \mathrm{gr}(TM)\oplus B^0\oplus(\oplus_j\mathfrak{a}_j) \to B\oplus_{dR}(\oplus_{j>0}\mathfrak{a}_j)$ defined by C1 is invertible.*
(2) *The algebraic prolongation $\mathcal{G} = \mathcal{H}\times_R\exp(\oplus_{j>0}\mathfrak{a}_j)$ is a transitive, filtered Cartan geometry, with model data τ defined by C2 and Cartan connection D uniquely determined by*

C3′. $\quad \bar{D}_{(h,k)}\Phi(V,C,Z) = \Phi(\mathrm{Ad}^T_k(R_hV,\mathrm{Ad}_h C, r_hZ)).$ {cch03}

(3) *Every transitive, filtered, Cartan geometry $(\mathcal{G},D,\tau)$, with Lie algebroid A and anchor kernel A^0, is, up to isomorphism, the algebraic prolongation of some abstract Levi structure, with Cartan connection D and model data τ of the form just described.*

Proof. The preceding infinitesimal result, Theorem 7.4, applies with R there replaced with dR. In particular, (1) is just a restatement of 7.4(1).

The isomorphism Φ intertwines the representation R of $\mathcal{H}$ on $\mathrm{gr}(TM)\oplus B^0\oplus(\oplus_j\mathfrak{a}_j)$ with some representation $\rho^{\mathcal{H}\times\mathrm{id}}$ on $\mathcal{H}\times\mathrm{id}\subset\mathcal{G}$ on $A=B\oplus_{dR}(\oplus_{j>0}\mathfrak{a}_j)$:

(4) {ret01}
$$\rho^{\mathcal{H}\times\mathrm{id}}_{(h,\mathrm{id})}\Phi(V,C,Z)=\Phi\Big(R_h((V,C,Z))\Big).$$

To apply the global case of Proposition 6.2 to $\mathcal{G}$, with τ defined by C2, and $\mathcal{G}'=\mathrm{id}\times\exp(\oplus_{j>0}\mathfrak{a}_j)$, we need to establish 6.2(4). Here this means verifying that

$$\rho^{\mathcal{H}\times\mathrm{id}}_{(h,\mathrm{id})}\Phi(0,C,Z)=\mathrm{Ad}_{(h,\mathrm{id})}\,\Phi(0,C,Z),$$

for each arrow $h\in\mathcal{G}$ beginning as some $m\in M$, $C\in B^0|_m$, $Z\in\oplus_{j>0}\mathfrak{a}_j|_m$. With the help of (4), and the definition of Ad for semi-direct products, this is readily checked. We conclude that there exists a unique τ-compatible Cartan connection D on $\mathcal{G}$ such that

(5) {mask1}
$$\bar{D}_{(h,k)}\Phi(V,C,Z)=\mathrm{Ad}^{\tau}_{(\mathrm{id},k)}\,\rho^{\mathcal{H}\times\mathrm{id}}_{(h,\mathrm{id})}\Phi(V,C,Z),$$

where Ad^{τ} is the representation of $\mathrm{id}\times\exp(\oplus_{j>0}\mathfrak{a}_j)\subset\mathcal{G}$ on A integrating the representation ad^{τ} of $A^1=0\oplus(\oplus_{j>0}\mathfrak{a}_j)$ on A defined by $\mathrm{ad}^{\tau}_{(0,Y)}W=\tau((0,Y),W)$. By the definition of τ, the isomorphism Φ intertwines the representation ad^{τ} of A^1 with the representation $Y\mapsto T((0,0,Y),\,\cdot\,)$ of $\oplus_{j>0}\mathfrak{a}_j$. Consequently, as $\exp(\oplus_{j>0}\mathfrak{a}_j)$ has connected fibres, Φ intertwines the representations Ad^{τ} and Ad^{T}:

(6) {mask2}
$$\mathrm{Ad}^{\tau}_{(\mathrm{id},k)}\,\Phi(V,C,Z)=\Phi(\mathrm{Ad}^{T}_{k}((V,C,Z)).$$

Combining (4), (5) and (6), we obtain C3′.

It is easy to see that the infinitesimalization ∇ of D must be the Cartan connection delivered by Theorem 7.4. Since ∇ is *strongly* compatible with τ, D is also.

Turning to (3), let $(\mathcal{G},D,\tau)$ be a transitive, filtered, Cartan geometry with Lie algebroid A and anchor kernel A^0. By Theorem 4.3(8), we may suppose, without loss of generality, that $\mathcal{G}=\mathcal{H}\times_R\exp(\oplus_{j>0}\mathfrak{a}_j)$, for some filtered Lie groupoid $\mathcal{H}$ and graded Lie algebra bundle $\oplus_{j>0}\mathfrak{a}_j$, where $\mathcal{H}$ acts on $\oplus_{j>0}\mathfrak{a}_j$ by grading-preserving fibre-wise derivations, an action that integrates to an action of $\mathcal{H}$ on $\exp(\oplus_{j>0}\mathfrak{a}_j)$ by fibre-wise Lie group isomorphisms, both actions to be denoted R. Moreover, the Lie algebroid B of $\mathcal{H}$ will satisfy $B^j=0$ for $j>0$ and, by our hypotheses on A, B is transitive with B^0 being the anchor kernel.

Evidently, A is identifiable with $B\oplus_{dR}(\oplus_{j>0}\mathfrak{a}_j)$, which the infinitesimalization ∇ of D makes into a transitive, infinitesimal, filtered Cartan geometry. Consequently, the analysis of 7.3 applies, with R in that analysis replaced with dR. Define $\mathfrak{a}'_j$, $j>0$, and Φ as in that analysis. Then τ can be recovered from $T=\mathrm{gr}_0(\tau)$ according to C2, and, according to Proposition 6.3(3), the grading $A=\oplus_i\mathfrak{a}'_i$ (there denoted $\oplus_i\mathfrak{a}_i$) is $\bar{D}_{(h,\mathrm{id})}$-invariant, for every $h\in\mathcal{H}$. It follows that Φ intertwines some grading-preserving representation R of $\mathcal{H}$ on $\mathrm{gr}(A)=\mathrm{gr}(TM)\oplus B^0\oplus(\oplus_{j>0}\mathfrak{a}_j)$ with the $\bar{D}$-representation of $\mathcal{H}\times\mathrm{id}$ on A:

(7)
$$\bar{D}_{(h,\mathrm{id})}\Phi(V,C,Z)=\Phi\Big(R_h(V,C,Z)\Big);\quad h\in\mathcal{H}.$$

Since $\Phi(0,C,Z)=(C,Z)$ and $\bar{D}$ is just the canonical representation on sections of A^0, R must extend the canonical representation of $\mathcal{H}$ on $\mathrm{gr}(A^0)=B^0\oplus(\oplus_{j>0}\mathfrak{a}_j)$,

which gives

(8) {cab1}
$$\bar{D}_{(h,\mathrm{id})}\Phi(V,C,Z)\Phi(R_hV,\mathrm{Ad}_h\,C,R_hZ);\quad h\in\mathcal{H},$$

where the "R" in R_hZ has the meaning given at the beginning of this discussion.

On the other hand, 7.3(2) implies

$$\bar{\nabla}_{(0,Y)}\Phi(\xi)=\Phi(\mathrm{ad}_Y^T\,\xi),\qquad Y\in\Gamma(\oplus_{j>0}\mathfrak{a}_j),\ \xi\in\Gamma(\mathrm{gr}(A)).$$

Because $\exp(\oplus_{j>0}\mathfrak{a}_j)$ has simply-connected fibres, this intertwining by Φ of infinitesimal actions is an intertwining of their corresponding globalizations:

(9) {cab2}
$$\bar{D}_{(\mathrm{id},k)}\Phi(V,C,Z)=\Phi(\mathrm{Ad}_k^T(V,C,Z)).$$

The formula C3′ follows from (8) and (9) and the identity $(h,k)=(\mathrm{id},k)(h,\mathrm{id})$ for the semi-direct product. □

8. Deforming model data and Cartan connections {deform}

We need a way to relate different choices of model data in a filtered Cartan geometry using transformations of the underlying Lie algebroid or integrating Lie groupoid. Proposition 8.1 below addresses this need. {deformations}

8.1. **ψ-deformations.** For each $\psi\in\Gamma(T^*M\otimes A)$, define $\psi_{TM}:TM\to TM$ by

(1) {opal1}
$$\psi_{TM}V=V+\#\,\psi\,V,$$

and define $\mathrm{Ad}_\psi\colon A\to A$ by

(2) {opal2a}
$$\mathrm{Ad}_\psi\,X=X+\psi(\#X).$$

Here $\#\colon A\to TM$ denotes the anchor. The natural context giving rise to these definitions is provided in 8.3. For the moment, it suffices to observe that: (i) If ψ_{TM} is invertible, then so is Ad_ψ, with

(3) {opal2}
$$\mathrm{Ad}_\psi^{-1}=\mathrm{Ad}_{-\psi\circ(\psi_{TM})^{-1}};$$

and (ii) If A is filtered and transitive and ψ is homogeneous of degree one, i.e., $\psi(A^k)\subset A^{k+1}$ for any k, then ψ_{TM} is invertible, because in that case ψ_{TM} is also homogeneous of degree one with $\mathrm{gr}_1(\psi_{TM})=\mathrm{id}$.

If ψ above is homogeneous of degree $l\geqslant 1$, then $\mathrm{Ad}_\psi-\mathrm{id}$ is too, and the following easily follows:

Lemma. If A is transitive then the push-forward $\tau'=(\mathrm{Ad}_\psi)_*\tau$ of any model data τ on A compatible with a filtration will share the same reduction to a Lie algebra bundle structure on $\mathrm{gr}(A)$, that is, $\mathrm{gr}_0(\tau')=\mathrm{gr}_0(\tau)$.

Conversely, we have:

Proposition. Let A be a transitive filtered Lie algebroid with anchor kernel A^0, and τ, τ' model data compatible with the filtration. Assume $T:=\mathrm{gr}_0(\tau)$ admits a fibre-wise grading section E, and that $\mathrm{gr}_0(\tau')=T$. Then: {saturn1}

(4) There exists a unique $\psi\in\Gamma(T^*M\otimes A)$, necessarily homogeneous of degree one, such that $\tau'=(\mathrm{Ad}_\psi)_*\tau$.

{saturn2}

(5) If τ is compatible with some infinitesimal Cartan connection ∇, then τ' is compatible with the infinitesimal Cartan connection ∇' defined implicitly by

$$\bar{\nabla}'_X Y = \mathrm{Ad}_\psi(\bar{\nabla}_X(\mathrm{Ad}_\psi^{-1} Y)).$$

{saturn3}

(6) If $\mathcal{G}$ is a Lie groupoid integrating A and τ is compatible with some Cartan connection D, then τ' is compatible with the Cartan connection D' defined implicitly by

$$\bar{D}'_g X = \mathrm{Ad}_{\psi(gm)}\, \bar{D}_g\, \mathrm{Ad}_{\psi(m)}^{-1}\, X,$$

for each arrow $g \in \mathcal{G}$ beginning at some $m \in M$, and each $X \in A|_m$, and whose infinitesimalization is ∇' above,

{saturn5}

(7) If $(\mathcal{G}, D, \tau)$ is a transitive filtered Cartan geometry, then $(\mathcal{G}, D', \tau')$ is another transitive filtered Cartan geometry with the same Levi reduction. An analogous statement holds in the infinitesimal case.

When $\tau' = (\mathrm{Ad})_*\tau$, and D' and ∇' are defined (implicitly) by (6) and (5), then the primed quantities are called the *ψ-deformations* of τ, D and ∇. The proof of the proposition is given in 8.2–8.6 below. In 8.7 we connect ψ-deformations to Weyl gauge transformations.

{existence_deformations}

8.2. **The existence of deformations.** The existence part of Proposition 8.1(4) has an elementary proof proceeding as follows. Let τ and τ' be model data as in the proposition and fix a lift $\epsilon \in \Gamma(A^0)$ through $A^0 \to A^0/A^1$ of E. Then, as we have described in 4.1, we get two corresponding splittings $\Phi, \Phi' \colon \mathrm{gr}(A) \to A$ of the Lie algebra bundles (A, τ) and (A, τ'). By definition, we have $\Phi_* T = \tau$ and $\Phi'_* T = \tau'$, and, because τ and τ' agree on A^0 (by the definition of model data) $\Phi' \circ \Phi^{-1}$ is the identity on A^0. It follows from the transitivity of A that $\tau' = \Psi_*\tau$, where $\Psi := \Phi' \circ \Phi^{-1}$ satisfies $\Psi(X) = X + \psi(\#X) = \mathrm{Ad}_\psi X$ for some vector bundle morphism $\psi \colon TM \to A$.

Also following from the definition of splittings, we have, for any $X \in \Gamma(A^k)$ and $k \in \mathbb{Z}$,

{opal3}

(1) $$X \equiv \Phi(\mathrm{gr}_k(X)) \bmod A^{k+1},$$

{opal4}

(2) $$\Phi'(\mathrm{gr}_k(X)) \equiv X \bmod A^{k+1}.$$

From (1) we conclude that $(\Phi' \circ \Phi^{-1})(X) \equiv \Phi'(\mathrm{gr}_k(X)) \bmod A^{k+1}$, because $\Phi' \circ \Phi^{-1}$ is filtration-preserving. Combining this with (2), we conclude that $\Phi' \circ \Phi^{-1}(X) \equiv X \bmod A^{k+1}$, i.e., ψ is homogeneous of degree one.

{kernel}

8.3. **The kernel of $J^1\mathcal{G} \to \mathcal{G}$.** Let A be any Lie algebroid. The sections ψ of $T^*M \otimes A$ for which ψ_{TM} are invertible are sections of an open neighborhood of the zero section of $T^*M \otimes A$ that we denote $\mathrm{Aut}(TM, A)$. The proposition below shows that this set is a model for the kernel of the projection $J^1\mathcal{G} \to \mathcal{G}$ for *any* Lie groupoid $\mathcal{G}$ integrating A. In particular, it is a totally intransitive Lie groupoid (a bundle of Lie groups). We now describe the corresponding groupoid structure directly.

The zero section of $T^*M \otimes A$ constitutes the set of identity elements of $\mathrm{Aut}(TM, A)$. Multiplication is given by

{clost}

(1) $$\psi\varphi = \psi + \varphi + \psi \circ \# \circ \varphi = \varphi + \psi \circ \varphi_{TM}.$$

Inversion is given by

$$\psi \mapsto -\psi \circ (\psi_{TM})^{-1}.$$

A convenient consequence we mention in passing is the formula

(2) {clost2}
$$\psi\varphi^{-1} = (\psi - \varphi) \circ \varphi_{TM}^{-1}.$$

With respect to these groupoid structures, the map

$$\psi \mapsto \psi_{TM} \colon \operatorname{Aut}(TM, A) \to \operatorname{Aut}(TM)$$

is a morphism of totally intransitive Lie groupoids and the commutative kernel consists of those $\psi \colon TM \to A$ with image lying in the kernel of the anchor.

The reader will readily see that the maps defined by 8.3(1) and 8.3(2) define representations of $\operatorname{Aut}(TM, A)$ on A and TM respectively. With respect to these representations the anchor is equivariant: $\# \operatorname{Ad}_\psi X = \psi_{TM} \# X$.

Although the proposition below provides useful context, the proof of Proposition 8.1 will not actually require it. However, the result is needed in Appendix C.

Let $\mathcal{G}$ be any Lie groupoid with Lie algebroid A. Represent elements of $J^1\mathcal{G}$ by maps tangent to some local bisection of $\mathcal{G}$, i.e., by certain vector space morphisms $T_m M \to T_g \mathcal{G}$, where $g \in \mathcal{G}$ is some arrow beginning at some $m \in M$. Recall that the adjoint representation for a Lie groupoid $\mathcal{G}$ is actually a representation of $J^1\mathcal{G}$ on A [6], and that $J^1\mathcal{G}$ also acts naturally on TM. Both representations are denoted Ad, and with respect to these the anchor is equivariant.

Proposition ([6])**.** The map $\psi \mapsto \psi^\wedge \colon \operatorname{Aut}(TM, A) \to J^1 G$, defined by $\psi^\wedge V = V + \psi V$, is a Lie groupoid morphism and embedding, whose image is the kernel of the natural projection $J^1 G \to G$.[9] This map is equivariant with respect to the natural representations of both groupoids on A and TM: $\operatorname{Ad}_{\psi^\wedge} V = \psi_{TM} V$, $\operatorname{Ad}_{\psi^\wedge} X = \operatorname{Ad}_\psi X$, for $V \in T_m M$, $X \in A|_m$, and $m \in M$.

8.4. **The deformed Cartan connections.** In passing, let us explain how the deformed connections ∇' and D' appearing in Proposition 8.1 arise very naturally. Recall that a Cartan connection D is the n-plane field determined by some Lie groupoid right-inverse $S \colon \mathcal{G} \to J^1\mathcal{G}$ for the natural projection $J^1\mathcal{G} \to \mathcal{G}$. Given some $\psi \in \Gamma(\operatorname{Aut}(TM, A))$, we may, according to the preceding proposition, view it as bisection of $J^1\mathcal{G}$ and obtain, via conjugation, a corresponding automorphism K_ψ of $J^1\mathcal{G}$, sending an arrow $\mu \in J^1_g\mathcal{G}$, beginning at some $m \in M$ with projection $g \in \mathcal{G}$, to $\psi(gm)^\wedge \mu (\psi(m)^\wedge)^{-1}$. It is easy to see that $S' := K_\psi \circ S \colon \mathcal{G} \to J^1\mathcal{G}$ is another right-inverse, defining a new Cartan connection D'. From the definition of the corresponding representations $\bar{D}$ and $\bar{D}'$, one immediately derives the formula in 8.1(6).

Now $\bar{\nabla}$ is the infinitesimalization of the representation $\bar{D}$, and it therefore follows from the formula in 8.1(6), and a simple computation, that the infinitesimalization $\bar{\nabla}'$ of $\bar{D}'$ is given by the formula in 8.1(5).

We now return to the proof of Proposition 8.1. Regardless of whether A is integrated by some Lie groupoid, the formula for $\bar{\nabla}'$ given in 8.1(5) must define an infinitesimal Cartan connection, by the characterization of such connections given in Proposition 5.2, which applies because we assume A is transitive. The main point here is that 8.1(5) indeed defines a representation. The details are not difficult and

[9] [6] adopts a different sign convention for the embedding and definition of $\operatorname{Aut}(TM, A)$.

are left to the reader. It is a simple computation to show that $\tau' = (\mathrm{Ad}_\psi)_*\tau$ must be $\bar\nabla'$-invariant (∇'-compatible) whenever τ is $\bar\nabla$-invariant (∇-compatible).

Similarly, the formula for $\bar D'$ given in 8.1(6) must define a global Cartan connection, by the characterization of such connections given in Proposition 5.1, and checking that $\tau' = (\mathrm{Ad}_\psi)_*\tau$ must be $\bar D'$-invariant (D'-compatible) whenever τ is $\bar D$-invariant (D-compatible) is a simple computation.

This completes the proof of parts (5) and (6) of Proposition 8.1.

{deformation_uniqueness}

8.5. **The uniqueness of ψ-deformations.** Having seen that $\psi \mapsto \mathrm{Ad}_\psi$ defines a representation of $\mathrm{Aut}(TM, A)$ on A, to prove the uniqueness of ψ in Proposition 8.1(4), it evidently suffices to show that $(\mathrm{Ad}_\psi)^*\tau = \tau$ implies $\psi = 0$, whenever ψ is homogeneous of degree one, because $\mathrm{Ad}_0 = \mathrm{id}$.

Let τ and E be as in Proposition 8.1 and $\psi\colon TM \to A$ be homogeneous of degree one. If we temporarily write $\hat\psi(X) = \psi(\#X)$, then $(\mathrm{Ad}_\psi)^*\tau = \tau$ means

$$\tau(\hat\psi X, Y) + \tau(X, \hat\psi Y) + \tau(\hat\psi X, \hat\psi Y) = \hat\psi\tau(X, Y),$$

for all $X, Y \in \Gamma(A)$. In particular, if ϵ is any lift of E through $A^0 \to A^0/A^1$, then

(1) {opal5}
$$\tau(\hat\psi X, \epsilon) = \hat\psi(\tau(X, \epsilon)); \qquad X \in \Gamma(A).$$

For an inductive argument, suppose ψ (and therefore $\hat\psi$) is homogeneous of degree $l \geqslant 1$. Then (1) gives $T(\mathrm{gr}_l(\hat\psi)\xi, E) = \mathrm{gr}_l(\hat\psi)(T(\xi, E))$, for any $\xi \in \mathrm{gr}(A)$. In particular, if $j < 0$ and $\xi \in \mathrm{gr}(A)_j$, then $(j + l)\,\mathrm{gr}_l(\hat\psi)\xi = j\,\mathrm{gr}_l(\hat\psi)\xi$, which can only be true if $\mathrm{gr}_l(\hat\psi)\xi = 0$. Since this last is trivially true in the other cases $j \geqslant 0$, we conclude that $\hat\psi$, and hence ψ, is homogeneous of degree $l + 1$. Induction on l shows $\psi = 0$. This completes the proof of Proposition 8.1(4).

{same_levi}

8.6. **Levi reductions are unchanged by ψ-deformations.** Conclusion 8.1(7) follows fairly easily from the definitions and the statement preceding Lemma 8.1 and is left to the reader.

{gauge_transforms}

8.7. **Weyl gauge transformations and ψ-deformations.** If we use a Weyl gauge transformation to transform model data, then we can say something about the ψ-deformation realizing the same change:

Proposition. Let A be a transitive filtered Lie algebroid with anchor kernel A^0, τ model data compatible with the filtration, and assume $T := \mathrm{gr}_0(\tau)$ admits a fibre-wise grading section E. Then for any $Z \in \Gamma(A^l)$, $l \geqslant 1$, there exists unique $\psi \in \Gamma(T^*M \otimes A)$, homogeneous of degree l, such that $(e^{-\mathrm{ad}_Z})_*\tau = (\mathrm{Ad}_\psi)_*\tau$. In that case, for any $X \in A^k$, $k \in \mathbb{Z}$,

(1) {opal7}
$$\psi(\#X) \equiv \tau(Z, X) \bmod A^{k+l+1}.$$

Proof. By Proposition 8.1, there exists unique $\psi \in \Gamma(T^*M \otimes A)$ such that $(e^{-\mathrm{ad}_Z})_*\tau = (\mathrm{Ad}_\psi)_*\tau$, but *a priori* we know only that ψ is homogeneous of degree *one*. If $k \geqslant 0$, then (1) is trivially true, because $\#X = 0$ in that case. So we suppose $k < 0$. Fixing some lift ϵ through $A^0 \to A^0/A^1$ of E, we obtain corresponding splittings $A = \oplus_j \oplus \mathfrak{a}_j$ and $A = \oplus_j \mathfrak{a}'_j$ with respect to the Lie algebra bundle structures τ and

$\tau' := (\mathrm{Ad}_\psi)_*\tau$. To prove (1), it suffices to prove it, for any $k < 0$, under the more restrictive hypothesis $X \in \mathfrak{a}_k$, in which case

(2) {opal0}
$$\mathrm{Ad}_\psi X \in \mathfrak{a}'_k.$$

Since $\mathrm{Ad}_\psi(X) = X + \psi(\#X)$ and ψ is homogeneous of degree one, we may define $X_k := X$ and write $\mathrm{Ad}_\psi(X) = \sum_{m\geqslant k} X_m$, for some $X_{k+1} \in \mathfrak{a}_{k+1}, X_{k+2} \in \mathfrak{a}_{k+2}$, etc., finite in number. These preliminaries set up a simple but tedious computation, tracking the relevant degrees of homogeneity. For completeness, details now follow.

The requirement (2) gives

(3) {opal8}
$$\tau'\big(\epsilon, \sum_{m\geqslant k} X_m\big) = k \sum_{m\geqslant k} X_m.$$

Now, using the fact that $\tau(Z,\epsilon) = -lZ$, we have

$$\begin{aligned} e^{\mathrm{ad}_Z}\epsilon &= \epsilon + [Z,\epsilon] + \frac{1}{2}[Z,[Z,\epsilon]] + \cdots \\ &= \epsilon + \tau(Z,\epsilon) + \frac{1}{2}\tau(Z,\tau(Z,\epsilon)) + \cdots = \epsilon - lZ, \end{aligned}$$

More generally, F1 and the expansion of e^{ad_Z} show that

(4) {ruby0}
$$e^{\mathrm{ad}_Z} Y \equiv Y \bmod A^l; \qquad Y \in \Gamma(A).$$

Regarding the left-hand side of (3), we now compute

$$\tau'\big(\epsilon, \sum_{m\geqslant k} X_m\big) = e^{-\mathrm{ad}_Z}\tau\Big(e^{\mathrm{ad}_Z}\epsilon, e^{\mathrm{ad}_Z}\big(\sum_{m\geqslant k} X_m\big)\Big)$$

(5) {ruby5}
$$= e^{-\mathrm{ad}_Z}\tau\big(\epsilon - lZ, \sum_{m\geqslant k} X_m\big) + e^{-\mathrm{ad}_Z}\tau(\epsilon - lZ, W),$$

where $W := e^{-\mathrm{ad}_Z}(\sum_{m\geqslant k} X_m) - \sum_{m\geqslant k} X_m$, which lies in A^l, by (4). In particular, the second term on the right-hand side of (5) must lie in A^l also. Developing (5) further:

$$\begin{aligned} \tau'\big(\epsilon, \sum_{m\geqslant k} X_m\big) &\equiv e^{-\mathrm{ad}_Z}\tau\big(\epsilon - lZ, \sum_{m\geqslant k} X_m\big) \bmod A^l \\ &\equiv e^{-\mathrm{ad}_Z}\Big(\sum_{m\geqslant k} mX_m - l\sum_{m\geqslant k}\tau(Z,X_m)\Big) \bmod A^l \\ &\equiv \sum_{m\geqslant k} mX_m - l\sum_{m\geqslant k}\tau(Z,X_m) \bmod A^l, \end{aligned}$$

where in the last step we have again applied identity (4). Combining this with (3), we conclude

(6) {ruby6}
$$\sum_{m>k}(m-k)X_m \equiv \sum_{m\geqslant k}\tau(Z,X_m) \bmod A^l.$$

Since $\sum_{m\geqslant k}\tau(Z,X_m)$ lies in A^{k+l}, this shows that $X_m = 0$ whenever $k+1 \leqslant m \leqslant k+l-1$. In particular, this shows that $\psi(\#X) = \sum_{m\geqslant k+1} X_m$, lies in A^{k+l}, completing

the proof that ψ is homogeneous of degree l. But now we may rewrite (6) as

$$\sum_{m\geqslant l} mX_{m+k} \equiv l \sum_{m\geqslant k} \tau(Z, X_m) \bmod A^l.$$

Since $l \geqslant l+k+1$, because $k<0$, we have $A^l \subset A^{l+k+1}$, and it follows that

$$\begin{aligned} \sum_{m\geqslant l} mX_{m+k} &\equiv l \sum_{m\geqslant k} \tau(Z, X_m) \bmod A^{l+k+1} \\ \implies lX_{l+k} &\equiv l\tau(Z, X_k) \bmod A^{l+k+1} \\ \implies \psi(\#X) &\equiv \tau(Z, X) \bmod A^{l+k+1} \end{aligned}$$

□

9. Model curvature

{modelcurvature}

Model curvature, defined for parabolic geometries in 3.10, makes sense for an arbitrary infinitesimal filtered Cartan geometry (A, ∇, τ), so long as we restrict to the transitive case. However, it now depends on both τ and the compatible connection ∇, which is not generally determined uniquely by τ. The main goal of this section is to understand how model curvature transforms under Weyl gauge transformations (Proposition 9.2) and ψ-deformations (Proposition 9.4). We additionally define the "oid" analogue of tractor bundles [2], [14, §1.5.7] to explain the relationship between infinitesimal Cartan connections on Lie algebroids and the adjoint tractor connections appearing in the principal bundle approach to parabolic geometries. Tractor bundles will not be discussed further in the present article.

{model_curvature}

9.1. **Model curvature and consequences of its vanishing.** Given a transitive global or infinitesimal filtered Cartan geometry with underlying Lie algebroid $\#\colon A \to TM$, let τ denote the model data and ∇ the compatible infinitesimal Cartan connection. With Theorem 5.3 in mind, the difference between between $\operatorname{tor}\bar\nabla$ (a "deformed bracket") and τ (the model bracket) is the *model curvature* of the geometry. Specifically, this is the section κ of $\wedge^2 T^*M \otimes A$ defined by

$$\kappa(\#X, \#Y) = \operatorname{tor}\bar\nabla(X, Y) - \tau(X, Y); \qquad X, Y \in \Gamma(A),$$

which makes sense, on account of the strong compatibility property 6.1(1) and equation 5.2(4) defining $\bar\nabla$.

Proposition. The model curvature κ determines the curvature of ∇ according to

$$\operatorname{curv}\nabla(U, V)X = (\bar\nabla_X \kappa)(U, V).$$

In particular, if $\kappa = 0$, then $\operatorname{curv}\nabla = 0$ and $\operatorname{tor}\bar\nabla = \tau$. If the geometry is regular, then the model curvature is homogeneous of degree one, i.e., $\kappa(U, V) \in \Gamma(A^{j+k+1})$ for $U \in \Gamma(T^jM)$ and $V \in \Gamma(T^kM)$; it is otherwise homogeneous of degree zero.

The consequences of $\operatorname{curv}\nabla = 0$ have already been discussed in 5.3. At the very least, according to Theorem 5.3, A will locally be a transitive action algebroid, with the acting Lie algebra isomorphic to the Lie algebra model for τ.

Proof. For any Cartan connection ∇ on a transitive Lie algebroid, one has the easily derived identity,

(1) {solidarity}
$$\operatorname{curv}\nabla(\#X,\#Y)Z = \bar\nabla_Z \operatorname{tor}\bar\nabla(X,Y).$$

See, for example, [3, §5.4]. From the definition of κ,

(2) {poland}
$$(\bar\nabla_Z\kappa)(\#X,\#Y) = (\bar\nabla_Z \operatorname{tor}\bar\nabla)(X,Y) - (\bar\nabla_Z\tau)(X,Y).$$

The stated identity follows from (1), (2) and the $\bar\nabla$-invariance of τ (compatibility of τ and ∇).

By definition,

(3) {pot}
$$\kappa(\#X,\#Y) = \bar\nabla_X Y - \bar\nabla_Y X - ([X,Y] + \tau(X,Y))$$

Suppose $X \in \Gamma(A^j)$ and $Y \in \Gamma(A^k)$ for some $j,k<0$. As we argued in the proof of Proposition 5.5, the non-positivity of j and k means that $A^j \subset A^{j+k+1}$ and $A^k \subset A^{j+k+1}$. By $\bar\nabla$-invariance of the filtration of A, the first two terms on the right-hand side of (3) lie in $\Gamma(A^{j+k+1})$. The last bracketed term lies in $\Gamma(A^{j+k})$, by axiom F1 for Lie algebroid filtrations, and compatibility of τ with the filtration of A. In the regular case, this bracketed term lies in A^{j+k+1}, by the definition of regularity. □ {curv_deform_gauge}

9.2. **Model curvature under gauge transformations.**

Proposition. Let τ be model data on a firmly filtered Lie algebroid A, ∇ a compatible Cartan connection on A, and $q\colon A \to A$ an infinitesimal Weyl gauge transformation, which is necessarily of the form $q = e^{-\operatorname{ad}_Z}$, for some $Z \in \Gamma(A^1)$. Then the pushforward $\tau' = q_*\tau$ is also model data, with $\operatorname{gr}_0(\tau') = \operatorname{gr}_0(\tau)$, and

(1) {tarragon1}
$$\bar\nabla'_X Y = q\nabla_{q^{-1}X}(q^{-1}Y)$$

implicitly defines a Cartan connection ∇' compatible with τ'. If, additionally, $\mathcal{G}$ is a firmly filtered Lie groupoid integrating A, ∇ the infinitesimalization of a Cartan connection D on $\mathcal{G}$, and $Q\colon \mathcal{G} \to \mathcal{G}$ is the global Weyl transformation integrating q (conjugation by $\exp(Z)$), then

(2) {tarragon2}
$$\bar D'_g = q \circ \bar D_{Q^{-1}(g)} \circ q^{-1}$$

implicitly defines a Cartan connection D' on $\mathcal{G}$ compatible with τ'. In any case, if κ is the model curvature of (τ,∇) and κ' the model curvature of (τ',∇'), then

(3) {tarragon5}
$$\kappa'(V_1,V_2) = q\kappa(V_1,V_2); \qquad V_1,V_2 \in \Gamma(TM).$$

Corollary. *If $|A|=1$ and τ is normal, then so is τ'. That is, normality is invariant under Weyl gauge transformations.*

Proof. Regard κ and κ' as sections of $\wedge^2 A^1 \otimes A$. In the special case $|A|=1$, we have $qY = Y$, for any $Y \in \Gamma(A^1)$ and, applying the proposition, we deduce that $\kappa' = q_*\kappa$. But then we have $\partial^{\tau'}\kappa' = \partial^{\tau'} q_*\kappa = q_*\partial^\tau\kappa$, so that normality of κ' follows from the normality of τ. □

Proof of proposition. Since q maps the anchor kernel of A to itself, and q is a Lie algebroid morphism, it is easy to see τ' is model data, given that τ is. That $\mathrm{gr}_0(\tau') = \mathrm{gr}_0(\tau)$ follows from the fact that $q - \mathrm{id}$ is homogeneous of degree one. To show ∇' is well-defined, we need to apply Proposition 5.2, which means showing: (i) $\bar\nabla'$, as defined by (1), is a representation, and supposing $\#Y = 0$, that (ii) $\bar\nabla'_X Y = [X, Y]$. Verifying (i) is is an easy computation appealing to the fact that $\bar\nabla$ is a representation and that q is a Lie algebroid morphism. Regarding (ii), we have

$$\bar\nabla'_X Y = q\bar\nabla_{q^{-1}X}(q^{-1}Y) = q[q^{-1}X, q^{-1}Y] = [X, Y],$$

where the second equality follows from the fact that $\bar\nabla'_{X'}Y' = [X', Y']$ whenever $\#Y' = 0$, and that $\#q^{-1}Y = \#Y$ for any $Y \in \Gamma(A)$. Since τ is $\bar\nabla$-invariant by hypothesis, another easy computation shows that τ' is $\bar\nabla'$-invariant, i.e., is compatible with $\bar\nabla'$.

Regarding the global case, we prove a little more, indicating the origins of formulas (1) and (2). Note first that because Q is a groupoid morphism, we have the identity

(4) {tarragon3}
$$q \circ \mathrm{Ad}_\mu = \mathrm{Ad}_{(J^1Q)\mu} \circ q; \qquad \mu \in J^1\mathcal{G},$$

where Ad is the adjoint representation of $J^1\mathcal{G}$ on $\mathcal{G}$ (see [6]). For a proof of (4), suppose $\mu = J^1_m b$, for some bisection b of $\mathcal{G}$. Then for an arbitrary element of A, represented by source-fibre path $t \mapsto g(t)$ beginning at m, we have

$$\mathrm{Ad}_{J^1_m b}\left(\frac{d}{dt}g(t)\big|_{t=0}\right) = \frac{d}{dt}b(g(t)m)g(t)b(m)^{-1}\big|_{t=0},$$

so that

$$\begin{aligned}(q \circ \mathrm{Ad}_{J^1_m b})\left(\frac{d}{dt}g(t)\big|_{t=0}\right) &= \frac{d}{dt}Q(b(g(t)m))Q(g(t))Q(b(m))^{-1}\big|_{t=0}\\ &= \mathrm{Ad}_{(J^1Q)(J^1_m b)}\,\frac{d}{dt}Q(g(t))\big|_{t=0} = \mathrm{Ad}_{(J^1Q)(J^1_m b)}\,q\left(\frac{d}{dt}g(t)\big|_{t=0}\right).\end{aligned}$$

and (4) is proven. If $S\colon \mathcal{G} \to J^1\mathcal{G}$ is the splitting associated with D, then associated with the splitting

(5) {tarragon4}
$$S' = J^1Q \circ S \circ Q^{-1}$$

is a new Cartan connection D', and applying the identity (4) and the definition of representations $\bar D$ and $\bar D'$ associated with these connections, we obtain the formula (2). One can either show directly that $\bar\nabla'$ is the infinitesimalization of $\bar D'$ or argue as follows. Let $s, s'\colon A \to J^1A$ be the splittings associated with the infinitesimalizations ∇ and ∇' of D and D'. Then, differentiating (5), we have $s' = J^1q \circ s \circ q^{-1}$. Now we compute

$$\begin{aligned}\bar\nabla'_X Y = \mathrm{ad}_{s'X}\,Y = \mathrm{ad}_{J^1q(s(q^{-1}X))}\,Y &= \mathrm{ad}_{J^1q(J^1(q^{-1}X)+\nabla(q^{-1}X))}\,Y\\ &= \mathrm{ad}_{J^1X+q\circ\nabla(q^{-1}X)}\,Y = [X, Y] + q(\nabla(q^{-1}X))(\#Y)\\ &= q\big([q^{-1}X, q^{-1}Y] + \nabla_{\#q^{-1}Y}(q^{-1}X)\big) = q\bar\nabla_{q^{-1}X}(q^{-1}Y),\end{aligned}$$

using $\#Y = \#q^{-1}Y$ in the second last step. This recovers (2). The $\bar D'$-invariance of τ' readily follows from the $\bar D$-invariance of τ, i.e., τ' is indeed compatible with D'.

Regarding (3), we compute, for any $X, Y \in \Gamma(A)$,

$$\begin{aligned}\kappa'(\#X, \#Y) &= \bar{\nabla}'_X Y - \bar{\nabla}'_Y X - [X, Y] - \tau'(X, Y) \\ &= q\bar{\nabla}_{q^{-1}X}(q^{-1}Y) - q\bar{\nabla}_{q^{-1}Y}(q^{-1}X) - q[q^{-1}X, q^{-1}Y] - q(\tau(q^{-1}X, q^{-1}Y)) \\ &= q\kappa(\#q^{-1}X, \#q^{-1}Y) = q\kappa(\#X, \#Y).\end{aligned}$$

□

As remarked in 3.10, in the parabolic case we can view model curvature as a section of $\wedge^2 A^1 \otimes A$, which appears in a natural chain complex. We examine this complex in more detail next. {homology}

9.3. **The Kostant codifferential and its homology.** Let A be a filtered Lie algebroid with filtration-compatible model data τ. For the moment, the Lie algebra model for τ is arbitrary; it need not be semisimple, and we need not suppose A is transitive. However we will assume A admits a τ-compatible Weyl gauge.

Recall from 3.10 that we have a natural chain complex

(1) {chain1}
$$0 \xleftarrow{\partial^\tau} A \xleftarrow{\partial^\tau} A^1 \otimes A \xleftarrow{\partial^\tau} \wedge^2 A^1 \otimes A \xleftarrow{\partial^\tau} \cdots,$$

where ∂^τ is defined by 3.10(2). In the proposition below, we will establish that the corresponding bundles of homology groups have realizations depending only on $T = \mathrm{gr}_0(\tau)$, and are canonically *graded*, and not just canonically filtered.

Notice that the bracket $T = \mathrm{gr}_0(\tau)$ makes $\mathrm{gr}(A)$ into a representation of the Lie algebra bundle $\mathrm{gr}(A^1)$ and we obtain a graded version of (1),

(2) {chain2}
$$0 \xleftarrow{\partial^*} \mathrm{gr}(A) \xleftarrow{\partial^*} \mathrm{gr}(A^1) \otimes \mathrm{gr}(A) \xleftarrow{\partial^*} \wedge^2 \mathrm{gr}(A^1) \otimes \mathrm{gr}(A) \xleftarrow{\partial^*} \cdots,$$

where ∂^* is defined by the same formula 3.10(2), with τ there replaced with T.

The filtration of A induces a filtration of each term in the complex (1); for example decomposable elements of $(\wedge^2 A^1 \otimes A)^l$ are of the form $Y \wedge Z \otimes X$, for some $Y \in A^i$, $Z \in A^j$ and $X \in A^k$ with $i, j \geqslant 1$ and $i + j + k = l$. Similarly, each term of the complex (2) is graded. The differential ∂^τ is filtration-preserving, while ∂^* is grading-preserving. There is no canonical chain map from complex (1) to (2). However, for each fixed $l \in \mathbb{Z}$, we have subcomplexes

(3) {chain3}
$$0 \xleftarrow{\partial^\tau} A^l \xleftarrow{\partial^\tau} (A^1 \otimes A)^l \xleftarrow{\partial^\tau} (\wedge^2 A^1 \otimes A)^l \xleftarrow{\partial^\tau} \cdots$$

(4) {chain4}
$$0 \xleftarrow{\partial^*} \mathrm{gr}(A)_l \xleftarrow{\partial^*} (\mathrm{gr}(A^1) \otimes \mathrm{gr}(A))_l \xleftarrow{\partial^*} (\wedge^2 \mathrm{gr}(A^1) \otimes \mathrm{gr}(A))_l \xleftarrow{\partial^*} \cdots,$$

and gr is a chain map from (3) to (4).

We let $H_n(A^1, A)$, $H_n(\mathrm{gr}(A^1), \mathrm{gr}(A))$, $H_n(A^1, A)^l$, and $H_n(\mathrm{gr}(A^1), \mathrm{gr}(A))_l$ denote the degree n homologies of the complexes (1), (2), (3), and (4), understood as vector bundles over M. Because ∂^τ is filtration-preserving and ∂^* is grading-preserving, for each $l \in \mathbb{Z}$, the natural morphisms

$$H_n(A^1, A)^l \to H_n(A^1, A)$$
$$H_n(\mathrm{gr}(A^1), \mathrm{gr}(A))_l \to H_n(\mathrm{gr}(A^1), \mathrm{gr}(A))$$

are injective, and we see easily that this allows us to view $H_n(A^1, A)$ as a filtered vector bundle, and $H_n(\mathrm{gr}(A^1), \mathrm{gr}(A))$ as a graded vector bundle.

Each term in the complex (1) is a representation of A^1, and ∂^τ is equivariant with respect to these representations, so that $H_n(A^1, A)$ is a also an A^1-representation. These integrate to representations of the bundle of simply-connected nilpotent Lie groups $\exp(A^1)$ integrating A^1. In the case A is integrated by some Lie groupoid $\mathcal{G}$ admitting a Weyl gauge, we may identify $\exp(A^1)$ with a normal subgroupoid $\mathcal{G}^1 \subset \mathcal{G}$, by Proposition 4.3.

Proposition. Let A be a filtered Lie algebroid and τ model data compatible with the filtration. Then:

(5) The actions of A^1 and $\exp(A^1)$ on $H_n(A^1, A)$ are trivial. In particular, if ∇ is a Cartan connection on A compatible with τ, then the representation $\bar{\nabla}$ of A on itself makes $H_n(A^1, A)$ into an A/A^1-representation. In the global case, if D is a τ-compatible Cartan connection on $\mathcal{G}$, then the representation $\bar{D}$ of $\mathcal{G}$ on A makes $H_n(A^1, A)$ into $\mathcal{G}/\mathcal{G}^1$-representation. {hay1}

(6) For each Weyl gauge ϵ compatible with τ, the canonical isomorphism associated with the splitting $\Phi_\epsilon\colon \operatorname{gr}(A) \to A$,

$$\tilde{\Phi}_\epsilon\colon \wedge^n \operatorname{gr}(A^1) \otimes \operatorname{gr}(A) \to \wedge^n A^1 \otimes A$$

is a chain map between the complexes (2) and (1). {hay2}

(7) The corresponding isomorphism in homology

$$(\tilde{\Phi}_\epsilon)_*\colon H_n(\operatorname{gr}(A^1), \operatorname{gr}(A)) \to H_n(A^1, A)$$

is independent of the choice of ϵ. In particular, if A admits a Weyl gauge, then the filtered vector bundle $H_n(A^1, A)$ has a canonical splitting. {hay3}

Lemma. Let Ad^τ denote the representation of of $\exp(A^1)$ on A and let ϵ_1, ϵ_2 be two Weyl gauges compatible with τ. Then there exists a smooth section k of $\exp(A^1)$ such that $\Phi_{\epsilon_2}(\xi) = \operatorname{Ad}^\tau_{k(m)} \Phi_{\epsilon_1}(\xi)$, for any $\xi \in \operatorname{gr}(A)|_m$.

Proof. Applying Lemma 4.1(1) fibre-wise, there is a unique section k of the bundle of simply-connected nilpotent Lie groups $\exp(A^1)$ such that $\epsilon_2(m) = \operatorname{Ad}^\tau_{k(m)} \epsilon_1(m)$. Since we suppose ϵ_1 and ϵ_2 are smooth, k must be also. The desired result now follows from the fibre-wise application of 4.1(2). □

Proof of proposition. In proving (5), it suffices to prove the claim for A^1, since $\exp(A^1)$ has connected source-fibres. We need to show that A^1 maps the kernel of $\partial^\tau\colon \wedge^n A^1 \otimes A \to \wedge^{n-1} A^1 \otimes A$ into the image of $\partial^\tau\colon \wedge^{n+1} A^1 \otimes A \to \wedge^n A^1 \otimes A$. Following [14, p. 263], we see this follows from the identity

$$-\operatorname{ad}^\tau_Y \omega = \partial^\tau(Y \wedge \omega) + Y \wedge \partial^\tau \omega; \qquad \omega \in \Gamma(\wedge^n A^1 \otimes A),\ Y \in \Gamma(A^1).$$

To prove the identity, it suffices to prove it for decomposable ω. For example, if $\omega = Z \otimes X$ $(n = 1)$, then we compute

$$\partial^\tau(Y \wedge \omega) = -Z \otimes \tau(Y, X) + Y \otimes \tau(Z, X) - \tau(Y, Z) \otimes X$$
$$Y \wedge \partial^\tau \omega = Y \wedge (-\tau(Z, X)),$$

so that $\quad \partial^\tau(Y \wedge \omega) + Y \wedge \partial^\tau \omega = -Z \otimes \tau(Y, X) - \tau(Y, Z) \otimes X = -\operatorname{ad}^\tau_Y \omega.$

See op. cited for the general argument.

The definition of $\tilde{\Phi}_\epsilon$ is clear. For example, we have $\tilde{\Phi}_\epsilon(Z_1 \wedge Z_2 \otimes X) = \Phi_\epsilon(Z_1) \wedge \Phi_\epsilon(Z_2) \wedge \Phi_\epsilon(X)$. To prove (6), it suffices to show that $\partial^\tau \tilde{\Phi}_\epsilon \omega = \tilde{\Phi}_\epsilon \partial^* \omega$ for all $\omega \in \wedge^{n+1} \operatorname{gr}(A^1) \otimes \operatorname{gr}(A)$ that is decomposable, which is readily accomplished: Take

ω to be of the form $\mathrm{gr}(Y_0) \wedge \ldots \wedge \mathrm{gr}(Y_n) \otimes \mathrm{gr}(X)$, where, without loss of generality, we suppose $Y_0, \ldots, Y_n$ and X each lie in some component $\mathfrak{a}_j$, in the decomposition $A = \oplus_i \mathfrak{a}_i$ associated with ϵ ($j \geqslant 1$ in the case of $Y_0, \ldots, Y_n$). Part (7) now follows from preceding parts and the lemma above. □

{curv_deform}

9.4. **Model curvature under ψ-deformations.** Under the same blanket assumptions as 9.3, consider the *cochain* complex

{chain5a}

$$0 \xrightarrow{\partial} \mathrm{gr}(A) \xrightarrow{\partial} \mathrm{gr}(A)^*_- \otimes \mathrm{gr}(A) \xrightarrow{\partial} \wedge^2 \mathrm{gr}(A)^*_- \otimes \mathrm{gr}(A) \xrightarrow{\partial} \cdots, \tag{1}$$

where $\mathrm{gr}(A)_- := \oplus_{j<0} \mathrm{gr}(A)_j$ is a Lie algebra subbundle, which accordingly acts on $\mathrm{gr}(A)$. Over each point $m \in M$, (1) is the classical cochain complex associated with the representation ad^T of $\mathrm{gr}(A)_-|_m$ on $\mathrm{gr}(A)|_m$. Here ∂ is defined by

$$\begin{aligned}\partial\omega(V_0, \ldots, V_n) = &\sum_{i=0}^{n} (-1)^i T(V_i, \omega(V_0, \ldots, \hat{V}_i, \ldots V_n)) \\ &+ \sum_{i<j} (-1)^{i+j} \omega(T(V_i, V_j), V_0, \ldots, \hat{V}_i, \ldots \hat{V}_j, \ldots V_n),\end{aligned}$$

where hats indicate omission. In the semisimple case we will see in 9.5 that (1) can be viewed as the "adjoint" of 9.3(2).

The formula shows that ∂ is grading-preserving if we equip $\wedge^n \mathrm{gr}(A)^*_- \otimes \mathrm{gr}(A)$ with the grading in which a section ω has degree l if it is homogeneous of degree l, by which we mean that $\omega(V_1, \ldots V_n) \in \Gamma(\mathrm{gr}(A)_{l+i_1+i_2+\ldots+i_n})$ whenever $V_j \in \Gamma(\mathrm{gr}(A)_{i_j})$, for each $j \in \{1, \ldots n\}$.

Our interest in the cochain complex (1) is following:

Proposition. Let (A, ∇, τ) be an infinitesimal filtered Cartan geometry. Furthermore, suppose that A is transitive and that A^0 is the anchor kernel of A, so that $\mathrm{gr}(A)_- \cong \mathrm{gr}(TM)$ and ∂_n is a map

$$\partial_n \wedge^n \mathrm{gr}(T^*M) \otimes \mathrm{gr}(A) \to \wedge^{n+1} \mathrm{gr}(T^*M) \otimes \mathrm{gr}(A).$$

For some $\psi \in \Gamma(T^*M \otimes A)$ that is homogeneous of degree $l \geqslant 1$, let ∇' and τ' be the ψ-deformations of ∇ and τ, as defined in 8.1, and let κ, κ' denote the model curvatures of the infinitesimal parabolic geometries (A, ∇, τ) and (A, ∇', τ'). Then $\kappa' - \kappa$ is homogeneous of degree l with $\mathrm{gr}_l(\kappa' - \kappa) = -\partial\, \mathrm{gr}_l(\psi)$.

Proof. It is not hard to see that the claim follows from the following two facts proven below:

{plat1}

{plat2}

(2) The restriction of $\mathrm{tor}\,\bar{\nabla}' - \mathrm{tor}\,\bar{\nabla}$ to a section of $\wedge^2 \mathrm{gr}(A)^*_- \otimes A$ is homogeneous of degree $l+1$.
(3) The restriction $r = \tau' - \tau$ to a section of $\wedge^2 \mathrm{gr}(A)^*_- \otimes A$ is homogeneous of degree l, with $\mathrm{gr}_l(r) = -\partial\, \mathrm{gr}_l(\psi)$.

Let $X \in \Gamma(A^j)$ and $Y \in \Gamma(A^k)$ arbitrary sections, with $j, k < 0$. Here are some *Preliminary Observations:* Since ψ is homogeneous of degree $l \geqslant 1$, $\psi_{TM} \colon TM \to TM$ is filtration-preserving, with $\mathrm{gr}_0(\psi_{TM}) = \mathrm{id} = \mathrm{gr}_0(\psi_{TM}^{-1})$; $\bar{\nabla}_X$ and $\bar{\nabla}_Y$ are also filtration-preserving, because ∇ is required to be compatible with filtration, by the definition of infinitesimal filtered Cartan geometries; because $j, k < 0$, we always have $A^m \subset A^{m+j+1}$ and $A^m \subset A^{m+k+1}$ for any integer m; because $l \geqslant 1$, we always have $A^{m+l} \subset A^m$, for any m; and $\mathrm{gr}_0(\#) \colon \mathrm{gr}(A)_- \to \mathrm{gr}(TM) \cong \mathrm{gr}(A)_-$ is the identity.

On expanding the definition of $\bar\nabla'$ (Equation 8.1(5)) and the definition of Ad_ψ (Equation 8.1(2)) and its inverse (Equation 8.1(3)) one obtains

$$\bar\nabla'_X Y - \bar\nabla_X Y = \bar\nabla_X(\psi(\psi_{TM}^{-1}(\#Y))) + \psi(\#\bar\nabla_X Y) + \psi(\#\bar\nabla_X(\psi(\psi_{TM}^{-1}(\#Y)))).$$

By our preliminary observations, the terms on the right-hand side lie in A^{k+l}, A^{k+l} and A^{k+2l}, respectively, and accordingly,

$$\bar\nabla'_X Y - \bar\nabla_X Y \in \Gamma(A^{j+k+l}) \ni \bar\nabla'_Y X - \bar\nabla_Y X,$$

where the second part follows by symmetry. Claim (2) follows immediately.

Similarly expanding $\tau' = (\mathrm{Ad}_\psi)_*\tau$, one derives

$$\begin{aligned}(4)\quad \tau'(X,Y) - \tau(X,Y) = {} & -\tau(\psi(\psi_{TM}^{-1}(\#X)),Y) - \tau(X,\psi(\psi_{TM}^{-1}(\#Y))) \\ & + \tau(\psi(\psi_{TM}^{-1}(\#X)),\psi(\psi_{TM}^{-1}(\#Y))) \\ & + \psi(\#\tau(X,Y)) \\ & - \psi(\#\tau(\psi(\psi_{TM}^{-1}(\#X))),Y) - \psi(\#\tau(X,\psi(\psi_{TM}^{-1}(\#Y)))) \\ & + \psi(\#(\psi(\psi_{TM}^{-1}(\#X)),\psi(\psi_{TM}^{-1}(\#Y)))).\end{aligned}$$

By our preliminary observations, the terms on the second and fourth lines lie in A^{j+k+2l}, while that on the last line lies in A^{j+k+3l}, which means both lie in $A^{j+k+l+1}$. The remaining terms lie in A^{j+k+l}, which shows that r is indeed homogeneous of degree l, and

$$\begin{aligned}&\tau'(X,Y) - \tau(X,Y) \equiv \\ &\quad - \tau(\psi(\psi_{TM}^{-1}(\#X)),Y) - \tau(X,\psi(\psi_{TM}^{-1}(\#Y))) + \psi(\#\tau(X,Y)) \bmod A^{j+k+l+1},\end{aligned}$$

from which (3) now readily follows. □

{hodge}

9.5. **Hodge theory.** To the blanket assumptions of 9.3 we now add the following assumptions: τ is semisimple, A^0 is a fibre-wise parabolic subalgebra of (A,τ), and the filtration of A is the associated canonical one. Then $T = \mathrm{gr}(\tau)$ is semisimple and $|A| \leqslant k$ for some $k \in \mathbb{Z}$ (Lemma A.3). In this case, as we now explain, ∂^* and ∂ enjoy a relationship analogous to that between the differential and codifferential operators of classical Hodge theory. Here we are trivially extending to semisimple Lie algebra *bundles* observations about semisimple Lie *algebras* originating in Kostant [21], and detailed in [14, §3.3].

Fixing some multiple of the Killing form for T, we may identify $\mathrm{gr}(A^1) = \mathrm{gr}(A)_1 \oplus \mathrm{gr}(A)_2 \oplus \cdots \mathrm{gr}(A)_k$ with $\mathrm{gr}(A)^*_- = (\mathrm{gr}(A)_{-k} \oplus \cdots \oplus \mathrm{gr}(A)_{-2} \oplus \mathrm{gr}(A)_{-1})^*$. Under this identification, we can combine 9.4(1) and 9.3(2) into one diagram

$$(1)\quad \overset{\{\mathrm{chain5}\}}{0} \underset{\partial^*}{\overset{\partial}{\rightleftarrows}} \mathrm{gr}(A) \underset{\partial^*}{\overset{\partial}{\rightleftarrows}} \mathrm{gr}(A^1)\otimes\mathrm{gr}(A) \underset{\partial^*}{\overset{\partial}{\rightleftarrows}} \wedge^2\mathrm{gr}(A^1)\otimes\mathrm{gr}(A) \underset{\partial^*}{\overset{\partial}{\rightleftarrows}} \cdots .$$

As we shall see, the following non-trivial decomposition, together with the calculation summarised in Proposition 9.4, is what makes normalization of curvature in parabolic geometry possible:

$$(2)\quad \overset{\{\mathrm{nontrivial}\}}{\mathrm{im}\,\partial_{n-1} \oplus \ker\partial_n^*} = \boxed{\wedge^n\mathrm{gr}(A^1)\otimes\mathrm{gr}(A) = \ker\partial_n \oplus \mathrm{im}\,\partial_{n+1}^*}.$$

As in Hodge theory, this follows from the existence (over each point $m \in M$) of an inner product with respect to which ∂_n is the adjoint of ∂_{n+1}^*, for any n; in the

construction, omitted here, the non-degenerate Killing form of T plays an essential role. See [14, §3.3] for details. And, as in Hodge theory, we can deduce from (2):

Theorem. *If* $\Box := \partial^*\partial + \partial\partial^*$, *then*

(3) {hodge1}
$$\ker \Box_n = \ker_n \partial_n \cap \ker_n \partial_n^*$$

(4) {hodge2} *and*
$$\wedge^n \operatorname{gr}(A^1) \otimes \operatorname{gr}(A) = \operatorname{im} \partial_{n-1} \oplus \ker \Box_n \oplus \operatorname{im} \partial_{n+1}^*.$$

In particular, the projection $\ker \partial_n \to H^n(\operatorname{gr}(A^1), \operatorname{gr}(A))$ *restricts to an isomorphism* $\ker \Box_n \cong H^n(\operatorname{gr}(A^1), \operatorname{gr}(A))$, *while the projection* $\ker \partial_n^* \to H_n(\operatorname{gr}(A^1), \operatorname{gr}(A))$ *restricts to an isomorphism* $\ker \Box_n \cong H_n(\operatorname{gr}(A^1), \operatorname{gr}(A))$.

Here $H^n(\operatorname{gr}(A^1), \operatorname{gr}(A))$ is the degree-n ∂-cohomology of (1), which by the theorem, is isomorphic to the ∂^*-homology $H_n(\operatorname{gr}(A^1), \operatorname{gr}(A))$, which is canonically isomorphic to the ∂^τ-homology $H_n(A^1, A)$, by Proposition 9.3. That is, the (co)homologies of the chain complexes 9.3(1), 9.3(2) and 9.4(1) are all canonically isomorphic.

The next result is an immediate consequence of 9.3(7) and transitivity. It connects us to the point of view in the principal bundle approach, but is also important because, following from the work of [21], the cohomology groups $H^n(\mathfrak{g}_-, \mathfrak{g})$ are known in the cases we will need them [14, §3.3].

Proposition. Let $(\mathcal{G}, \tau)$ be a parabolic geometry with Lie algebroid A, and let $\mathfrak{g}$ be a filtered Lie algebra constituting a model for the semisimple Lie algebra bundle structure τ; without loss of generality, we may suppose $\mathfrak{g}$ is actually graded (Lemma A.2). Then the mutually isomorphic (co)homology vector bundles

$$H_n(A^1, A), \quad H_n(\operatorname{gr}(A^1), \operatorname{gr}(A)), \quad H^n(\operatorname{gr}(A)_-, \operatorname{gr}(A))$$

are associated bundles for a representation of the vertex group of $\mathcal{G}/\mathcal{G}^1$ over some point $m_0 \in M$ on the Lie algebra cohomology $H^n(\mathfrak{g}_-, \mathfrak{g})$, where $\mathfrak{g}_- = \oplus_{j<0}\mathfrak{g}_j$.

Remark. By Remark 3.6, the vertex group is reductive, its Lie algebra being the reductive Levi quotient $\mathfrak{p}/\mathfrak{g}^0$, where $\mathfrak{p} = \oplus_{j<0}\mathfrak{g}_j$ and $\mathfrak{g}^0 = \oplus_{j\geqslant 0}\mathfrak{g}_j$.

{tractors}

9.6. **Tractor bundles.** This section, not referenced in the sequel, is included to explain the relationship between the infinitesimal Cartan connections of the current approach, and the adjoint tractor connections of the principal bundle approach. Tractor bundles are relevant to the problem of constructing invariant differential operators, which is beyond the scope of the present article.

Let $(\mathcal{G}, D, \tau)$ be a transitive, filtered, Cartan geometry and ∇ the corresponding infinitesimal Cartan connection on the Lie algebroid A of $\mathcal{G}$. Recall that strong compatibility means that the representation $\bar\nabla$ of A on itself as a Lie algebroid, and the representation of A on itself as a Lie algebra bundle, $X \mapsto \operatorname{ad}^\tau(X) := \tau(X, \cdot)$, restrict to the same representation of the anchor kernel A^0. So, we have

$$\operatorname{ad}^\tau_Y X = \bar\nabla_Y X; \qquad Y \in \Gamma(A^0),\ X \in \Gamma(A)$$
$$\bar D_g \operatorname{ad}^\tau_Y X = \operatorname{ad}^\tau_{\bar D_g Y} \bar D_g X; \qquad Y,\ X \in A|_m,\ m \in M,\ g \in \mathcal{G},$$

where the second identity is just the $\bar D$-invariance of τ, and we there understand g to be any arrow beginning at m.

More generally, a *tractor bundle* is any representation E of $\mathcal{G}$, together with some infinitesimal representation ad^τ of the Lie algebra bundle (A, τ) on E, satisfying

(1) {hj1}
$$\mathrm{ad}^\tau_Y \sigma = Y\sigma; \qquad Y \in \Gamma(A^0),\ \sigma \in \Gamma(E)$$

(2) {hj2}
$$g \cdot \mathrm{ad}^\tau_Y \sigma = \mathrm{ad}^\tau_{\bar{D}_g Y}(g \cdot \sigma); \qquad Y \in A|_m,\ m \in M,\ \sigma \in E,\ g \in \mathcal{G}.$$

Here we use juxtaposition to denote the action of A on E as a Lie algebroid (the infinitesimalization of the representation of $\mathcal{G}$ on E).

The significance of tractor bundles is that they come equipped with Koszul connections, called *tractor connections*, well-defined by

$$\nabla^E_{\#X}\sigma = X\sigma - \mathrm{ad}^\tau_X \sigma.$$

Here # denotes anchor. Returning to the special case $E = A$, where ∇^A is known as the *adjoint tractor connection*, we readily derive the formula

(3) {tractor1}
$$\nabla^A_V Y = \nabla_V Y + \kappa(V, \#Y),$$

which proves the following:

> *The infinitesimal Cartan connection is the adjoint tractor connection if and only if the model curvature vanishes.*

For an explicit construction of tractor bundles, connecting them with definition given in [14], we adopt the principal bundle point of view. Fix some arbitrary $m_0 \in M$, let $\Pi \subset \mathcal{G}$ denote all arrows beginning at m_0, $P \subset \Pi$ the Lie group of arrows beginning and ending at m_0, $\mathfrak{g}$ the semisimple Lie algebra $(A|_{m_0}, \tau)$, and $\mathfrak{g}^0$ the Lie algebra of P. Transitivity means Π will be a principal P-bundle. Now let V be a finite dimensional representation of P and *suppose the corresponding infinitesimal representation* $\mathfrak{g}^0 \to \mathfrak{gl}(V)$ *is the restriction of a representation* $\rho\colon \mathfrak{g} \to \mathfrak{gl}(V)$. Then, we have:

Proposition. The associated bundle $E(\rho) = \Pi \times_P V$ is a tractor bundle, and all tractor bundles are of the form $E(\rho)$ for some $\rho\colon \mathfrak{g} \to \mathfrak{gl}(V)$ as above.

Proof. For elements $g_1, g_2 \in \Pi$ and $\xi \in \mathfrak{g}$, let us write

$$\langle g_1, g_2\rangle = g_1 g_2^{-1} \in \mathcal{G}$$
$$\langle g_2, \xi\rangle = \bar{D}_{g_2}\xi \in A.$$

Transitivity means arbitrary elements of $\mathcal{G}$ and A are of this form, and we have

(4) {hj3}
$$\bar{D}_{\langle g_1, g_2\rangle}\langle g_2, \xi\rangle = \langle g_1, \xi\rangle.$$

If, for each $[g_2, v] \equiv (g_2, v) \bmod P \in E(\rho)$, we define

$$\mathrm{ad}^\tau_{\langle g_2, \xi\rangle}[g_2, v] = [g_2, \rho_\xi v],$$

then, with the help of (4), it is easy to see that (1) and (2) hold. The proof of the second assertion is left to the reader. □

10. Parabolic geometries. Part II

{parabolic_proofs}

This section proves many results stated in Section 3, where (global) parabolic geometries were defined. An *infinitesimal* parabolic geometry consists of a transitive, firmly filtered Lie algebroid A, and filtration-compatible model data τ, satisfying the same conditions given there, but with the Cartan connection D in (3) replaced with an infinitesimal Cartan connection ∇. Evidently, every parabolic geometry has a corresponding infinitesimalization.

{parabolic}

10.1. On the definition of parabolic geometries.

Proposition. If M is connected then, in the global groupoid case, and in the case of an integrable Lie algebroid, it suffices to check that 3.5(2) holds over a single point $m \in M$.

Proof. We begin with the global case. Assume the conditions hold at $m_1 \in M$ and let $m_2 \in M$ be arbitrary. By transitivity, there exists an arrow $g \in \mathcal{G}$ from m_1 to m_2. By firmness of the filtration, Ad_g maps $A^0|_{m_1}$ to $A^0_{m_2}$; since the adjoint representation preserves the anchor kernel, condition 3.5(2) must hold at m_2.

In the integrable, infinitesimal case, we may suppose A is integrated by some Lie groupoid $\mathcal{G}$ with simply connected source fibres, so that the representation $\bar{\nabla}$ integrates to a representation $\bar{D}$, which must leave τ invariant, because $\bar{\nabla}$ does. Also, the adjoint representation of $\mathcal{G}$ will preserve $\mathrm{gr}(A^0)$ because this is already ad-invariant. We can accordingly apply the preceding arguments for the global case. □

{uniquely_determined}

10.2. Cartan connections are uniquely determined by model data.

Theorem. *In a global or infinitesimal parabolic geometry, a compatible Cartan connection is uniquely determined by the model data τ, with which it is strongly compatible.*

The proof requires these facts about semisimple Lie algebras: All derivations are inner; Every parabolic subalgebra (as we have defined it) contains a Borel subalgebra [10]; The centralizer of a Borel subalgebra is always trivial; The only automorphism of the Lie algebra fixing elements of a Borel subalgebra is the identity. The last two facts are readily proved by readers familiar with the basic structure theory of semisimple Lie algebras.

Proof. In the infinitesimal case, suppose that ∇^1 and ∇^2 are Cartan connections compatible with model data τ on A, and write $\delta_X Y = \bar{\nabla}^1_X Y - \bar{\nabla}^2_X Y$. For each $X \in A|_m$, $m \in M$, we may regard δ_X as a linear map $A|_m \to A|_m$ and, because τ is both $\bar{\nabla}^1$ and $\bar{\nabla}^2$-invariant (by compatibility) it follows that δ_X is even a Lie algebra derivation, with respect to the bracket τ. Moreover, as $\bar{\nabla}^j$ extends the canonical representation of A on its anchor kernel A^0, it follows that $\delta_X Y = 0$ for all Y in the parabolic subalgebra $A^0|_m \subset A|_m$. By semisimplicity, there exists $Z \in A|_m$ such that $\delta_X(\,\cdot\,) = \tau(Z, \,\cdot\,)$, and in particular such that $\tau(Z, Y) = 0$ for all $Y \in A^0|_m$. So Z is in the centralizer of a parabolic subalgebra of a semisimple Lie algebra, and so is in the centralizer of some Borel subalgebra, and consequently zero. This shows $\delta_X = 0$ and we conclude $\bar{\nabla}^1 = \bar{\nabla}^2$. Since A is transitive, $\nabla^1 = \nabla^2$, by Proposition 5.2.

To prove strong compatibility, suppose $\bar{\nabla}$ is a τ-compatible Cartan connection (the infinitesimalization of some Cartan connection D in the global case). Fix some $X \in A^0|_m$, $m \in M$. Then as A^0 lies in the anchor kernel, we may view $\bar{\nabla}_X$ as a

linear map $A|_m \to A|_m$. In fact, with respect to the bracket τ, this is a Lie algebra derivation, because τ is $\bar\nabla$-invariant (∇ and τ are compatible). By the semisimplicity, there is $X' \in A|_m$ such that $\bar\nabla_X Y = \tau(X', Y)$ for all $Y \in A|_m$. In the special case $Y \in A^0|_m$, we have $\bar\nabla_X Y = [X, Y] = \tau(X, Y)$, the first equality following from 5.2(4), the second because τ extends the canonical Lie algebra structure of A^0. Therefore, $\tau(X' - X, Y) = 0$ for all such Y. In other words, $X' - X$ must lie in the centralizer of a parabolic subalgebra, which is trivial, as noted above. We conclude $X' = X$ and whence $\bar\nabla_X Y = \tau(X, Y)$ for all $Y \in A|_m$.

In the global case, which is proven similarly, suppose D^1 and D^2 are Cartan connections compatible with model data τ and write $\Delta_g Y = \bar D^1_{g^{-1}} \bar D^2_g Y$. For each arrow $g \in \mathcal{G}$ beginning at some $m \in M$, we may regard Δ_g as a vector space automorphism $A|_m \to A|_m$ and, because τ is both $\bar D^1$ and $\bar D^2$-invariant, it follows that Δ_g is even a Lie algebra automorphism, with respect to the bracket τ. Since $\bar D^j$ extends the canonical representation of $\mathcal{G}$ on A^0, $\Delta_g Y = Y$ for all $Y \in A^0|_m$. So Δ_g is an automorphism of a semisimple Lie algebra fixing all points in a parabolic subalgebra, and whence in some Borel subalgebra, implying $\Delta_g = \mathrm{id}$. We conclude $\bar D^1 = \bar D^2$ and, by Proposition 5.1, that $D^1 = D^2$. □

{all}

10.3. **All parabolic geometries admit compatible Weyl gauges.** Suppose τ is the model data associated with a parabolic geometry (global or infinitesimal). Then, according to Lemma A.3, $T = \mathrm{gr}_0(\tau)$ is a semisimple Lie algebra bundle structure on $\mathrm{gr}(A)$, where A is the underlying filtered Lie algebroid. We accordingly have a *unique* section E of $\mathrm{gr}(A)_0$ that is a fibre-wise grading section of $(\mathrm{gr}(A), T)$, what we shall call the *canonical grading section*.

Proposition. For any global or infinitesimal parabolic geometry, any lift ϵ of the canonical grading section E is a Weyl gauge compatible with the model data τ, and all Weyl gauges compatible with τ are lifts of this form.

The existence of Weyl gauges, combined with Corollary 6.3, delivers:

Corollary. *In a global or infinitesimal parabolic geometry, any Cartan connection compatible with the model data is automatically compatible with the Lie algebroid filtration.*

Proof of proposition. This time we begin with the global case which is simpler. Let $(\mathcal{G}, \tau)$ be a parabolic geometry, M the base manifold, and D a Cartan connection compatible with τ and the filtration of the Lie algebroid A of $\mathcal{G}$. Since τ is $\bar D$-invariant (D and τ are compatible) T is invariant with respect to the associated representation of $\mathcal{G}$ on $\mathrm{gr}(A)$, that we have also been denoting $\bar D$ (see 3.7). In particular, for any arrow $g \in \mathcal{G}$ beginning at some $m \in M$, $\bar D_g$ must map grading elements of $\mathrm{gr}(A)|_m$ to grading elements of $\mathrm{gr}(A)|_{gm}$, which, by the aforementioned uniqueness, means $\bar D_g E(m) = E(gm)$, i.e., E is $\bar D$-invariant, which is the same as Ad-invariance, since $\bar D$ restricts to the canonical representation on A^0. It follows that any lift ϵ of E through $A^0 \to A^0/A^1$ will be a Weyl gauge for $\mathcal{G}$. It is not hard to see that every Weyl connection is of this form.

Turning to the infinitesimal case, let (A, τ) be an infinitesimal parabolic geometry and ∇ a Cartan connection compatible with τ and the filtration of A. Since τ is $\bar\nabla$-invariant (compatibility) T is invariant with respect to the associated representation of A on $\mathrm{gr}(A)$, which we have also been denoting by $\bar\nabla$. Let $X \in \Gamma(A)$, $j \in \mathbb{Z}$ and

$Y \in \Gamma(\mathrm{gr}(A)_j)$ be arbitrary. Then $T(E,Y) = jY$ and, applying $\bar{\nabla}_X$ to both sides, we obtain, courtesy of the $\bar{\nabla}$-invariance of T,

(1) {won}
$$T(\bar{\nabla}_X E, Y) + T(E, \bar{\nabla}_X Y) = j\bar{\nabla}_X Y.$$

On the other hand, the grading of $\mathrm{gr}(A)$ is $\bar{\nabla}$-invariant (the filtration of A is $\bar{\nabla}$-invariant) so that $\bar{\nabla}_X Y \in \Gamma(\mathrm{gr}(A)_j)$, implying $T(E, \bar{\nabla}_X Y) = j\bar{\nabla}_X Y$. From (1) we now conclude that $T(\bar{\nabla}_X E, Y) = 0$. From the arbitrariness of our choice of j and Y above, we conclude that $\bar{\nabla}_X E$ is fibre-wise in the center of a semisimple Lie algebra, and consequently vanishes. As $\bar{\nabla}$ restricts to the canonical representation on $\mathrm{gr}(A^0)$, this shows that E is ad-invariant. Accordingly, any lift ϵ of E through $A^0 \to A^0/A^1$ will be a Weyl gauge for A. It is not hard to see that every Weyl connection is of this form. □

Having already established, in Theorem 10.2, that the (infinitesimal) Cartan connections associated with (infinitesimal) parabolic geometries are automatically strongly compatible with the model data, we obtain:

Corollary. *Every (infinitesimal) parabolic geometry is an (infinitesimal) filtered Cartan geometry.*

{subgroups}

10.4. **On the Lie subgroupoid $\mathcal{G}^1$ integrating A^1.**

Proposition. Let $(\mathcal{G}, \tau)$ be a parabolic geometry, and let $\mathcal{G}^1 \subset \mathcal{G}$ denote the source-connected subgroupoid integrating A^1. Then $\mathcal{G}^1$ is a bundle of simply-connected nilpotent Lie groups; it is also a normal subgroupoid of $\mathcal{G}$, so that $\mathcal{G}/\mathcal{G}^1$ is a Lie groupoid, also transitive.

Proof. The first result claim follows from Proposition 4.3(3), the second from Theorem 4.3(4). □

{existence _}

10.5. **The existence of parabolic geometries with given Levi reduction.** The preceding corollary shows that that following result is a special case of Theorem 7.5(3):

Theorem. *Every Levi structure $(\mathcal{H}, T, R)$ is the reduction of some parabolic geometry $(\mathcal{G}, \tau)$. If the Levi structure is regular, then the parabolic geometry can be chosen to be regular.*

{normalization _}

10.6. **Normalizing model curvature.** We turn now to the proof of the following central result:

Theorem. *Let $(\mathcal{G}, \tau)$ be a regular parabolic geometry. Then there exists normal model data τ' such that $(\mathcal{G}, \tau')$ is a regular parabolic geometry with the same Levi reduction.*

Let A be the Lie algebroid of $\mathcal{G}$.

Lemma A. Suppose $\sigma \in \Gamma(\wedge^n A^1 \otimes A)$ and $\partial^\tau \sigma$ is homogeneous of degree l. Then there exists $\tilde{\sigma} \in \Gamma(\wedge^n A^1 \otimes A)$, homogeneous of degree l, such that $\partial^\tau \tilde{\sigma} = \partial^\tau \sigma$.

Proof. Any lift ϵ of the canonical grading section E of $\mathrm{gr}(A)$ defines a splitting $A = \oplus_{j\in\mathbb{Z}} A_j$ for the filtration of A, which in turn gives us a splitting of $C^n := \wedge^n A^1 \otimes A$, which is filtered, namely $C^n = \oplus_{m\in\mathbb{Z}} C^n_m$, where $C^n_m = \oplus_{j\geqslant 1; j+k=m} \wedge^n A_j \otimes A_k$. In particular, writing $\sigma = \sum_m \sigma_m$ (finite sum) for some $\sigma_m \in C^n_m$, we have $\sum_m \partial^\tau \sigma_m \in$

$(C^{n-1})^l$, by hypothesis. But from this is follows that $\partial^\tau \sigma_j = 0$ for all $j < l$, because ∂^τ maps C^n_m to C^{n-1}_m, for all m. We may therefore take $\tilde{\sigma} = \sum_{j \geqslant l} \sigma_j$. □

If τ' is also model data on A, and if $\tau' = (\mathrm{Ad}_\psi)_* \tau$, for some $\psi \in T^*M \otimes A$ homogeneous of degree $l \geqslant 1$, then we will write $\tau \overset{l}{\sim} \tau'$.

Lemma B. If $\sigma \in \Gamma(\wedge^n A^1 \otimes A)$ and $\tau' \overset{l}{\sim} \tau$, then $\partial^{\tau'}\sigma - \partial^\tau \sigma$ is homogeneous of degree $l+1$

Proof. Expanding the definition of Ad_ψ (Equation 8.1(2)) it is not difficult to see that $\tau' \overset{l}{\sim} \tau$ implies that $\tau' - \tau$ is homogeneous of degree l. The lemma follows easily from this fact and the definition of ∂^τ (Equation 3.10(2)). □

Proof of theorem. Recall that *normal* means $\partial^\tau \kappa = 0$. To prove existence of normal model data it suffices to prove the following:

> *If $\partial^\tau \kappa$ is homogeneous of degree $l \geqslant 1$, then there exists model data τ' with $\tau' \overset{l}{\sim} \tau$ such that $\partial^{\tau'}\kappa$ is homogeneous of degree $l+1$, where κ' is the model curvature of τ'.*

For then the theorem follows from an induction on l, an induction anchored by the fact that κ (and hence $\partial^\tau \kappa$)) is homogeneous of degree one (Proposition 9.1).

So, suppose $\partial^\tau \kappa$ is homogeneous of degree $l \geqslant 1$. Recall that gr is a chain map between the chain complexes 9.3(3) and 9.3(4), so that we have the identity

(1) {wood0}
$$\partial^* \circ \mathrm{gr}_l = \mathrm{gr}_l \circ \partial^\tau.$$

Identifying T^*M with A^1 using the Killing form of τ, there exists, by Lemma A, $\tilde{\kappa} \in \Gamma(\wedge^2 T^*M \otimes A)$, homogeneous of degree l, such that $\partial^\tau \tilde{\kappa} = \partial^\tau \kappa$. From the boxed equation in 9.5(2), there exists $\psi \in \Gamma(T^*M \otimes A)$, homogeneous of degree l, such that $\mathrm{gr}_l(\tilde{\kappa}) - \partial\, \mathrm{gr}_l(\psi)$ lies in $\ker \partial^*$, so that $\partial^* \mathrm{gr}_l(\tilde{\kappa}) = \partial^* \partial\, \mathrm{gr}_l(\psi)$. Applying (1) to the left-hand side, we can write this as

(2) {wood1}
$$\mathrm{gr}_l(\partial^\tau \kappa) = \partial^* \partial\, \mathrm{gr}_l(\psi),$$

which eliminates the auxiliary variable $\tilde{\kappa}$. On the other hand, if we take $\tau' = (\mathrm{Ad}_\psi)_* \tau$, then Proposition 9.4 tells us that $\kappa' - \kappa$ is homogeneous of degree l with

(3) {wood2}
$$\mathrm{gr}_l(\kappa' - \kappa) = -\partial\, \mathrm{gr}_l(\psi).$$

Now as both $\kappa' - \kappa$ and $\partial^\tau \kappa$ are homogeneous of degree l, so is $\partial^\tau \kappa'$. But, applying ∂^* to both sides of (3) and applying (1) again on the left, we obtain $\mathrm{gr}_l(\partial^\tau \kappa' - \partial^\tau \kappa) = -\partial^* \partial\, \mathrm{gr}_l(\psi)$, which can accordingly be written

$$\mathrm{gr}_l(\partial^\tau \kappa') = \mathrm{gr}_l(\partial^\tau \kappa) - \partial^* \partial\, \mathrm{gr}_l(\psi)$$

Substituting (2), we obtain $\mathrm{gr}_l(\partial^\tau \kappa') = 0$ which, on account of Lemma B, shows $\mathrm{gr}_l(\partial^{\tau'} \kappa') = 0$, as required.

By construction, the normal model data τ' is a ψ-deformation of τ. Keeping Theorem 10.2 in mind, it follows, as a special case of Proposition 8.1(7), that (A', τ') is an infinitesimal parabolic geometry with the same Levi reduction as (A, τ), and in the global case that $(\mathcal{G}, \tau')$ is a parabolic geometry with the same Levi reduction as $(\mathcal{G}, \tau)$.

□

Appendix A. On filtrations of semisimple Lie algebras

{lie_algebra_facts}

While the results proven in this section are well known, we do not know a place where they are established without advanced structure theory, as we do here. We do need Cartan's criterion for semisimplicity, and the fact that all derivations of a semisimple Lie algebra are inner [20]. Among other things we prove Proposition 3.2, restated below for convenience.

{lpp_}

Proposition. (1) Let $\mathfrak{g}$ be a finite-dimensional semisimple Lie algebra over $\mathbb{C}$ or $\mathbb{R}$ and $\mathfrak{p} \subset \mathfrak{g}$ a parabolic subalgebra ($\mathfrak{p}^\perp$ is a nilpotent subalgebra of $\mathfrak{p}$). Define $\mathfrak{g}^0 = \mathfrak{p}$ and $\mathfrak{g}^1 = \mathfrak{p}^\perp$, and inductively define $\mathfrak{g}^j = [\mathfrak{g}^1, \mathfrak{g}^{j-1}]$, for $j \geqslant 2$, and $\mathfrak{g}^j = \{X \in \mathfrak{g} \mid [X, \mathfrak{g}^1] \subset \mathfrak{g}^{j+1}\}$, for $j \leqslant -1$. Then this makes $\mathfrak{g}$ into a filtered Lie algebra with $|\mathfrak{g}| \leqslant k$ for some $k \geqslant 0$. This is called the *canonical filtration* associated with $\mathfrak{p}$.

{lqq_}

(2) In any filtration of $\mathfrak{g}$ defined by a grading $\mathfrak{g} = \oplus_{j\in\mathbb{Z}}\mathfrak{g}_j$, $\mathfrak{g}^0 = \oplus_{j\geqslant 0}\mathfrak{g}_j$ is a parabolic subalgebra. Moreover, this filtration is the precisely the associated canonical filtration, provided $\mathfrak{g}_1$ generates $\mathfrak{g}^1 = \oplus_{j\geqslant 1}\mathfrak{g}_j$.

Proof of first part. By definition, we $\mathfrak{g}^1 \perp \mathfrak{g}^0$, which anchors an inductive proof that $\mathfrak{g}^j \perp \mathfrak{g}^k$, whenever $j + k = 1$. By a similar induction, the reader will also see that $[\mathfrak{g}^j, \mathfrak{g}^k] \subset \mathfrak{g}^{j+k}$. Now $\cap_{j\in\mathbb{Z}}\mathfrak{g}^j = 0$, because $\mathfrak{g}^1 = \mathfrak{p}^\perp$ is nilpotent. So there is $k > 0$ such that $\mathfrak{g}^k \neq 0$, but $\mathfrak{g}^{k+1} = 0$, so that $\mathrm{gr}(\mathfrak{g})_k \neq 0$ but $\mathrm{gr}(\mathfrak{g})_{k+1} = 0$. We have $\mathfrak{g}^{-k} = (\mathfrak{g}^{k+1})^\perp \neq \mathfrak{g}$, i.e., $\mathrm{gr}(\mathfrak{g})_{-k} \neq 0$, and $\mathfrak{g}^{-k} = (\mathfrak{g}^{k+1})^\perp = 0^\perp = \mathfrak{g}$, i.e., $\cup_{j\in\mathbb{Z}}\mathfrak{g}_j = \mathfrak{g}$, and $\mathrm{gr}(\mathfrak{g})_{-k-1} = 0$. □

The converse part is proven in A.3.

{grading_elements}

A.1. Grading elements.

Grading elements have been defined in 1.4.

Lemma. Every graded semisimple Lie algebra has a grading element.

Proof. Define $D\colon \mathfrak{g} \to \mathfrak{g}$ by $DX = jX$ whenever $X \in \mathfrak{g}_j$ and extend by linearity. Then D is a derivation. Since $\mathfrak{g}$ is semisimple, all derivations are inner, so there is E, necessarily unique since $\mathfrak{g}$ has trivial centre, such that $DX = [E, X]$. □

{gradings_from_filtrations}

A.2. Canonical filtrations always split.

Lemma. Let $\mathfrak{g}$ be a finite-dimensional semisimple Lie algebra over $\mathbb{C}$ or $\mathbb{R}$ equipped with the canonical filtration associated with some parabolic subalgebra. Then $\mathrm{gr}(\mathfrak{g})$ has a unique grading element E and the splitting $\Phi_\epsilon\colon \mathrm{gr}(\mathfrak{g}) \to \mathfrak{g}$ defined by any lift $\epsilon \in \mathfrak{g}^0$ of E is a Lie algebroid isomorphism.

Splittings Φ_ϵ were defined in 4.1.

Sketch of proof. As argued above, we have $(\mathfrak{g}^j)^\perp = \mathfrak{g}^k$ whenever $j + k = 1$. This condition allows one to drop the Killing form to a nondegenerate bilinear form on $\mathrm{gr}(\mathfrak{g})$ in which $\mathrm{gr}(\mathfrak{g})_j \perp \mathrm{gr}(\mathfrak{g})_k$, unless $j + k = 0$. One shows that this must coincide with the Killing form on $\mathrm{gr}(\mathfrak{g})$, which means $\mathrm{gr}(\mathfrak{g})$ is semisimple, by Cartan's criterion. It follows from Lemma A.1 that $\mathrm{gr}(\mathfrak{g})$ has a unique grading element E, and consequently that Φ_ϵ is a Lie algebroid isomorphism, by the argument given in 4.1. For details in a more general setting, see [12, Propositions 2.14 & 3.9]. □

{parabolics_from_filtrations}

A.3. **Parabolics from gradings.**

Lemma. Let $\mathfrak{g}$ be a graded, finite-dimensional, semisimple Lie algebra. Then with respect to the Killing form B, we have:

{ft1}

{ft2}

(1) $\mathfrak{g}_j \perp \mathfrak{g}_k$ whenever $j + k \neq 0$.
(2) For all j, $\mathfrak{g}_j$ and $\mathfrak{g}_{-j}$ are B-dual. In particular, $\dim \mathfrak{g}_j = \dim \mathfrak{g}_{-j}$, and consequently $|\mathfrak{g}| \leqslant k$ for some $k \in \mathbb{Z}$.

Proof. Let $E \in \mathfrak{g}_0$ be the grading element. Then, for any $X \in \mathfrak{g}_j$ and $Y \in \mathfrak{g}_k$, we have $(j+k)B(X,Y) = B([E,X],Y) + B(X,[E,Y]) = 0$, by ad-invariance of the Killing form B. Equation (1) follows, while (2) is a necessary corollary, to ensure nondegeneracy of B. □

To prove part (2) of the proposition above, suppose $\mathfrak{g}$ is graded. Applying (1), we obtain $\mathfrak{g}^0 = \mathfrak{g}_0 \oplus \mathfrak{g}_1 \oplus \cdots \oplus \mathfrak{g}_k \subset (\mathfrak{g}_1 \oplus \cdots \oplus \mathfrak{g}_k)^\perp = (\mathfrak{g}^1)^\perp$. By (2), nondegeneracy of B and a dimension count, this inclusion is actually equality. In fact, similar arguments actually establish

(3) {trees}
$$\mathfrak{g}^j = (\mathfrak{g}^k)^\perp \quad \text{whenever } j + k = 1.$$

In any case, since $\mathfrak{g}^1$ is nilpotent, by the definition of Lie algebra filtrations, $\mathfrak{g}^0$ is parabolic, by the definition of parabolic given in 3.2.

Let us write the canonical filtration of $\mathfrak{g}$ associated with $\mathfrak{p} := \mathfrak{g}^0$ as $\mathfrak{g} = \tilde{\mathfrak{g}}^{-k} \supset \ldots \supset \tilde{\mathfrak{g}}^k$. So far it is clear that $\mathfrak{g}^0 = \tilde{\mathfrak{g}}^0$ and $\mathfrak{g}^1 = \tilde{\mathfrak{g}}^1$. For $j \geqslant 2$, $\mathfrak{g}^j = \tilde{\mathfrak{g}}^j$ follows the definition of $\tilde{\mathfrak{g}}^j$ and the assumption that $\mathfrak{g}_1$ generates $\mathfrak{g}^1$. For $j \leqslant -1$, equality follows from the $j \leqslant 1$ case and the fact that $\tilde{\mathfrak{g}}^j = (\tilde{\mathfrak{g}}^k)^\perp$ whenever $j + k = 1$ (see the proof of part (1) of the proposition) and its analogue (3).

Appendix B. The space of splittings of a filtered Lie algebra

{appendix_splittings}

In this section we prove Lemma 4.1, which applies to arbitrary graded Lie algebras, not just semisimple ones. While a known result, we have given a formulation fit for current purposes. For convenience, here is a restatement:

Lemma. Let $\mathfrak{g}$ be a filtered Lie algebra and let $\epsilon \in \mathfrak{g}^0$ be a lift of some grading element $E \in \mathrm{gr}(\mathfrak{g})$. Let G^0 be *any* Lie group integrating $\mathfrak{g}^0$, and let $G^1 \subset G^0$ be the connected Lie subgroup with Lie algebra $\mathfrak{g}^1$. Then:

(1) Under the adjoint action, G^1 acts freely and transitively on the affine space $\epsilon + \mathfrak{g}^1 \subset \mathfrak{g}^0$, and consequently on E-splittings of $\mathfrak{g}$. In particular, G^1 is simply connected (i.e., $G^1 \cong \exp(\mathfrak{g}^1)$) so that the infinitesimal adjoint action of $\mathfrak{g}^1$ on $\mathfrak{g}$ has a unique globalization $\mathrm{Ad}\colon G^1 \to \mathrm{Aut}(\mathfrak{g})$ extending the adjoint action just mentioned.

{splittings1_}

(2) For any $\epsilon_1, \epsilon_2 \in \epsilon + \mathfrak{g}^1$, we have $\Phi_{\epsilon_2} = \mathrm{Ad}_g \circ \Phi_{\epsilon_1}$, where $g \in G^1$ is the unique element for which $\epsilon_2 = \mathrm{Ad}_g\, \epsilon_1$.

{splittings2_}

Proof. Note first that the filtration property ensures that the action of $\mathfrak{g}^1$ on $\mathfrak{g}^0$ is filtration-preserving, and the corresponding action on $\mathrm{gr}(\mathfrak{g}^0)$ is trivial. The same must be true for adjoint action of G^1, as G^1 is connected. In particular $e + \mathfrak{g}^1$ must be G^1-invariant, for any lift $e \in \mathfrak{g}^0$ of E. Suppose that the action of G^1 on $e + \mathfrak{g}^1$ is infinitesimally free. Then, by a dimension count, this action is infinitesimally transitive, and so transitive, since G^1 is connected. Infinitesimal freeness means the

stabilizer subgroup H for the action is discrete, so that $G^1 \to G^1/H \cong e + \mathfrak{g}^1$ is a covering space. Since $e + \mathfrak{g}^1$ is simply connected, H is trivial and the remaining claims in (1) follow.

To show that the action of G^1 on $e + \mathfrak{g}^1$ is infinitesimally free, for each $X \in \mathfrak{g}^1$, let $X^\dagger$ denote the infinitesimal generator of X, a vector field on $e + \mathfrak{g}^1$. We have $X^\dagger(e') = [X, e']$, for $e' \in e + \mathfrak{g}^1$. Supposing $X^\dagger(e') = 0$, write $X = \sum_{j>0} X_j$ for some $X_j \in \mathfrak{g}_j$, where $\mathfrak{g} = \oplus_j \mathfrak{g}_j$ is the E-splitting corresponding to e'. Then we conclude that $\sum_{j>0} jX_j = 0$, from which it follows that $X = 0$.

To prove (2), observe that Φ_e is characterized among all isomorphisms $\mathrm{gr}(\mathfrak{g}) \to \mathfrak{g}$ by the requirement

$$[\Phi_e(X), e] = j\Phi_e(X) \quad \text{for all } X \in \mathrm{gr}_j(\mathfrak{g}),\ j \in \mathbb{Z}.$$

Taking $e = e_1$ and applying Ad_g to both sides, we obtain $[\mathrm{Ad}_g \Phi_{e_1} X, \mathrm{Ad}_g e_1] = j \,\mathrm{Ad}_g e_1$, i.e.,

$$[(\mathrm{Ad}_g \circ \Phi_{e_1})X, e_2] = je_2.$$

So, by the aforementioned characterization, we conclude $\mathrm{Ad}_g \circ \Phi_{e_1} = \Phi_{e_2}$. □

Appendix C. Cartan connections on transitive Lie groupoids from representations

{rut}

In this section we prove 5.1, restated here for convenience:

Proposition. Suppose $\mathcal{G}$ is a Lie groupoid with a transitive Lie algebroid A. Then for every representation ρ of $\mathcal{G}$ on A, extending the canonical representation of $\mathcal{G}$ on the anchor kernel, there exists a unique Cartan connection D on $\mathcal{G}$ such that $\bar{D} = \rho$.

First note that the kernel of the canonical projection $J^1\mathcal{G} \to \mathcal{G}$ is a totally intransitive Lie groupoid whose form we can show is independent of the particular integration $\mathcal{G}$ of A. This kernel is denoted $\mathrm{Aut}(TM, A)$ and described further in Section 8. For present purposes, the following properties suffice:

Lemma. If the Lie algebroid A of a Lie groupoid $\mathcal{G}$ is transitive, then:

(1) The adjoint representation of $J^1\mathcal{G}$ on A restricts to representation of $\mathrm{Aut}(TM, A) \subset J^1\mathcal{G}$ that is faithful. {kr1}

(2) For any $m \in M$ and each linear map $L\colon A|_m \to A|_m$, restricting to the identity on the anchor kernel, and whose corresponding factoring $T_mM \to T_mM$ through the anchor $\#\colon A \to TM$ is an isomorphism, there exists a unique $\psi \in \mathrm{Aut}(TM, A)|_m$ such that $\mathrm{Ad}_\psi = L$. {kr2}

Proof of lemma. Part (1) follows from Proposition 8.3, and part (2) from 2 and transitivity. □

Proof of proposition. We seek a Lie groupoid morphism $S\colon \mathcal{G} \to J^1\mathcal{G}$ that is a right-inverse for the projection $\pi\colon J^1\mathcal{G} \to \mathcal{G}$ and satisfies

$$\mathrm{Ad}_{S(g)} = \rho_g \qquad g \in \mathcal{G}. \tag{3}$$

{kr3}

To that end, let $\tilde{S}\colon U \to \pi^{-1}(U)$ be any smooth, local, right inverse for π, defined on some open neighborhood $U \subset \mathcal{G}$ (not necessarily respecting groupoid multiplication). Then, for each arrow $g \in U$ beginning at some $m \in M$, $L = \mathrm{Ad}^{-1}_{\tilde{S}(g)} \circ \rho_g$ satisfies the condition in (2), because ρ extends the adjoint representation of $\mathcal{G}$ on the anchor

kernel of A. Accordingly, there exists unique $\psi(g) \in \mathrm{Aut}(TM, A)$ such that $\mathrm{Ad}_{\psi(g)} = \mathrm{Ad}^{-1}_{\tilde{S}(g)} \circ \rho_g$, so that (3) holds if we define $S(g) = \tilde{S}(g)\psi(g)$. It is not hard to see that $S\colon U \to J^1\mathcal{G}$ is smooth. Moreover, from the uniqueness, it is clear that $S(g)$ does not actually depend on the choice of local right inverse $\tilde{S}$, which means we can collate local definitions of S as just described into a global map $S\colon \mathcal{G} \to J^1\mathcal{G}$. From (3), we will have $\mathrm{Ad}_{S(g)S(h)} = \mathrm{Ad}_{S(gh)}$ for all multipliable $g, h \in \mathcal{G}$. Using (1) it is not hard to see that $S(gh) = S(g)S(h)$ must follow, because $S(g)S(h)$ and $S(gh)$ share a common image under the projection $J^1\mathcal{G} \to \mathcal{G}$, namely gh. A similar argument shows that S is uniquely determined by requirement (3).

□

References

[1] Ender Abadoğlu, Ercüment Ortaçgil, and Ferit Öztürk. Klein geometries, parabolic geometries and differential equations of finite type. *J. Lie Theory*, 18(1):67–82, 2008.

[2] T. N. Bailey, M. G. Eastwood, and A. R. Gover. Thomas's structure bundle for conformal, projective and related structures. *Rocky Mountain J. Math.*, 24(4):1191–1217, 1994.

[3] A. D. Blaom. Geometric structures as deformed infinitesimal symmetries. *Trans. Amer. Math. Soc.*, 358:3651–3671, 2006.

[4] A. D. Blaom. Lie algebroids and Cartan's method of equivalence. *Trans. Amer. Math. Soc.*, 364:3071–3135, 2012.

[5] A. D. Blaom. The infinitesimalization and reconstruction of locally homogeneous manifolds. *SIGMA*, 9, 074:19 pages, 2013.

[6] A. D. Blaom. Cartan connections on Lie groupoids and their integrability. *SIGMA Symmetry Integrability Geom. Methods Appl.*, 12:Paper No. 114, 26, 2016.

[7] A. D. Blaom. Pseudogroups via pseudoactions: Unifying local, global, and infinitesimal symmetry. *J. Lie Theory*, 26(2):535–565, 2016.

[8] Robert L. Bryant. *Some Aspects of the Local and Global Theory of Pfaffian Systems*. ProQuest LLC, Ann Arbor, MI, 1979. Thesis (Ph.D.)–The University of North Carolina at Chapel Hill.

[9] Robert L. Bryant. Conformal geometry and 3-plane fields on 6-manifolds. In *"Developments of Cartan Geometry and Related Mathematical Problems", RIMS Symposium Proceedings*, volume 1502, 2006.

[10] F. E. Burstall. A remark on parabolic subalgebras, 2010. https://people.bath.ac.uk/feb/parabolic.pdf.

[11] David M. J. Calderbank and Tammo Diemer. Differential invariants and curved Bernstein-Gelfand-Gelfand sequences. *J. Reine Angew. Math.*, 537:67–103, 2001.

[12] David M. J. Calderbank and Passawan Noppakaew. Parabolic subalgebras, parabolic buildings and parabolic projection, 2016.

[13] A. Cannas da Silva and A. Weinstein. *Geometric Models for Noncommutative Algebras*, volume 10 of *Berkeley Mathematics Lecture Notes*. American Mathematical Society, Providence, RI, 1999.

[14] Andreas Čap and Jan Slovák. *Parabolic geometries. I*, volume 154 of *Mathematical Surveys and Monographs*. American Mathematical Society, Providence, RI, 2009. Background and general theory.

[15] M. Crainic and R. L. Fernandes. Lectures on integrability of Lie brackets. In *Lectures on Poisson geometry*, volume 17 of *Geom. Topol. Monogr.*, pages 1–107. Geom. Topol. Publ., Coventry, 2011.

[16] Marius Crainic, Maria Amelia Salazar, and Ivan Struchiner. Multiplicative forms and Spencer operators. *Math. Z.*, 279(3-4):939–979, 2015.

[17] Jean-Paul Dufour and Nguyen Tien Zung. *Poisson structures and their normal forms*, volume 242 of *Progress in Mathematics*. Birkhäuser Verlag, Basel, 2005.

[18] Charles Ehresmann. Les connexions infinitésimales dans un espace fibré différentiable. In *Colloque de topologie (espaces fibrés), Bruxelles, 1950*, pages 29–55. Georges Thone, Liège; Masson et Cie., Paris, 1951.

[19] Charles Ehresmann. Catégories topologiques et catégories différentiables. In *Colloque Géom. Diff. Globale (Bruxelles, 1958)*, pages 137–150. Centre Belge Rech. Math., Louvain, 1959.

[20] Alexander Kirillov, Jr. *An introduction to Lie groups and Lie algebras*, volume 113 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, Cambridge, 2008.

[21] Bertram Kostant. Lie algebra cohomology and the generalized Borel-Weil theorem. *Ann. of Math. (2)*, 74:329–387, 1961.

[22] K. C. H. Mackenzie. *General Theory of Lie Groupoids and Lie Algebroids*, volume 213 of *London Mathematical Society Lecture Note Series*. Cambridge University Press, Cambridge, 2005.

[23] I. Moerdijk and J. Mrčun. *Introduction to foliations and Lie groupoids*, volume 91 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, Cambridge, 2003.

[24] Tohru Morimoto. Geometric structures on filtered manifolds. *Hokkaido Math. J.*, 22(3):263–347, 1993.

[25] A. S. Salazar. *Pfaffian groupoids*. PhD thesis, University of Utrecht, The Netherlands, 2013. http://arxiv.org/abs/1306.1164.

[26] R. W. Sharpe. *Differential Geometry: Cartan's generalization of Klein's Erlangen program*, volume 166 of *Graduate Texts in Mathematics*. Springer, 1997.

[27] Giorgio Trentinaglia. Reduced smooth stacks? *Theory Appl. Categ.*, 30:Paper No. 31, 1032–1066, 2015.

[28] Andreas Čap, Jan Slovák, and Vladimír Souček. Bernstein-Gelfand-Gelfand sequences. *Ann. of Math. (2)*, 154(1):97–113, 2001.

[29] Andreas Čap and Vladimír Souček. Curved Casimir operators and the BGG machinery. *SIGMA Symmetry Integrability Geom. Methods Appl.*, 3:Paper 111, 17, 2007.

[30] Andreas Čap and Vladimír Souček. Subcomplexes in curved BGG-sequences. *Math. Ann.*, 354(1):111–136, 2012.

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